Supplementary Material for the Article "**A Heterogeneous Graph Metapath Attention-based Multi-Agent Collaborative Optimization Framework for Integrated Cloud-Edge Collaborative Scheduling**"

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This material includes seven supplementary parts of the article "**A Heterogeneous Graph Metapath Attention-based Multi-Agent Collaborative Optimization Framework for Integrated Cloud-Edge Collaborative Scheduling**".

PART 1. COMPLETE MODEL NOTATIONS

Table 1
NOTATIONS AND DEFINITIONS OF THE COMPLETE CEC-MCISP MODEL.

Symbol	Description
Index	
i, i'	Index of tasks, $i, i' = 1, 2, ..., n$.
j, j'	Index of subtasks, $j, j' = 1, 2, ..., n_i$.
k	Index of devices, $k = 1, 2, ..., N$.
l	Index of edge servers, $l = 1, 2, ..., L$.
l'	Index of cloud servers, $l' = 1, 2, ..., U$.
Set	
$\boldsymbol{M}$	Set of manufacturing tasks, $\boldsymbol{M} = \{M_1, M_2, \ldots, M_n\}$.
$\boldsymbol{C}$	Set of computing tasks, $\boldsymbol{C} = \{C_1, C_2, \ldots, C_n\}$.
$\boldsymbol{O}_i$	Set of manufacturing subtasks, $\boldsymbol{O}_i = \{O_{i,1}, O_{i,2}, \ldots, O_{i,n_i}\}$.
$\boldsymbol{Q}_k$	Set of waiting tasks on the *k*-th device, $\boldsymbol{Q}_k = \{Q_{k,1}, Q_{k,2}, \ldots, Q_{k,R}\}$.
$\mathcal{M}$	Set of manufacturing devices, $\mathcal{M} = \{\mathcal{M}_1, \mathcal{M}_2, ..., \mathcal{M}_N\}$.
$\mathcal{E}$	Set of edge servers, $\mathcal{E} = \{\mathcal{E}_1, \mathcal{E}_2, ..., \mathcal{E}_L\}$.
$\mathcal{C}$	Set of cloud servers, $\mathcal{C} = \{\mathcal{C}_1, \mathcal{C}_2, ..., \mathcal{C}_U\}$.
Λ_i	Set of predecessor computational tasks of computing task $\boldsymbol{C}_i$ before execution $\Lambda_i = \{\Lambda_{i,1}, \Lambda_{i,2}, ..., \Lambda_{i,n}\}$.
Parameters	
n	Number of manufacturing tasks and computing tasks.
n_i	Number of subtasks for the manufacturing task M_i .
N	Number of manufacturing devices.
L	Number of edge servers.
U	Number of cloud servers.
Z	Length of the waiting queue on a device.
$O_{i,j}$	Manufacturing subtask with index j of task M_i .
M_i	Manufacturing task with index i .
C_i	Computational task with index i .
$T_{M_i}^{start}$	Start time of manufacturing task M_i .
$T_{M_i}^{end}$	Completion time of manufacturing task M_i .
T^{end}	Completion time of all manufacturing tasks.
$T_{C_i}^{start}$	Start execution time of computational task C_i .
$T_{C_i}^{end}$	Completion time of computational task C_i .
$T_{C_i}^{process}$	Execution time of the computational task C_i .
$T_{\mathcal{M}_k}^{start}$	Start working time of device $\mathcal{M}_k$.
$T_{\mathcal{M}_k}^{end}$	End working time of device $\mathcal{M}_k$.
W_{C_i}	Energy consumption of computational task C_i .

Symbol	Description
W_{M_i}	Energy consumption of manufacturing task M_i.
$W^{(m)}$	Total energy consumption of all manufacturing tasks.
$T_{O_{i,j}}^{start}$	Start execution time of subtask $O_{i,j}$ of task M_i.
$T_{O_{i,j}}^{process}$	Processing time of subtask $O_{i,j}$ of task M_i.
$T_{O_{i,j}}^{end}$	Completion time of subtask $O_{i,j}$ of task M_i.
$T_{O_{i,j-1}}^{end}$	Completion time of subtask $O_{i,j-1}$ of subtask $O_{i,j}$.
$T_{\mathcal{M}_{j-1}\to\mathcal{M}_j}^{trans}$	Transportation time required to move from the device executing subtask $O_{i,j-1}$ to the device executing $O_{i,j}$.
$T_{edge,C_i}^{process}$	Processing time of computing task C_i on an edge server.
$T_{cloud,C_i}^{process}$	Processing time of computing task C_i on a cloud server.
T_x^{end}	Completion time of the predecessor computational tasks of a given computing task on the server.
$Y_{i',j',k}$	Priority of subtask $O_{i',j'}$ among other tasks in the task queue of device $\mathcal{M}_k$.
$Y_{i,j,k}$	Priority of subtask $O_{i,j}$ in the task queue of device $\mathcal{M}_k$.
$P_{\mathcal{M}_k}$	Manufacturing power of device $\mathcal{M}_k$.
$P_{\mathcal{E}_l}$	Computing power of edge server $\mathcal{E}_l$.
$P_{\mathcal{C}_{l'}}$	Computing power of cloud server $\mathcal{E}_{l'}$.
ω	Nominal basic channel bandwidth of the 5G network.
$\rho_{\mathcal{M}_k\to\mathcal{E}_l}(t)$	Available-bandwidth ratio of the link from device $\mathcal{M}_k$ to edge server $\mathcal{E}_l$ at time t.
ϖ_0	Communication noise in the 5G network.
$G_{\mathcal{M}_k}$	Transmission power of device $\mathcal{M}_k$.
$\tilde{G}_{\mathcal{M}_k\to\mathcal{E}_l}$	Channel gain from device $\mathcal{M}_k$ to edge server $\mathcal{E}_l$.
C_i^{memory}	Memory requirement of computational task C_i.
$\mathcal{E}_l^{memory}$	Memory capacity provided by edge server $\mathcal{E}_l$.
$\mathcal{C}_{l'}^{memory}$	Memory capacity provided by cloud server $\mathcal{C}_{l'}$.
C_i^{cpu}	CPU core requirement of computational task C_i.
$\mathcal{E}_l^{cpu}$	CPU cores provided by edge server $\mathcal{E}_l$.
$\mathcal{C}_{l'}^{cpu}$	CPU cores provided by cloud server $\mathcal{C}_{l'}$.
$T_{\mathcal{M}_k\to\mathcal{E}_l}^{up}(t)$	Upload time from device $\mathcal{M}_k$ to edge server $\mathcal{E}_l$ under the effective transmission rate at time t.
H_i^{up}	Total input-data volume to be uploaded for a computational task.
H^{down}	Amount of data downloaded from the cloud server to the edge server.
$v_{edge\to cloud}(t)$	Effective transmission rate from the edge server to the cloud server at time t.
$v_{cloud\to edge}(t)$	Effective transmission rate from the cloud server to the edge server at time t.
$v_{\mathcal{M}_k\to\mathcal{E}_l}(t)$	Transmission rate from device $\mathcal{M}_k$ to edge server $\mathcal{E}_l$ at time t.
α_1, α_2	Weighting factors for energy consumption associated with CPU and memory resource usage.
λ_1, λ_2	Weighting factors for time and energy consumption.
Variables	
$X_{i,j,k}$	1 if subtask $O_{i,j}$ is assigned to device $\mathcal{M}_k$; 0 otherwise
$F_{i,j,k}$	1 if executed with priority on $\mathcal{M}_k$; 0 otherwise.
B_i	1 if computational task C_i is executed on an edge server; 0 if it is executed on a cloud server.

PART 2. STATE PARAMETERS OF THE THREE AGENTS

The state of the task agent $Agent^{task}$ is represented as follows: Assume that the number of manufacturing tasks to be scheduled is n. For each manufacturing task, the state at the t-th time step, denoted as s_t^{task}, is described by Equation (1), and the corresponding local manufacturing subtask $\Psi_{i,t}$ is represented by Equation (2):

$$s_t^{task} = \left\{ \Psi_{1,t}, \Psi_{2,t}, ..., \Psi_{N,t} \right\}, \forall t \in \{0,1,...,T\} \tag{1}$$

$$\Psi_{i,t} = \left\{ \psi_t^{(1)}, \psi_t^{(2)}, \psi_t^{(3)}, \psi_t^{(4)}, \psi_t^{(5)}, \psi_t^{(6)}, \psi_t^{(7)}, \psi_t^{(8)}, \psi_t^{(9)}, \psi_t^{(10)} \right\}, \forall i \in \{1,2,...,n\} \tag{2}$$

where T denotes the terminal time step, and $\Psi_{i,t}$ represents the state feature vector of the i-th manufacturing task with a dimension of 10. Let $\mathcal{J}_t$ denote the index of the operation being processed at the t-th time step, and T_t denote the current timestamp at the t-th time step. The ten state elements of $\Psi_{i,t}$ are introduced in Table 2, where each state element is normalized in the form of a fraction. The time step advances from t to $t+1$ when at least one device is available to trigger a state transition.

Table 2
DEFINITIONS OF THE STATE FEATURES OF THE TASK AGENT.

Status Item	Definition	Description
$\psi_t^{(1)}$	$\psi_t^{(1)} \in \{0,1\}$	Indicates whether the current subtask $O_{i,j}$ is being processed by a device; it is 1 if yes, otherwise 0.
$\psi_t^{(2)}$	$= \frac{t_{O_{i,\mathcal{J}_t}} - \left(T_t - T_{O_{i,\mathcal{J}_t}}^{start}\right)}{maxt_{O_{i,j}}}$	The remaining processing time of the current operation of the manufacturing task, scaled by the maximum processing time among all operations in the initial instance.
$\psi_t^{(3)}$	$= \frac{T_t - T_{M_i}^{start}}{maxt_{O_{i,j}}}$	The percentage of completed operations for the i-th job.
$\psi_t^{(4)}$	$= \frac{t_{O_{i,\mathcal{J}_t}} - (T_t - T_{O_{i,\mathcal{J}_t}}^{(start)}) + \sum_{j=\mathcal{J}_t+1}^{M} t_{O_{i,j}}}{max\sum_{j=1}^{M} t_{O_{i,j}}}$	The remaining time before the completion of the i-th job, ignoring waiting time, scaled by the maximum job processing time among all jobs in the initial instance.
$\psi_t^{(5)}$	$= \frac{T_{\mathcal{M}_k}^{end} - T_t}{maxt_{O_{i,j}}}$	The time required until the next idle time of the device, scaled by the maximum operation processing time among all operations in the initial instance.
$\psi_t^{(6)}$	$= \frac{max_{O_{i',j'} \in \mathbf{Q}_k, Y_{i',j',k} > Y_{i,j,k}} T_{O_{i',j'}}^{end} - T_{O_{i,\mathcal{J}_t-1}}^{end}}{\sum_i \sum_j t_{O_{i,j}}}$	The idle time during which the job has not been processed since its last operation, ignoring waiting time, scaled by the sum of processing times of all operations in the initial instance.
$\psi_t^{(7)}$	$= \sum_{j=1}^{\mathcal{J}_t} \psi_t^{(6)}$	Cumulative idle time of the job.
$\psi_t^{(8)}$	$= \frac{\sigma_k}{maxto_{i,j}}$	The time threshold required for the next operation on the k-th device, scaled by the maximum operation processing time among all operations in the initial instance.
$\psi_t^{(9)}$	$\in \{0,1\}$	Computation task trigger: 1 if the current operation is the final one and its corresponding computation task is unallocated, else 0.
$\psi_t^{(10)}$	$= \begin{cases} C_i^{cpu} \\ C_i^{memory} \end{cases}$	Represents the resource demand intensity of the computing task corresponding to the manufacturing task.

The state of the device agent $Agent^{dev}$ is defined as follows: For each device node $\mathcal{M}_k$, the corresponding local state at time step t, denoted as s_t^{dev}, is used to reflect the scheduling status of the associated device. The feature set of each device node is represented by Equation (3):

$$s_t^{dev} = \left\langle \wp_t^{(1)}, \wp_t^{(2)}, \wp_t^{(3)}, \wp_t^{(4)}, \wp_t^{(5)}, \wp_t^{(6)} \right\rangle \tag{3}$$

Table 3

DEFINITIONS OF THE STATE FEATURES OF THE DEVICE AGENT.

Status Item	Definition	Description
$\wp_t^{(1)}$	$\in \{0,1\}$	Indicates whether the device is idle; it is 1 if idle, otherwise 0.
$\wp_t^{(2)}$	$= \frac{T_{O_{i,j}}^{end} - T_t}{T_{O_{i,j}}^{process}}$	The ratio of the remaining time until the completion of the current processing task on the device.
$\wp_t^{(3)}$	$= \frac{T_{\mathcal{M}_k}^{end}}{T^{end}}$	The utilization rate of the device resource.
$\wp_t^{(4)}$	$= \sum_{O_{i,j} \in Q_k} T_{O_{i,j}}^{process}$	The total remaining processing time of all operations in the current queue.
$\wp_t^{(5)}$	$= P_{\mathcal{M}_k} \cdot \left(T_t - T_{\mathcal{M}_k}^{start}\right)$	Represents the cumulative energy consumption of the current equipment.
$\wp_t^{(6)}$	$= \left\{d_{\mathcal{M}_k \to \mathcal{E}_1}, d_{\mathcal{M}_k \to \mathcal{E}_2}, ..., d_{\mathcal{M}_k \to \mathcal{E}_l}\right\}$	The distances from the device to each edge server.

The state of the server agent $Agent^{ser}$ is defined as follows: For each edge server node, the corresponding local state at time step t, denoted as s_t^{ser}, is used to reflect the utilization status of the associated computing resources. The feature set of each server node is represented by Equation (4):

$$s_t^{ser} = \left\langle \xi_t^{(1)}, \xi_t^{(2)}, \xi_t^{(3)}, \xi_t^{(4)} \right\rangle \tag{4}$$

Table 4

DEFINITIONS OF THE STATE FEATURES OF THE SERVER AGENT.

Status Item	Definition	Description
$\xi_t^{(1)}$	$\in \{0,1\}$	Node type indicator: 1 if the node is a cloud server, 0 if it is an edge server. It is used to model scheduling preferences for heterogeneous resources.
$\xi_t^{(2)}$	$= \frac{\bar{\mathcal{E}}_l^{cpu}}{\mathcal{E}_l^{cpu}} + \frac{\bar{\mathcal{E}}_l^{memory}}{\mathcal{E}_l^{memory}}$	Represents the current overall resource idle rate of the server; a larger value indicates more abundant resources.
$\xi_t^{(3)}$	$= P_{\mathcal{E}_l} \cdot \left(T_t - T_{\mathcal{E}_l}^{start}\right)$	Represents the cumulative energy consumption of this edge server.
$\xi_t^{(4)}$	$\rho_{\mathcal{M}_k \to \mathcal{E}_l}(t) \cdot \omega$	Effective bandwidth of the candidate device-to-edge link at decision time t, accounting for link-specific time-varying bandwidth availability.

PART 3. ABLATION EXPERIMENTS

A. Episode-Level Dynamic Objective Weighting

This part first presents the episode-level dynamic weighting mechanism used in HGMAPPO and then evaluates its contribution through ablation experiments. The multi-objective solution set is obtained by independently training policies with different preference coefficients $\beta \in [0.01, 0.99]$. For each policy, β specifies the intended preference between TEC and makespan, while the effective coefficient β_j is adjusted between episodes to compensate for differences in the magnitudes of the two reward components. For a transition in episode j, the TEC and makespan improvements are defined as $\Delta E_t = E(s_t) - E(s_{t+1})$, $\Delta T_t = T(s_t) - T(s_{t+1})$. N_j denote the number of decision steps in episode j. After this episode terminates, the average absolute magnitudes of the two reward components are calculated as $d_j^E = \frac{1}{N_j}\sum_{t=1}^{N_j}|\Delta E_t|$, $d_j^T = \frac{1}{N_j}\sum_{t=1}^{N_j}|\Delta T_t|$. The exponential moving-average magnitudes used by the subsequent episode are updated as $m_{j+1}^E = (1-\eta_w)m_j^E + \eta_w d_j^E$, $m_{j+1}^T = (1-\eta_w)m_j^T + \eta_w d_j^T$, where $\eta_w \in (0,1)$ is the smoothing coefficient. The statistics are initialized as $m_1^E = m_1^T = 1$, giving $\beta_1 = \beta$. At the beginning of episode j, the effective TEC coefficient is calculated as $\beta_j = \frac{\beta m_j^T}{\beta m_j^T + (1-\beta)m_j^E}$. The shared reward within episode j is therefore $r_t(s_t, a_t, s_{t+1}) = \beta_j \times (E(s_t) - E(s_{t+1})) + (1-\beta_j) \times (T(s_t) - T(s_{t+1}))$. The coefficient β_j remains fixed throughout episode j, and the statistics collected during that episode affect only the subsequent episode. When the TEC component has a larger moving-average magnitude, its effective coefficient is reduced; the makespan component is adjusted analogously. This mechanism preserves the preference specified by β while preventing either reward component from dominating solely because of its numerical scale. The following subsection evaluates this mechanism by comparing HGMAPPO with its static-weight variant, HGMAPPO-SW.

B. Ablation Experiments

To examine the contributions of the proposed components, HGMAPPO is compared with six ablation variants. HGMAPPO-SW keeps the effective coefficient fixed as $\beta_j = \beta$ throughout training, whereas the other variants remove or replace different decision-making and representation components. Tables 5-7 report the instance-level HV, GD, and IGD results, respectively.

Each entry in the table represents the mean and variance over 20 independent runs conducted on the same test instance. Each entry reports the mean value, with the standard deviation given in parentheses. Boldface indicates the best mean value in each row. The symbols "+", "≈", and "−" indicate that the compared variant is significantly worse than, statistically indistinguishable from, and significantly better than HGMAPPO, respectively. The last row reports the counts of "+ / ≈ / −" across all instances.

Table 5
STATISTICAL RESULTS OF HGMAPPO AND ITS VARIANTS ON HV METRIC.

Instances	HGMAPPO	HGMAPPO-SW	HGMAPPO-JR	HGMAPPO-MR	HGMAPPO-SR	HGMAPPO-NoHG	HGMAPPO-NoMP
10×5×4×2-2	**9.05e-01 (4.77e-02)**	**8.82e-01 (4.75e-02) (≈)**	3.91e-01 (1.31e-02) (+)	4.57e-01 (1.00e-02) (+)	2.34e-01 (1.05e-02) (+)	4.06e-01 (2.05e-02) (+)	5.27e-01 (2.69e-02) (+)
10×5×4×2-3	**7.74e-01 (1.85e-02)**	**7.86e-01 (1.96e-02) (≈)**	5.10e-01 (2.73e-02) (+)	3.55e-01 (1.11e-02) (+)	4.59e-01 (1.70e-02) (+)	4.40e-01 (1.33e-02) (+)	6.03e-01 (1.41e-02) (+)
10×5×4×2-5	**8.06e-01 (3.32e-02)**	**7.98e-01 (3.09e-02) (≈)**	5.21e-01 (1.44e-02) (+)	4.69e-01 (1.65e-02) (+)	3.22e-01 (8.22e-03) (+)	5.75e-01 (2.25e-02) (+)	6.82e-01 (2.68e-02) (+)
10×5×4×2-8	**7.82e-01 (3.84e-02)**	**7.73e-01 (3.57e-02) (≈)**	3.40e-01 (1.91e-02) (+)	4.04e-01 (1.07e-02) (+)	3.38e-01 (1.51e-02) (+)	5.74e-01 (1.68e-02) (+)	5.14e-01 (2.57e-02) (+)
10×5×4×2-10	**8.26e-01 (3.99e-02)**	**8.12e-01 (4.00e-02) (≈)**	4.49e-01 (2.02e-02) (+)	6.60e-01 (1.52e-02) (+)	7.47e-01 (3.12e-02) (+)	5.81e-01 (2.81e-02) (+)	6.53e-01 (3.38e-02) (+)
10×5×4×2-11	**8.35e-01 (3.43e-02)**	8.09e-01 (4.08e-02) (+)	5.24e-01 (2.45e-02) (+)	6.41e-01 (2.84e-02) (+)	5.78e-01 (1.22e-02) (+)	5.27e-01 (2.82e-02) (+)	5.81e-01 (2.58e-02) (+)
10×5×4×2-13	**8.13e-01 (2.41e-02)**	**8.07e-01 (2.21e-02) (≈)**	5.03e-01 (2.58e-02) (+)	6.56e-01 (2.54e-02) (+)	6.62e-01 (1.53e-02) (+)	6.82e-01 (2.38e-02) (+)	5.94e-01 (1.71e-02) (+)
10×5×4×2-18	**8.25e-01 (5.57e-02)**	**8.04e-01 (4.82e-02) (≈)**	5.13e-01 (2.97e-02) (+)	5.04e-01 (2.38e-02) (+)	3.95e-01 (1.74e-02) (+)	5.44e-01 (3.03e-02) (+)	6.54e-01 (3.55e-02) (+)
10×5×4×2-19	**9.26e-01 (3.69e-02)**	8.81e-01 (4.01e-02) (+)	4.17e-01 (8.51e-03) (+)	7.29e-01 (4.06e-02) (+)	3.03e-01 (1.27e-02) (+)	6.85e-01 (3.68e-02) (+)	6.54e-01 (1.88e-02) (+)
10×5×4×2-20	**8.27e-01 (3.67e-02)**	**8.09e-01 (3.11e-02) (≈)**	4.49e-01 (1.49e-02) (+)	4.90e-01 (2.33e-02) (+)	4.26e-01 (2.13e-02) (+)	7.17e-01 (2.10e-02) (+)	6.17e-01 (2.62e-02) (+)
20×15×10×5-2	**7.60e-01 (1.90e-02)**	7.33e-01 (2.07e-02) (+)	3.59e-01 (1.91e-02) (+)	4.07e-01 (1.07e-02) (+)	4.12e-01 (2.18e-02) (+)	3.73e-01 (2.13e-02) (+)	5.45e-01 (2.71e-02) (+)
20×15×10×5-4	**9.17e-01 (3.04e-02)**	**9.03e-01 (3.12e-02) (≈)**	5.58e-01 (2.25e-02) (+)	2.72e-01 (1.27e-02) (+)	5.81e-01 (2.93e-02) (+)	3.91e-01 (1.78e-02) (+)	6.79e-01 (2.31e-02) (+)
20×15×10×5-9	**7.89e-01 (1.86e-02)**	**7.81e-01 (1.73e-02) (≈)**	5.06e-01 (2.11e-02) (+)	3.57e-01 (1.02e-02) (+)	4.69e-01 (2.64e-02) (+)	4.56e-01 (2.46e-02) (+)	6.97e-01 (1.51e-02) (+)
20×15×10×5-11	**8.74e-01 (3.26e-02)**	**8.68e-01 (3.62e-02) (≈)**	4.07e-01 (2.08e-02) (+)	2.31e-01 (9.65e-03) (+)	4.43e-01 (2.21e-02) (+)	4.42e-01 (1.51e-02) (+)	7.24e-01 (3.44e-02) (+)
20×15×10×5-15	**8.12e-01 (2.27e-02)**	7.93e-01 (1.98e-02) (+)	4.79e-01 (2.59e-02) (+)	3.74e-01 (1.35e-02) (+)	7.18e-01 (1.98e-02) (+)	5.48e-01 (2.86e-02) (+)	6.98e-01 (3.00e-02) (+)
20×15×10×5-16	**8.49e-01 (2.47e-02)**	8.16e-01 (2.78e-02) (+)	4.91e-01 (1.81e-02) (+)	4.32e-01 (2.50e-02) (+)	5.08e-01 (2.86e-02) (+)	3.28e-01 (9.78e-03) (+)	5.32e-01 (1.82e-02) (+)
20×15×10×5-17	**7.48e-01 (4.15e-02)**	**7.54e-01 (3.85e-02) (≈)**	4.70e-01 (1.82e-02) (+)	4.60e-01 (2.18e-02) (+)	3.48e-01 (1.14e-02) (+)	3.11e-01 (1.70e-02) (+)	5.19e-01 (2.69e-02) (+)
20×15×10×5-18	**8.41e-01 (1.50e-02)**	8.05e-01 (1.65e-02) (+)	3.55e-01 (1.48e-02) (+)	3.89e-01 (2.30e-02) (+)	3.47e-01 (1.27e-02) (+)	3.17e-01 (1.16e-02) (+)	5.60e-01 (2.71e-02) (+)
20×15×10×5-20	**7.53e-01 (3.40e-02)**	**7.43e-01 (3.36e-02) (≈)**	4.03e-01 (1.77e-02) (+)	5.17e-01 (1.16e-02) (+)	4.86e-01 (1.57e-02) (+)	4.69e-01 (1.03e-02) (+)	6.19e-01 (3.28e-02) (+)
30×20×15×7-1	**8.92e-01 (5.71e-02)**	**8.81e-01 (5.07e-02) (≈)**	2.57e-01 (7.22e-03) (+)	6.18e-01 (2.76e-02) (+)	4.38e-01 (2.39e-02) (+)	5.24e-01 (2.80e-02) (+)	3.21e-01 (1.27e-02) (+)
30×20×15×7-2	**8.33e-01 (2.87e-02)**	**8.17e-01 (3.32e-02) (≈)**	6.50e-01 (2.50e-02) (+)	2.05e-01 (1.10e-02) (+)	4.39e-01 (2.30e-02) (+)	3.69e-01 (2.07e-02) (+)	1.52e-01 (6.42e-03) (+)
30×20×15×7-3	**7.89e-01 (2.48e-02)**	7.61e-01 (2.39e-02) (+)	5.17e-01 (3.35e-02) (+)	2.59e-01 (9.72e-03) (+)	3.25e-01 (1.46e-02) (+)	3.46e-01 (2.24e-02) (+)	7.80e-02 (4.53e-03) (+)
30×20×15×7-4	**8.89e-01 (2.52e-02)**	**8.75e-01 (2.40e-02) (≈)**	5.70e-01 (2.96e-02) (+)	3.66e-01 (1.92e-02) (+)	5.05e-01 (1.45e-02) (+)	3.98e-01 (1.93e-02) (+)	3.01e-01 (1.33e-02) (+)
30×20×15×7-8	**7.84e-01 (3.15e-02)**	**7.65e-01 (2.95e-02) (≈)**	6.35e-01 (3.04e-02) (+)	2.65e-01 (1.04e-02) (+)	3.55e-01 (8.04e-03) (+)	5.72e-01 (2.25e-02) (+)	3.36e-01 (1.65e-02) (+)
30×20×15×7-14	**8.66e-01 (4.93e-02)**	**8.62e-01 (4.40e-02) (≈)**	6.57e-01 (4.06e-02) (+)	2.81e-01 (1.02e-02) (+)	4.09e-01 (9.25e-03) (+)	4.51e-01 (1.87e-02) (+)	2.10e-01 (1.07e-02) (+)
30×20×15×7-15	**8.90e-01 (2.19e-02)**	8.68e-01 (2.52e-02) (+)	5.71e-01 (3.10e-02) (+)	2.64e-01 (1.19e-02) (+)	5.22e-01 (2.80e-02) (+)	4.98e-01 (1.18e-02) (+)	1.97e-01 (7.14e-03) (+)
30×20×15×7-17	**7.92e-01 (2.53e-02)**	**7.90e-01 (2.74e-02) (≈)**	5.05e-01 (2.41e-02) (+)	4.17e-01 (2.01e-02) (+)	5.73e-01 (1.43e-02) (+)	4.98e-01 (2.92e-02) (+)	3.75e-01 (1.63e-02) (+)
30×20×15×7-20	**8.23e-01 (3.72e-02)**	**8.11e-01 (4.25e-02) (≈)**	6.09e-01 (2.92e-02) (+)	3.38e-01 (1.51e-02) (+)	**8.20e-01 (1.23e-02) (≈)**	4.24e-01 (2.41e-02) (+)	4.44e-01 (1.79e-02) (+)
40×25×20×10-2	**7.93e-01 (4.10e-02)**	**7.68e-01 (4.25e-02) (≈)**	6.56e-01 (1.92e-02) (+)	5.35e-01 (2.24e-02) (+)	4.46e-01 (1.09e-02) (+)	5.58e-01 (3.07e-02) (+)	3.79e-01 (1.74e-02) (+)
40×25×20×10-3	**7.68e-01 (1.92e-02)**	7.31e-01 (1.67e-02) (+)	7.16e-01 (3.97e-02) (+)	3.52e-01 (1.56e-02) (+)	3.12e-01 (1.09e-02) (+)	4.70e-01 (1.09e-02) (+)	1.99e-01 (4.12e-03) (+)
40×25×20×10-4	**7.31e-01 (3.61e-02)**	**7.35e-01 (3.47e-02) (≈)**	5.83e-01 (1.53e-02) (+)	3.34e-01 (1.49e-02) (+)	3.18e-01 (1.59e-02) (+)	5.97e-01 (1.77e-02) (+)	3.40e-01 (9.40e-03) (+)
40×25×20×10-6	**7.75e-01 (3.70e-02)**	**7.69e-01 (3.62e-02) (≈)**	**7.56e-01 (3.26e-02) (≈)**	5.63e-01 (2.78e-02) (+)	4.89e-01 (1.81e-02) (+)	3.81e-01 (8.84e-03) (+)	3.97e-01 (8.14e-03) (+)
40×25×20×10-8	**7.29e-01 (3.30e-02)**	**7.35e-01 (2.85e-02) (≈)**	6.95e-01 (3.68e-02) (+)	4.78e-01 (3.02e-02) (+)	4.02e-01 (1.80e-02) (+)	6.33e-01 (1.58e-02) (+)	3.04e-01 (1.65e-02) (+)
40×25×20×10-9	**7.36e-01 (4.26e-02)**	**7.28e-01 (4.75e-02) (≈)**	7.03e-01 (1.38e-02) (+)	3.60e-01 (1.53e-02) (+)	3.90e-01 (1.27e-02) (+)	6.50e-01 (2.80e-02) (+)	3.78e-01 (2.08e-02) (+)
40×25×20×10-11	**8.15e-01 (2.85e-02)**	7.94e-01 (2.65e-02) (+)	6.74e-01 (2.75e-02) (+)	3.74e-01 (1.80e-02) (+)	4.14e-01 (2.16e-02) (+)	5.20e-01 (2.18e-02) (+)	3.81e-01 (1.30e-02) (+)
40×25×20×10-15	**8.71e-01 (4.26e-02)**	**8.57e-01 (3.78e-02) (≈)**	5.34e-01 (3.00e-02) (+)	4.35e-01 (1.72e-02) (+)	5.97e-01 (2.90e-02) (+)	5.14e-01 (1.86e-02) (+)	3.35e-01 (1.65e-02) (+)
+/ ≈ / −		**10/26/0**	**35/1/0**	**36/0/0**	**35/1/0**	**36/0/0**	**36/0/0**

Table 6
STATISTICAL RESULTS OF HGMAPPO AND ITS VARIANTS ON GD METRIC.

Instances	HGMAPPO	HGMAPPO-SW	HGMAPPO-JR	HGMAPPO-MR	HGMAPPO-SR	HGMAPPO-NoHG	HGMAPPO-NoMP
10×5×4×2-2	**0.00e+00 (0.00e+00)**	2.62e-02 (2.89e-03) (+)	3.09e-01 (8.88e-03) (+)	1.38e-01 (4.88e-03) (+)	4.11e-01 (8.96e-03) (+)	2.30e-01 (1.05e-02) (+)	7.10e-02 (3.67e-03) (+)
10×5×4×2-3	5.61e-02 (4.33e-03)	**0.00e+00 (0.00e+00) (-)**	2.65e-01 (1.32e-02) (+)	2.85e-01 (7.04e-03) (+)	3.62e-01 (1.02e-02) (+)	2.45e-01 (1.17e-02) (+)	1.82e-01 (7.05e-03) (+)
10×5×4×2-5	**5.15e-03 (5.71e-04)**	6.66e-03 (7.14e-04) (+)	2.92e-01 (1.34e-02) (+)	1.79e-01 (7.79e-03) (+)	3.40e-01 (1.19e-02) (+)	2.00e-01 (9.28e-03) (+)	4.74e-02 (3.61e-03) (+)
10×5×4×2-8	**3.68e-02 (1.66e-03)**	4.31e-02 (3.25e-03) (+)	3.24e-01 (9.56e-03) (+)	3.32e-01 (7.27e-03) (+)	3.80e-01 (2.19e-02) (+)	2.16e-01 (1.17e-02) (+)	1.86e-01 (6.84e-03) (+)
10×5×4×2-10	**0.00e+00 (0.00e+00)**	**0.14e-03 (4.36e-o3) (≈)**	3.51e-01 (2.01e-02) (+)	1.40e-01 (6.11e-03) (+)	3.51e-02 (2.30e-03) (+)	1.80e-01 (8.16e-03) (+)	1.72e-01 (3.90e-03) (+)
10×5×4×2-11	**1.65e-02 (1.91e-03)**	2.31e-02 (2.36e-03) (+)	1.62e-01 (5.17e-03) (+)	1.56e-01 (7.32e-03) (+)	8.98e-02 (4.84e-03) (+)	7.50e-02 (2.56e-03) (+)	2.33e-01 (8.25e-03) (+)
10×5×4×2-13	**0.00e+00 (0.00e+00)**	2.77e-02 (1.03e-03) (+)	2.42e-01 (5.18e-03) (+)	1.94e-01 (5.35e-03) (+)	1.48e-01 (4.33e-03) (+)	1.10e-01 (2.65e-03) (+)	1.75e-01 (1.10e-02) (+)
10×5×4×2-18	**2.03e-02 (1.14e-03)**	2.78e-02 (2.03e-03) (+)	3.86e-01 (1.68e-02) (+)	1.70e-01 (6.36e-03) (+)	3.70e-01 (1.04e-02) (+)	1.99e-01 (4.47e-03) (+)	1.94e-01 (4.05e-03) (+)
10×5×4×2-19	**1.19e-02 (1.47e-03)**	1.65e-02 (1.69e-03) (+)	3.20e-01 (1.21e-02) (+)	6.35e-02 (5.07e-03) (+)	4.30e-01 (2.04e-02) (+)	7.31e-02 (4.24e-03) (+)	1.59e-01 (7.46e-03) (+)
10×5×4×2-20	**4.43e-02 (5.27e-03)**	5.83e-02 (5.92e-03) (+)	2.80e-01 (1.02e-02) (+)	2.77e-01 (8.17e-03) (+)	3.17e-01 (2.04e-02) (+)	1.59e-01 (4.64e-03) (+)	1.90e-01 (5.44e-03) (+)
20×15×10×5-2	**0.00e+00 (0.00e+00)**	4.39e-02 (2.26e-03) (+)	3.73e-01 (1.84e-02) (+)	3.54e-01 (1.17e-02) (+)	4.05e-01 (1.28e-02) (+)	4.31e-01 (2.52e-02) (+)	1.25e-01 (2.75e-03) (+)
20×15×10×5-4	**2.96e-02 (3.04e-03)**	3.59e-02 (2.56e-03) (+)	1.13e-01 (8.54e-03) (+)	1.89e-01 (1.79e-02) (+)	2.18e-01 (1.34e-02) (+)	3.15e-01 (1.20e-02) (+)	1.79e-01 (5.81e-03) (+)
20×15×10×5-9	**1.26e-02 (8.09e-04)**	1.58e-02 (6.12e-04) (+)	2.82e-01 (6.64e-03) (+)	4.01e-01 (1.79e-02) (+)	3.13e-01 (1.45e-02) (+)	3.56e-01 (2.06e-02) (+)	4.74e-02 (2.84e-03) (+)
20×15×10×5-11	**6.94e-03 (3.19e-04)**	8.59e-03 (9.32e-04) (+)	3.60e-01 (1.58e-02) (+)	4.79e-01 (1.51e-02) (+)	2.21e-01 (1.18e-02) (+)	1.47e-01 (5.58e-03) (+)	5.02e-02 (4.02e-03) (+)
20×15×10×5-15	**0.00e+00 (0.00e+00)**	1.56e-02 (1.22e-03) (+)	3.73e-01 (2.09e-02) (+)	3.22e-01 (1.03e-02) (+)	4.69e-02 (3.76e-03) (+)	3.24e-01 (1.72e-02) (+)	5.64e-02 (2.08e-03) (+)
20×15×10×5-16	**3.50e-02 (2.39e-03)**	4.95e-02 (6.22e-03) (+)	2.86e-01 (1.82e-02) (+)	3.02e-01 (1.71e-02) (+)	2.74e-01 (7.52e-03) (+)	2.99e-01 (6.74e-03) (+)	1.40e-01 (7.49e-03) (+)
20×15×10×5-17	4.70e-02 (3.34e-03)	**3.27e-02 (2.72e-03) (-)**	2.19e-01 (1.25e-02) (+)	3.00e-01 (1.28e-02) (+)	3.07e-01 (1.57e-02) (+)	2.69e-01 (8.36e-03) (+)	1.22e-01 (2.88e-03) (+)
20×15×10×5-18	**3.54e-02 (4.30e-03)**	5.00e-02 (6.21e-03) (+)	2.02e-01 (1.47e-02) (+)	2.18e-01 (6.98e-03) (+)	4.34e-01 (1.36e-02) (+)	3.81e-01 (9.75e-03) (+)	2.48e-01 (6.13e-03) (+)
20×15×10×5-20	**2.26e-02 (1.99e-03)**	2.93e-02 (1.60e-03) (+)	2.56e-01 (6.49e-03) (+)	1.98e-01 (6.13e-03) (+)	3.41e-01 (9.30e-03) (+)	2.54e-01 (9.89e-03) (+)	1.01e-01 (2.64e-03) (+)
30×20×15×7-1	**0.00e+00 (0.00e+00)**	3.08e-02 (3.21e-03) (+)	5.34e-01 (1.52e-02) (+)	8.92e-02 (7.71e-03) (+)	3.24e-01 (1.48e-02) (+)	2.39e-01 (1.47e-02) (+)	1.74e-01 (1.01e-02) (+)
30×20×15×7-2	**3.70e-02 (3.14e-03)**	4.52e-02 (2.21e-03) (+)	1.61e-01 (8.31e-03) (+)	2.00e-01 (8.09e-03) (+)	1.65e-01 (7.86e-03) (+)	2.83e-01 (5.69e-03) (+)	4.28e-01 (2.06e-02) (+)
30×20×15×7-3	**1.46e-02 (5.62e-04)**	1.89e-02 (1.65e-03) (+)	8.50e-02 (6.26e-03) (+)	3.67e-01 (9.01e-03) (+)	2.85e-01 (1.11e-02) (+)	9.22e-02 (3.56e-03) (+)	2.30e-01 (1.07e-02) (+)
30×20×15×7-4	**4.55e-02 (3.60e-03)**	5.94e-02 (6.59e-03) (+)	2.76e-01 (4.91e-03) (+)	3.04e-01 (1.13e-02) (+)	3.85e-01 (1.67e-02) (+)	4.22e-01 (1.28e-02) (+)	5.19e-01 (1.23e-02) (+)
30×20×15×7-8	**0.00e+00 (0.00e+00)**	8.89e-03 (5.03e-04) (+)	5.58e-02 (2.64e-03) (+)	4.28e-01 (7.54e-03) (+)	2.17e-01 (8.04e-03) (+)	1.60e-01 (8.76e-03) (+)	3.92e-01 (8.54e-03) (+)
30×20×15×7-14	**2.33e-02 (2.92e-03)**	2.73e-02 (2.32e-03) (+)	7.88e-02 (3.40e-03) (+)	3.66e-01 (9.44e-03) (+)	2.05e-01 (8.58e-03) (+)	3.20e-01 (1.46e-02) (+)	4.19e-01 (1.24e-02) (+)
30×20×15×7-15	**2.17e-02 (1.00e-03)**	2.77e-02 (2.36e-03) (+)	8.52e-02 (4.61e-03) (+)	3.61e-01 (1.23e-02) (+)	1.70e-01 (4.02e-03) (+)	1.74e-01 (7.37e-03) (+)	3.31e-01 (1.01e-02) (+)
30×20×15×7-17	**3.91e-02 (2.74e-03)**	**3.99e-02 (2.99e-03) (≈)**	1.41e-01 (4.70e-03) (+)	1.80e-01 (8.60e-03) (+)	2.11e-01 (4.25e-03) (+)	3.54e-01 (1.91e-02) (+)	2.52e-01 (9.97e-03) (+)
30×20×15×7-20	**0.00e+00 (0.00e+00)**	1.83e-03 (7.46e-05) (+)	1.07e-01 (5.53e-03) (+)	2.41e-01 (5.41e-03) (+)	7.07e-03 (4.78e-04) (+)	2.25e-01 (6.44e-03) (+)	2.66e-01 (1.66e-02) (+)
40×25×20×10-2	**8.55e-03 (7.67e-04)**	1.08e-02 (5.34e-04) (+)	6.50e-02 (2.92e-03) (+)	1.78e-01 (6.42e-03) (+)	3.92e-01 (1.04e-02) (+)	2.12e-01 (8.22e-03) (+)	3.23e-01 (2.03e-02) (+)
40×25×20×10-3	**2.56e-03 (2.62e-04)**	3.71e-03 (1.62e-04) (+)	2.17e-02 (9.13e-04) (+)	1.61e-01 (1.36e-02) (+)	2.76e-01 (1.63e-02) (+)	1.11e-01 (2.78e-03) (+)	4.27e-01 (2.30e-02) (+)
40×25×20×10-4	1.63e-02 (1.72e-03)	**1.29e-02 (9.30e-04) (-)**	7.78e-02 (6.87e-03) (+)	2.11e-01 (8.35e-03) (+)	4.30e-01 (2.34e-02) (+)	7.38e-02 (5.43e-03) (+)	3.93e-01 (8.46e-03) (+)
40×25×20×10-6	**3.51e-03 (4.10e-04)**	4.34e-03 (4.19e-04) (+)	1.57e-02 (9.25e-04) (+)	1.86e-01 (7.33e-03) (+)	2.08e-01 (5.56e-03) (+)	1.80e-01 (6.97e-03) (+)	2.07e-01 (6.97e-03) (+)
40×25×20×10-8	5.93e-03 (7.46e-04)	**4.83e-03 (5.98e-04) (-)**	2.45e-02 (2.21e-03) (+)	1.30e-01 (5.80e-03) (+)	3.81e-01 (1.58e-02) (+)	1.28e-01 (5.27e-03) (+)	4.68e-01 (8.97e-03) (+)
40×25×20×10-9	**0.00e+00 (0.00e+00)**	3.68e-03 (4.65e-04) (+)	2.14e-02 (1.74e-03) (+)	1.61e-01 (8.33e-03) (+)	2.73e-01 (1.33e-02) (+)	4.59e-02 (1.96e-03) (+)	3.64e-01 (1.72e-02) (+)
40×25×20×10-11	**1.78e-02 (9.31e-04)**	2.22e-02 (2.60e-03) (+)	1.03e-01 (5.28e-03) (+)	1.69e-01 (4.80e-03) (+)	2.97e-01 (1.46e-02) (+)	9.26e-02 (5.21e-03) (+)	3.00e-01 (1.77e-02) (+)
40×25×20×10-15	**2.41e-02 (2.80e-03)**	2.98e-02 (2.83e-03) (+)	1.33e-01 (4.97e-03) (+)	2.79e-01 (1.53e-02) (+)	2.27e-01 (7.58e-03) (+)	3.37e-01 (1.79e-02) (+)	3.05e-01 (1.73e-02) (+)
+/ ≈ / –		30/2/4	36/0/0	36/0/0	36/0/0	36/0/0	36/0/0

Table 7
STATISTICAL RESULTS OF HGMAPPO AND ITS VARIANTS ON IGD METRIC.

Instances	HGMAPPO	HGMAPPO-SW	HGMAPPO-JR	HGMAPPO-MR	HGMAPPO-SR	HGMAPPO-NoHG	HGMAPPO-NoMP
10×5×4×2-2	**0.00e+00 (0.00e+00)**	6.04e-02 (6.85e-03) (+)	1.89e-01 (5.88e-03) (+)	3.38e-01 (7.62e-03) (+)	4.14e-01 (1.81e-02) (+)	2.31e-01 (8.90e-03) (+)	2.75e-01 (1.37e-02) (+)
10×5×4×2-3	6.35e-02 (4.76e-03)	**0.00e+00 (0.00e+00) (-)**	2.86e-01 (1.06e-02) (+)	3.41e-01 (1.73e-02) (+)	3.29e-01 (2.00e-02) (+)	1.48e-01 (3.87e-03) (+)	1.19e-01 (5.57e-03) (+)
10×5×4×2-5	**1.46e-02 (9.96e-04)**	1.87e-02 (9.84e-04) (+)	3.86e-01 (2.15e-02) (+)	2.68e-01 (1.16e-02) (+)	4.30e-01 (1.64e-02) (+)	2.59e-01 (1.65e-02) (+)	6.99e-02 (3.33e-03) (+)
10×5×4×2-8	**4.96e-02 (5.97e-03)**	5.42e-02 (5.34e-03) (+)	4.34e-01 (1.90e-02) (+)	3.46e-01 (8.15e-03) (+)	2.25e-01 (5.00e-03) (+)	1.73e-01 (8.56e-03) (+)	1.88e-01 (4.51e-03) (+)
10×5×4×2-10	8.72e-03 (8.11e-04)	**8.56e-03 (6.68e-04) (≈)**	3.10e-01 (8.53e-03) (+)	1.81e-01 (1.11e-02) (+)	4.66e-02 (4.26e-03) (+)	1.83e-01 (9.56e-03) (+)	2.06e-01 (6.20e-03) (+)
10×5×4×2-11	**5.53e-02 (2.25e-03)**	7.71e-02 (3.06e-03) (+)	1.78e-01 (8.22e-03) (+)	2.60e-01 (5.99e-03) (+)	2.04e-01 (1.11e-02) (+)	1.84e-01 (3.69e-03) (+)	1.89e-01 (5.93e-03) (+)
10×5×4×2-13	**1.04e-02 (6.20e-04)**	3.03e-02 (2.95e-03) (+)	1.64e-01 (5.33e-03) (+)	2.12e-01 (1.14e-02) (+)	1.20e-01 (4.41e-03) (+)	6.45e-02 (3.15e-03) (+)	1.38e-01 (7.06e-03) (+)
10×5×4×2-18	**3.03e-02 (3.58e-03)**	3.61e-02 (1.58e-03) (+)	2.08e-01 (7.94e-03) (+)	2.42e-01 (6.61e-03) (+)	4.65e-01 (2.66e-02) (+)	2.07e-01 (7.01e-03) (+)	2.10e-01 (4.60e-03) (+)
10×5×4×2-19	**2.08e-02 (2.62e-03)**	2.80e-02 (3.11e-03) (+)	3.27e-01 (1.83e-02) (+)	2.44e-01 (1.06e-02) (+)	3.99e-01 (1.98e-02) (+)	1.94e-01 (9.73e-03) (+)	7.97e-02 (1.84e-03) (+)
10×5×4×2-20	**4.74e-02 (3.90e-03)**	6.62e-02 (2.90e-03) (+)	2.45e-01 (1.19e-02) (+)	2.67e-01 (1.41e-02) (+)	3.10e-01 (1.06e-02) (+)	2.02e-01 (7.60e-03) (+)	7.91e-02 (3.69e-03) (+)
20×15×10×5-2	**0.00e+00 (0.00e+00)**	5.48e-02 (6.67e-03) (+)	2.39e-01 (8.31e-03) (+)	3.09e-01 (1.43e-02) (+)	3.20e-01 (1.65e-02) (+)	3.64e-01 (2.18e-02) (+)	1.70e-01 (4.77e-03) (+)
20×15×10×5-4	**3.94e-02 (5.03e-03)**	4.69e-02 (3.67e-03) (+)	3.07e-01 (1.19e-02) (+)	4.88e-01 (2.11e-02) (+)	1.99e-01 (4.09e-03) (+)	3.69e-01 (1.52e-02) (+)	1.02e-01 (3.46e-03) (+)
20×15×10×5-9	**1.95e-02 (1.44e-03)**	2.29e-02 (8.70e-04) (+)	4.12e-01 (1.84e-02) (+)	3.91e-01 (8.16e-03) (+)	2.34e-01 (9.47e-03) (+)	3.14e-01 (7.54e-03) (+)	6.54e-02 (3.82e-03) (+)
20×15×10×5-11	**3.14e-02 (3.86e-03)**	3.56e-02 (4.35e-03) (+)	3.08e-01 (1.31e-02) (+)	4.44e-01 (2.60e-02) (+)	3.44e-01 (1.02e-02) (+)	1.82e-01 (4.82e-03) (+)	1.76e-01 (5.32e-03) (+)
20×15×10×5-15	**3.61e-02 (4.14e-03)**	4.88e-02 (5.06e-03) (+)	2.25e-01 (1.24e-02) (+)	3.97e-01 (1.11e-02) (+)	1.15e-01 (4.27e-03) (+)	2.61e-01 (1.55e-02) (+)	1.14e-01 (3.91e-03) (+)
20×15×10×5-16	**5.30e-02 (5.80e-03)**	7.67e-02 (9.13e-03) (+)	3.48e-01 (1.37e-02) (+)	2.46e-01 (6.41e-03) (+)	3.67e-01 (1.22e-02) (+)	3.39e-01 (1.84e-02) (+)	1.04e-01 (3.86e-03) (+)
20×15×10×5-17	6.90e-02 (6.63e-03)	**3.03e-02 (3.40e-03) (-)**	2.95e-01 (7.96e-03) (+)	3.10e-01 (1.28e-02) (+)	2.61e-01 (8.87e-03) (+)	2.64e-01 (1.23e-02) (+)	1.76e-01 (5.57e-03) (+)
20×15×10×5-18	**3.76e-02 (4.27e-03)**	5.74e-02 (7.17e-03) (+)	3.79e-01 (2.18e-02) (+)	2.09e-01 (1.04e-02) (+)	3.34e-01 (1.20e-02) (+)	3.83e-01 (2.09e-02) (+)	1.22e-01 (5.56e-03) (+)
20×15×10×5-20	**4.61e-02 (2.04e-03)**	5.91e-02 (5.60e-03) (+)	3.78e-01 (1.87e-02) (+)	3.45e-01 (1.67e-02) (+)	2.36e-01 (1.25e-02) (+)	2.84e-01 (1.46e-02) (+)	1.67e-01 (6.66e-03) (+)
30×20×15×7-1	**0.00e+00 (0.00e+00)**	3.83e-02 (4.75e-03) (+)	4.79e-01 (1.62e-02) (+)	2.54e-01 (1.21e-02) (+)	3.33e-01 (1.31e-02) (+)	3.27e-01 (1.92e-02) (+)	3.84e-01 (8.75e-03) (+)
30×20×15×7-2	**5.41e-02 (3.61e-03)**	6.71e-02 (3.06e-03) (+)	1.91e-01 (1.03e-02) (+)	3.93e-01 (1.08e-02) (+)	3.75e-01 (7.54e-03) (+)	1.78e-01 (6.90e-03) (+)	4.03e-01 (2.41e-02) (+)
30×20×15×7-3	**1.96e-02 (1.47e-03)**	2.23e-02 (1.30e-03) (+)	4.75e-02 (1.54e-03) (+)	3.57e-01 (1.84e-02) (+)	3.60e-01 (1.18e-02) (+)	2.98e-01 (6.96e-03) (+)	5.86e-01 (3.14e-02) (+)
30×20×15×7-4	**5.74e-02 (2.60e-03)**	7.25e-02 (8.33e-03) (+)	2.11e-01 (7.06e-03) (+)	4.08e-01 (8.02e-03) (+)	3.31e-01 (6.79e-03) (+)	4.26e-01 (2.40e-02) (+)	2.61e-01 (1.28e-02) (+)
30×20×15×7-8	**1.78e-02 (8.76e-04)**	2.21e-02 (1.01e-03) (+)	1.13e-01 (5.58e-03) (+)	5.02e-01 (1.86e-02) (+)	3.17e-01 (6.32e-03) (+)	1.92e-01 (5.38e-03) (+)	3.70e-01 (1.33e-02) (+)
30×20×15×7-14	**3.11e-02 (3.82e-03)**	3.47e-02 (3.68e-03) (+)	8.29e-02 (1.91e-03) (+)	4.09e-01 (1.96e-02) (+)	2.33e-01 (1.24e-02) (+)	1.93e-01 (4.34e-03) (+)	5.31e-01 (2.48e-02) (+)
30×20×15×7-15	**2.53e-02 (2.04e-03)**	4.27e-02 (5.49e-03) (+)	1.01e-01 (4.88e-03) (+)	3.57e-01 (1.04e-02) (+)	2.24e-01 (7.76e-03) (+)	2.25e-01 (1.17e-02) (+)	3.77e-01 (9.23e-03) (+)
30×20×15×7-17	**4.73e-02 (3.31e-03)**	**4.82e-02 (3.62e-03) (≈)**	1.05e-01 (5.51e-03) (+)	2.42e-01 (8.52e-03) (+)	3.05e-01 (1.58e-02) (+)	2.43e-01 (1.16e-02) (+)	2.86e-01 (1.10e-02) (+)
30×20×15×7-20	**0.00e+00 (0.00e+00)**	2.33e-03 (1.20e-04) (+)	2.20e-01 (9.91e-03) (+)	2.79e-01 (1.10e-02) (+)	1.02e-02 (5.09e-04) (+)	2.16e-01 (8.27e-03) (+)	2.07e-01 (7.31e-03) (+)
40×25×20×10-2	**1.65e-02 (1.65e-03)**	2.25e-02 (2.41e-03) (+)	9.48e-02 (3.37e-03) (+)	1.76e-01 (4.24e-03) (+)	2.40e-01 (6.70e-03) (+)	1.94e-01 (6.49e-03) (+)	4.49e-01 (1.31e-02) (+)
40×25×20×10-3	**4.69e-03 (3.04e-04)**	6.48e-03 (5.76e-04) (+)	2.92e-02 (1.57e-03) (+)	2.69e-01 (1.09e-02) (+)	3.53e-01 (8.87e-03) (+)	1.80e-01 (1.13e-02) (+)	3.35e-01 (1.37e-02) (+)
40×25×20×10-4	2.00e-02 (1.80e-03)	**1.79e-02 (2.18e-03) (-)**	6.76e-02 (3.57e-03) (+)	3.92e-01 (1.44e-02) (+)	2.93e-01 (7.53e-03) (+)	1.59e-01 (7.57e-03) (+)	3.53e-01 (7.13e-03) (+)
40×25×20×10-6	**5.30e-03 (6.17e-04)**	6.28e-03 (2.56e-04) (+)	2.10e-02 (1.91e-03) (+)	1.67e-01 (5.25e-03) (+)	1.51e-01 (7.35e-03) (+)	2.06e-01 (1.21e-02) (+)	2.57e-01 (1.09e-02) (+)
40×25×20×10-8	8.51e-03 (3.00e-04)	**6.88e-03 (5.81e-04) (-)**	2.97e-02 (1.21e-03) (+)	3.71e-01 (1.73e-02) (+)	2.45e-01 (1.08e-02) (+)	8.84e-02 (4.45e-03) (+)	4.40e-01 (1.34e-02) (+)
40×25×20×10-9	**3.27e-03 (4.15e-04)**	5.64e-03 (4.53e-04) (+)	2.85e-02 (2.08e-03) (+)	2.69e-01 (7.33e-03) (+)	1.96e-01 (5.85e-03) (+)	1.44e-01 (7.15e-03) (+)	4.72e-01 (1.91e-02) (+)
40×25×20×10-11	**2.52e-02 (2.99e-03)**	2.92e-02 (1.48e-03) (+)	1.75e-01 (1.00e-02) (+)	4.00e-01 (1.72e-02) (+)	2.59e-01 (1.18e-02) (+)	9.69e-02 (2.82e-03) (+)	2.82e-01 (1.25e-02) (+)
40×25×20×10-15	**6.76e-02 (5.71e-03)**	8.60e-02 (8.75e-03) (+)	2.60e-01 (5.23e-03) (+)	4.10e-01 (9.61e-03) (+)	3.52e-01 (2.05e-02) (+)	3.60e-01 (1.16e-02) (+)	4.35e-01 (1.28e-02) (+)
+/ ≈ / −		31/2/3	36/0/0	36/0/0	36/0/0	36/0/0	36/0/0

PART 4. COMPARISON EXPERIMENTS

This part reports the complete instance-level comparison between HGMAPPO and the eight baseline algorithms under the unified experimental environment. Tables 8-10 present the HV, GD, and IGD results, respectively. Each entry reports the mean over 20 independent runs, with the standard deviation in parentheses. All indicators are recalculated from the final nondominated solution sets obtained under the revised scheduling and evaluation protocol.

Table 8
STATISTICAL RESULTS OF HGMAPPO VERSUS OTHER COMPARISON ALGORITHMS ON HV METRIC.

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
10×5×4×2-1	**8.51e-01 (4.62e-02)**	3.70e-01 (1.18e-02) (+)	3.31e-01 (9.72e-03) (+)	3.82e-01 (1.94e-02) (+)	5.61e-01 (1.49e-02) (+)	2.26e-01 (6.27e-03) (+)	2.28e-01 (7.96e-03) (+)	5.05e-01 (1.37e-02) (+)	1.13e-01 (2.95e-03) (+)
10×5×4×2-2	**8.73e-01 (4.75e-02)**	4.72e-01 (1.77e-02) (+)	3.76e-01 (1.13e-02) (+)	3.13e-01 (1.50e-02) (+)	4.24e-01 (2.57e-02) (+)	1.59e-01 (6.15e-03) (+)	1.74e-01 (5.03e-03) (+)	5.58e-01 (1.28e-02) (+)	2.76e-01 (1.33e-02) (+)
10×5×4×2-3	**7.91e-01 (1.96e-02)**	5.04e-01 (2.12e-02) (+)	4.05e-01 (1.05e-02) (+)	4.28e-01 (1.09e-02) (+)	4.89e-01 (1.50e-02) (+)	2.40e-01 (1.24e-02) (+)	1.42e-01 (4.21e-03) (+)	6.31e-01 (2.25e-02) (+)	1.79e-01 (1.16e-02) (+)
10×5×4×2-4	**8.54e-01 (1.69e-02)**	3.55e-01 (1.97e-02) (+)	5.73e-01 (2.67e-02) (+)	5.86e-01 (3.37e-02) (+)	5.39e-01 (3.27e-02) (+)	3.10e-01 (1.08e-02) (+)	2.28e-01 (9.65e-03) (+)	5.24e-01 (1.18e-02) (+)	1.60e-01 (8.20e-03) (+)
10×5×4×2-5	**8.17e-01 (3.09e-02)**	3.99e-01 (2.10e-02) (+)	5.69e-01 (1.59e-02) (+)	3.81e-01 (1.78e-02) (+)	4.92e-01 (3.05e-02) (+)	2.03e-01 (6.44e-03) (+)	1.59e-01 (8.13e-03) (+)	6.55e-01 (2.88e-02) (+)	1.00e-01 (3.38e-03) (+)
10×5×4×2-6	**7.81e-01 (3.05e-02)**	3.88e-01 (1.48e-02) (+)	3.03e-01 (9.96e-03) (+)	4.35e-01 (1.28e-02) (+)	5.04e-01 (3.24e-02) (+)	2.44e-01 (6.78e-03) (+)	1.32e-01 (5.15e-03) (+)	5.67e-01 (2.78e-02) (+)	1.52e-01 (9.61e-03) (+)
10×5×4×2-7	**8.72e-01 (1.94e-02)**	3.59e-01 (2.24e-02) (+)	3.40e-01 (7.27e-03) (+)	4.46e-01 (1.55e-02) (+)	4.91e-01 (2.18e-02) (+)	2.27e-01 (1.27e-02) (+)	1.32e-01 (3.83e-03) (+)	5.58e-01 (2.16e-02) (+)	1.53e-01 (4.63e-03) (+)
10×5×4×2-8	**7.82e-01 (3.57e-02)**	3.58e-01 (8.35e-03) (+)	3.38e-01 (1.22e-02) (+)	3.91e-01 (2.14e-02) (+)	5.02e-01 (1.62e-02) (+)	2.16e-01 (1.33e-02) (+)	1.27e-01 (6.71e-03) (+)	5.03e-01 (2.50e-02) (+)	1.63e-01 (5.04e-03) (+)
10×5×4×2-9	**9.10e-01 (3.40e-02)**	4.90e-01 (1.55e-02) (+)	7.07e-01 (3.38e-02) (+)	6.23e-01 (3.08e-02) (+)	6.79e-01 (2.22e-02) (+)	4.41e-01 (1.73e-02) (+)	3.60e-01 (1.59e-02) (+)	5.70e-01 (3.35e-02) (+)	2.79e-01 (1.69e-02) (+)
10×5×4×2-10	**8.31e-01 (4.00e-02)**	5.40e-01 (2.40e-02) (+)	6.24e-01 (3.90e-02) (+)	6.92e-01 (2.29e-02) (+)	6.18e-01 (2.03e-02) (+)	4.01e-01 (9.40e-03) (+)	3.92e-01 (1.52e-02) (+)	5.87e-01 (2.12e-02) (+)	3.10e-01 (7.41e-03) (+)
10×5×4×2-11	**8.36e-01 (4.08e-02)**	4.94e-01 (1.12e-02) (+)	7.19e-01 (1.63e-02) (+)	6.77e-01 (2.44e-02) (+)	6.79e-01 (2.64e-02) (+)	4.62e-01 (1.76e-02) (+)	3.77e-01 (1.02e-02) (+)	5.95e-01 (1.47e-02) (+)	2.86e-01 (1.29e-02) (+)
10×5×4×2-12	**8.92e-01 (4.37e-02)**	4.88e-01 (2.19e-02) (+)	6.95e-01 (1.74e-02) (+)	**8.78e-01 (2.21e-02) (≈)**	6.64e-01 (2.80e-02) (+)	4.11e-01 (1.29e-02) (+)	4.18e-01 (2.37e-02) (+)	5.66e-01 (1.42e-02) (+)	2.92e-01 (1.24e-02) (+)
10×5×4×2-13	**8.13e-01 (2.21e-02)**	5.12e-01 (1.66e-02) (+)	6.38e-01 (2.74e-02) (+)	6.96e-01 (2.39e-02) (+)	6.87e-01 (3.59e-02) (+)	4.73e-01 (2.25e-02) (+)	4.33e-01 (8.97e-03) (+)	6.25e-01 (2.41e-02) (+)	3.21e-01 (1.57e-02) (+)
10×5×4×2-14	**8.85e-01 (2.79e-02)**	5.79e-01 (1.31e-02) (+)	6.54e-01 (3.67e-02) (+)	8.01e-01 (2.37e-02) (+)	6.96e-01 (2.30e-02) (+)	4.38e-01 (2.59e-02) (+)	3.83e-01 (2.17e-02) (+)	5.80e-01 (1.63e-02) (+)	3.22e-01 (1.58e-02) (+)
10×5×4×2-15	**7.78e-01 (2.40e-02)**	3.89e-01 (1.15e-02) (+)	5.02e-01 (2.88e-02) (+)	3.46e-01 (1.17e-02) (+)	4.83e-01 (2.20e-02) (+)	2.03e-01 (5.83e-03) (+)	1.75e-01 (8.94e-03) (+)	6.68e-01 (2.73e-02) (+)	9.77e-02 (3.22e-03) (+)
10×5×4×2-16	**8.55e-01 (4.12e-02)**	4.08e-01 (1.52e-02) (+)	5.40e-01 (1.45e-02) (+)	3.33e-01 (1.61e-02) (+)	4.78e-01 (1.89e-02) (+)	2.06e-01 (4.96e-03) (+)	1.46e-01 (5.46e-03) (+)	5.57e-01 (1.97e-02) (+)	1.09e-01 (2.47e-03) (+)
10×5×4×2-17	**7.76e-01 (4.09e-02)**	3.98e-01 (1.11e-02) (+)	5.18e-01 (1.29e-02) (+)	3.19e-01 (1.15e-02) (+)	5.45e-01 (1.87e-02) (+)	1.96e-01 (1.13e-02) (+)	1.76e-01 (7.87e-03) (+)	5.41e-01 (3.02e-02) (+)	1.16e-01 (2.31e-03) (+)
10×5×4×2-18	**8.34e-01 (4.82e-02)**	4.35e-01 (1.59e-02) (+)	5.37e-01 (1.31e-02) (+)	4.15e-01 (1.24e-02) (+)	5.28e-01 (2.16e-02) (+)	2.41e-01 (8.60e-03) (+)	1.91e-01 (8.25e-03) (+)	6.26e-01 (2.96e-02) (+)	1.53e-01 (8.58e-03) (+)
10×5×4×2-19	**8.79e-01 (4.01e-02)**	4.44e-01 (2.78e-02) (+)	6.30e-01 (2.80e-02) (+)	4.16e-01 (1.06e-02) (+)	6.45e-01 (2.99e-02) (+)	2.19e-01 (7.99e-03) (+)	2.29e-01 (8.21e-03) (+)	6.02e-01 (2.11e-02) (+)	1.87e-01 (4.77e-03) (+)
10×5×4×2-20	**8.16e-01 (3.11e-02)**	4.54e-01 (2.37e-02) (+)	5.69e-01 (2.82e-02) (+)	3.98e-01 (1.97e-02) (+)	7.15e-01 (4.26e-02) (+)	2.51e-01 (6.20e-03) (+)	2.40e-01 (1.21e-02) (+)	6.79e-01 (1.60e-02) (+)	2.38e-01 (1.25e-02) (+)
20×15×10×5-1	**7.30e-01 (3.40e-02)**	4.35e-01 (2.53e-02) (+)	3.53e-01 (1.66e-02) (+)	3.61e-01 (1.78e-02) (+)	4.05e-01 (1.62e-02) (+)	1.98e-01 (1.08e-02) (+)	1.58e-01 (5.01e-03) (+)	5.84e-01 (1.34e-02) (+)	6.58e-02 (2.01e-03) (+)
20×15×10×5-2	**7.56e-01 (2.07e-02)**	3.55e-01 (1.40e-02) (+)	3.69e-01 (2.33e-02) (+)	3.85e-01 (2.27e-02) (+)	3.56e-01 (1.05e-02) (+)	1.98e-01 (9.43e-03) (+)	1.98e-01 (5.80e-03) (+)	5.68e-01 (2.64e-02) (+)	1.18e-01 (4.67e-03) (+)
20×15×10×5-3	8.01e-01 (2.73e-02)	4.16e-01 (1.60e-02) (+)	3.24e-01 (1.46e-02) (+)	2.65e-01 (1.36e-02) (+)	3.15e-01 (1.45e-02) (+)	1.64e-01 (3.34e-03) (+)	1.26e-01 (7.29e-03) (+)	7.05e-01 (1.93e-02) (+)	**8.75e-01 (1.83e-03) (-)**
20×15×10×5-4	**9.17e-01 (3.12e-02)**	4.61e-01 (1.15e-02) (+)	2.67e-01 (9.79e-03) (+)	5.03e-01 (2.89e-02) (+)	5.20e-01 (2.02e-02) (+)	2.55e-01 (9.90e-03) (+)	1.35e-01 (5.16e-03) (+)	6.38e-01 (3.79e-02) (+)	5.54e-02 (2.65e-03) (+)
20×15×10×5-5	**7.78e-01 (1.99e-02)**	4.66e-01 (2.81e-02) (+)	4.52e-01 (1.48e-02) (+)	3.74e-01 (1.81e-02) (+)	3.53e-01 (2.05e-02) (+)	1.68e-01 (6.80e-03) (+)	2.46e-01 (4.70e-03) (+)	5.73e-01 (1.77e-02) (+)	1.81e-01 (1.01e-02) (+)
20×15×10×5-6	**6.91e-01 (2.41e-02)**	5.76e-01 (2.38e-02) (+)	3.62e-01 (1.44e-02) (+)	4.15e-01 (2.25e-02) (+)	3.89e-01 (1.27e-02) (+)	3.51e-01 (1.02e-02) (+)	2.41e-01 (1.30e-02) (+)	6.25e-01 (2.23e-02) (+)	3.17e-01 (1.32e-02) (+)
20×15×10×5-7	**8.27e-01 (2.42e-02)**	4.38e-01 (2.67e-02) (+)	1.98e-01 (6.03e-03) (+)	2.26e-01 (1.24e-02) (+)	3.91e-01 (9.63e-03) (+)	4.96e-02 (1.47e-03) (+)	3.45e-01 (2.50e-03) (+)	3.54e-01 (2.03e-02) (+)	1.56e-01 (7.85e-03) (+)
20×15×10×5-8	**8.18e-01 (4.41e-02)**	4.59e-01 (1.24e-02) (+)	2.87e-01 (1.48e-02) (+)	5.98e-01 (1.93e-02) (+)	5.13e-01 (1.82e-02) (+)	2.43e-01 (9.07e-03) (+)	1.69e-01 (3.77e-03) (+)	7.02e-01 (1.81e-02) (+)	7.04e-02 (3.64e-03) (+)
20×15×10×5-9	**7.92e-01 (1.73e-02)**	4.44e-01 (1.66e-02) (+)	2.68e-01 (1.09e-02) (+)	4.65e-01 (1.22e-02) (+)	4.39e-01 (2.39e-02) (+)	2.34e-01 (8.73e-03) (+)	1.48e-01 (3.74e-03) (+)	5.99e-01 (2.77e-02) (+)	5.60e-02 (2.54e-03) (+)
20×15×10×5-10	**8.00e-01 (3.22e-02)**	4.13e-01 (1.85e-02) (+)	2.85e-01 (1.52e-02) (+)	5.52e-01 (1.27e-02) (+)	5.05e-01 (1.40e-02) (+)	2.26e-01 (5.87e-03) (+)	1.66e-01 (8.66e-03) (+)	6.97e-01 (1.55e-02) (+)	6.17e-02 (3.59e-03) (+)
20×15×10×5-11	**8.77e-01 (3.62e-02)**	4.65e-01 (1.69e-02) (+)	3.04e-01 (1.39e-02) (+)	5.52e-01 (1.25e-02) (+)	5.26e-01 (2.79e-02) (+)	2.64e-01 (5.07e-03) (+)	1.67e-01 (7.17e-03) (+)	7.24e-01 (1.93e-02) (+)	6.21e-02 (1.49e-03) (+)
20×15×10×5-12	**8.93e-01 (2.05e-02)**	5.12e-01 (2.67e-02) (+)	3.89e-01 (1.06e-02) (+)	5.58e-01 (1.66e-02) (+)	5.95e-01 (2.99e-02) (+)	3.45e-01 (1.05e-02) (+)	2.42e-01 (9.49e-03) (+)	7.20e-01 (3.81e-02) (+)	7.21e-01 (8.59e-03) (+)
20×15×10×5-13	7.98e-01 (2.02e-02)	5.57e-01 (1.36e-02) (+)	3.45e-01 (9.45e-03) (+)	**8.18e-01 (2.61e-02) (-)**	5.22e-01 (2.32e-02) (+)	3.32e-01 (1.29e-02) (+)	2.35e-01 (1.22e-02) (+)	6.86e-01 (2.24e-02) (+)	1.41e-01 (5.48e-03) (+)
20×15×10×5-14	**8.07e-01 (2.14e-02)**	5.83e-01 (3.07e-02) (+)	3.58e-01 (2.08e-02) (+)	6.31e-01 (2.72e-02) (+)	6.32e-01 (2.55e-02) (+)	3.33e-01 (1.02e-02) (+)	2.87e-01 (1.04e-02) (+)	7.20e-01 (3.14e-02) (+)	1.52e-01 (8.22e-03) (+)
20×15×10×5-15	**7.89e-01 (1.98e-02)**	5.13e-01 (2.93e-02) (+)	4.12e-01 (1.53e-02) (+)	5.25e-01 (2.53e-02) (+)	5.68e-01 (2.59e-02) (+)	3.61e-01 (1.40e-02) (+)	2.99e-01 (1.27e-02) (+)	7.28e-01 (1.43e-02) (+)	1.89e-01 (8.50e-03) (+)
20×15×10×5-16	**8.22e-01 (2.78e-02)**	3.71e-01 (1.41e-02) (+)	4.22e-01 (1.94e-02) (+)	3.94e-01 (1.78e-02) (+)	4.01e-01 (1.67e-02) (+)	1.80e-01 (8.30e-03) (+)	2.21e-01 (6.00e-03) (+)	5.30e-01 (2.47e-02) (+)	1.28e-01 (7.33e-03) (+)
20×15×10×5-17	**7.61e-01 (3.85e-02)**	3.97e-01 (2.03e-02) (+)	3.71e-01 (1.94e-02) (+)	4.14e-01 (1.05e-02) (+)	3.74e-01 (1.62e-02) (+)	1.95e-01 (9.29e-03) (+)	2.31e-01 (9.51e-03) (+)	5.29e-01 (2.33e-02) (+)	1.42e-01 (4.38e-03) (+)
20×15×10×5-18	**8.13e-01 (1.65e-02)**	3.88e-01 (1.40e-02) (+)	4.14e-01 (9.24e-03) (+)	3.89e-01 (2.58e-02) (+)	4.29e-01 (1.92e-02) (+)	2.04e-01 (5.78e-03) (+)	2.10e-01 (9.44e-03) (+)	6.01e-01 (2.14e-02) (+)	1.54e-01 (7.49e-03) (+)
20×15×10×5-19	**7.72e-01 (2.45e-02)**	3.77e-01 (1.73e-02) (+)	3.92e-01 (1.77e-02) (+)	4.42e-01 (1.76e-02) (+)	4.15e-01 (1.10e-02) (+)	2.39e-01 (9.38e-03) (+)	2.18e-01 (1.18e-02) (+)	5.43e-01 (1.75e-02) (+)	1.52e-01 (8.90e-03) (+)
20×15×10×5-20	**7.52e-01 (3.36e-02)**	4.28e-01 (1.67e-02) (+)	4.25e-01 (1.05e-02) (+)	4.17e-01 (1.71e-02) (+)	4.70e-01 (1.48e-02) (+)	2.51e-01 (1.35e-02) (+)	2.01e-01 (6.79e-03) (+)	5.51e-01 (2.84e-02) (+)	1.73e-01 (8.34e-03) (+)
30×20×15×7-1	**8.84e-01 (5.07e-02)**	3.36e-01 (2.08e-02) (+)	5.46e-01 (2.03e-02) (+)	4.21e-01 (2.20e-02) (+)	4.47e-01 (2.37e-02) (+)	9.66e-02 (1.84e-03) (+)	8.06e-01 (2.62e-03) (+)	2.70e-01 (9.78e-03) (+)	8.48e-02 (3.41e-03) (+)
30×20×15×7-2	**8.17e-01 (3.32e-02)**	6.68e-01 (2.05e-02) (+)	2.60e-01 (1.21e-02) (+)	4.54e-01 (1.90e-02) (+)	4.57e-01 (9.97e-03) (+)	1.53e-01 (3.85e-03) (+)	2.14e-01 (9.81e-03) (+)	2.58e-01 (1.47e-02) (+)	1.82e-01 (3.69e-03) (+)
30×20×15×7-3	**7.61e-01 (2.39e-02)**	6.33e-01 (1.48e-02) (+)	3.10e-01 (1.33e-02) (+)	**7.53e-01 (1.30e-02) (≈)**	4.36e-01 (1.71e-02) (+)	1.26e-01 (3.34e-03) (+)	1.77e-01 (7.92e-03) (+)	1.45e-01 (3.17e-03) (+)	9.58e-02 (5.68e-03) (+)
30×20×15×7-4	**8.75e-01 (2.40e-02)**	6.56e-01 (2.59e-02) (+)	4.69e-01 (1.10e-02) (+)	4.23e-01 (9.61e-03) (+)	3.79e-01 (1.79e-02) (+)	2.25e-01 (4.98e-03) (+)	1.74e-01 (7.20e-03) (+)	4.05e-01 (1.79e-02) (+)	2.32e-01 (9.72e-03) (+)
30×20×15×7-5	**6.79e-01 (2.65e-02)**	5.85e-01 (2.10e-02) (+)	3.71e-01 (1.95e-02) (+)	3.80e-01 (1.84e-02) (+)	3.98e-01 (1.45e-02) (+)	1.16e-01 (5.91e-03) (+)	2.76e-01 (5.36e-03) (+)	5.18e-01 (9.34e-03) (+)	2.20e-01 (1.39e-02) (+)
30×20×15×7-6	**7.82e-01 (2.65e-02)**	5.99e-01 (3.91e-02) (+)	3.64e-01 (1.55e-02) (+)	4.31e-01 (2.78e-02) (+)	4.02e-01 (2.11e-02) (+)	1.17e-01 (4.92e-03) (+)	3.12e-01 (1.67e-02) (+)	3.50e-01 (8.69e-03) (+)	2.59e-01 (9.26e-03) (+)
30×20×15×7-7	**6.90e-01 (1.51e-02)**	5.39e-01 (2.61e-02) (+)	3.77e-01 (1.63e-02) (+)	4.58e-01 (1.42e-02) (+)	3.89e-01 (1.06e-02) (+)	1.78e-01 (1.02e-02) (+)	2.78e-01 (6.89e-03) (+)	3.74e-01 (7.08e-03) (+)	2.34e-01 (1.34e-02) (+)
30×20×15×7-8	**7.65e-01 (2.95e-02)**	6.24e-01 (2.43e-02) (+)	2.81e-01 (1.51e-02) (+)	4.65e-01 (1.98e-02) (+)	5.09e-01 (1.32e-02) (+)	1.49e-01 (5.34e-03) (+)	2.11e-01 (1.07e-02) (+)	2.69e-01 (9.24e-03) (+)	1.71e-01 (7.71e-03) (+)
30×20×15×7-9	**7.25e-01 (2.24e-02)**	6.48e-01 (3.04e-02) (+)	2.75e-01 (1.51e-02) (+)	3.95e-01 (2.23e-02) (+)	4.58e-01 (1.22e-02) (+)	1.12e-01 (2.94e-03) (+)	2.19e-01 (6.29e-03) (+)	2.26e-01 (8.30e-03) (+)	1.62e-01 (4.25e-03) (+)
30×20×15×7-10	**7.51e-01 (3.02e-02)**	6.13e-01 (1.79e-02) (+)	3.23e-01 (1.59e-02) (+)	4.39e-01 (1.95e-02) (+)	5.21e-01 (1.23e-02) (+)	1.27e-01 (5.52e-03) (+)	2.32e-01 (9.04e-03) (+)	2.79e-01 (9.15e-03) (+)	1.74e-01 (1.12e-02) (+)
30×20×15×7-11	**8.11e-01 (1.72e-02)**	5.65e-01 (1.49e-02) (+)	2.76e-01 (9.08e-03) (+)	4.79e-01 (9.91e-03) (+)	5.00e-01 (2.07e-02) (+)	1.18e-01 (4.63e-03) (+)	1.97e-01 (1.26e-02) (+)	2.43e-01 (1.08e-02) (+)	1.66e-01 (7.94e-03) (+)
30×20×15×7-12	**8.06e-01 (2.24e-02)**	5.63e-01 (3.17e-02) (+)	3.19e-01 (1.40e-02) (+)	4.79e-01 (1.13e-02) (+)	4.97e-01 (1.54e-02) (+)	1.25e-01 (4.20e-03) (+)	2.01e-01 (1.14e-02) (+)	2.42e-01 (1.24e-02) (+)	1.72e-01 (5.58e-03) (+)
30×20×15×7-13	**7.43e-01 (3.61e-02)**	5.69e-01 (1.46e-02) (+)	3.25e-01 (1.34e-02) (+)	4.71e-01 (1.44e-02) (+)	4.57e-01 (1.01e-02) (+)	1.29e-01 (6.47e-03) (+)	1.97e-01 (8.78e-03) (+)	2.52e-01 (1.28e-02) (+)	1.62e-01 (8.35e-03) (+)
30×20×15×7-14	**8.62e-01 (4.40e-02)**	6.03e-01 (3.30e-02) (+)	3.80e-01 (1.13e-02) (+)	6.39e-01 (1.13e-02) (+)	5.22e-01 (3.06e-02) (+)	1.49e-01 (8.68e-03) (+)	1.82e-01 (7.96e-03) (+)	2.74e-01 (1.24e-02) (+)	1.43e-01 (4.65e-03) (+)
30×20×15×7-15	**8.68e-01 (2.52e-02)**	5.87e-01 (1.74e-02) (+)	3.41e-01 (9.13e-03) (+)	4.50e-01 (1.78e-02) (+)	5.08e-01 (2.63e-02) (+)	1.41e-01 (6.63e-03) (+)	1.97e-01 (6.83e-03) (+)	2.46e-01 (1.43e-02) (+)	1.41e-01 (6.69e-03) (+)
30×20×15×7-16	**6.98e-01 (2.49e-02)**	5.52e-01 (1.84e-02) (+)	3.94e-01 (2.13e-02) (+)	4.07e-01 (1.72e-02) (+)	3.66e-01 (1.95e-02) (+)	1.83e-01 (7.06e-03) (+)	3.00e-01 (8.92e-03) (+)	3.19e-01 (1.93e-02) (+)	3.52e-01 (7.87e-03) (+)

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
30×20×15×7-17	**7.90e-01 (2.74e-02)**	6.08e-01 (2.60e-02) (+)	3.99e-01 (7.99e-03) (+)	4.30e-01 (1.66e-02) (+)	3.90e-01 (1.42e-02) (+)	2.10e-01 (1.22e-02) (+)	2.87e-01 (1.34e-02) (+)	3.29e-01 (7.74e-03) (+)	2.47e-01 (1.40e-02) (+)
30×20×15×7-18	**6.96e-01 (2.44e-02)**	6.48e-01 (1.70e-02) (+)	4.26e-01 (2.18e-02) (+)	4.38e-01 (2.76e-02) (+)	3.93e-01 (2.07e-02) (+)	2.27e-01 (1.06e-02) (+)	2.94e-01 (6.75e-03) (+)	3.33e-01 (1.50e-02) (+)	2.61e-01 (1.27e-02) (+)
30×20×15×7-19	**7.27e-01 (2.39e-02)**	5.92e-01 (1.97e-02) (+)	4.26e-01 (1.06e-02) (+)	4.89e-01 (2.68e-02) (+)	4.10e-01 (2.22e-02) (+)	2.29e-01 (5.37e-03) (+)	3.24e-01 (6.69e-03) (+)	3.83e-01 (1.23e-02) (+)	2.95e-01 (1.05e-02) (+)
30×20×15×7-20	**8.11e-01 (4.25e-02)**	5.95e-01 (2.02e-02) (+)	4.34e-01 (1.00e-02) (+)	4.54e-01 (2.67e-02) (+)	4.63e-01 (2.23e-02) (+)	2.16e-01 (5.77e-03) (+)	**8.05e-01 (1.41e-02) (≈)**	3.94e-01 (8.71e-03) (+)	2.75e-01 (1.73e-02) (+)
40×25×20×10-1	**8.25e-01 (2.30e-02)**	7.44e-01 (3.91e-02) (+)	4.46e-01 (2.57e-02) (+)	3.26e-01 (1.00e-02) (+)	5.51e-01 (2.62e-02) (+)	1.02e-01 (6.50e-03) (+)	1.21e-01 (7.02e-03) (+)	3.04e-01 (7.01e-03) (+)	1.89e-01 (1.06e-02) (+)
40×25×20×10-2	**7.68e-01 (4.25e-02)**	7.15e-01 (2.61e-02) (+)	4.21e-01 (1.92e-02) (+)	3.17e-01 (1.11e-02) (+)	4.77e-01 (1.84e-02) (+)	1.05e-01 (2.33e-03) (+)	1.03e-01 (4.98e-03) (+)	3.24e-01 (1.70e-02) (+)	1.79e-01 (8.26e-03) (+)
40×25×20×10-3	**7.31e-01 (1.67e-02)**	6.85e-01 (2.46e-02) (+)	4.05e-01 (2.35e-02) (+)	3.03e-01 (1.78e-02) (+)	4.69e-01 (1.61e-02) (+)	1.18e-01 (2.51e-03) (+)	1.15e-01 (4.68e-03) (+)	3.15e-01 (1.60e-02) (+)	2.00e-01 (1.01e-02) (+)
40×25×20×10-4	**7.35e-01 (3.47e-02)**	6.89e-01 (2.18e-02) (+)	4.20e-01 (1.24e-02) (+)	2.88e-01 (1.35e-02) (+)	5.00e-01 (1.63e-02) (+)	1.69e-01 (6.33e-03) (+)	1.62e-01 (7.76e-03) (+)	3.51e-01 (1.97e-02) (+)	2.18e-01 (1.09e-02) (+)
40×25×20×10-5	**7.96e-01 (3.71e-02)**	5.63e-01 (1.23e-02) (+)	3.81e-01 (1.77e-02) (+)	5.00e-01 (1.34e-02) (+)	4.27e-01 (1.02e-02) (+)	2.18e-01 (1.10e-02) (+)	2.43e-01 (7.00e-03) (+)	3.74e-01 (9.66e-03) (+)	1.88e-01 (9.71e-03) (+)
40×25×20×10-6	**7.69e-01 (3.62e-02)**	6.08e-01 (2.75e-02) (+)	4.85e-01 (2.67e-02) (+)	5.20e-01 (1.10e-02) (+)	4.02e-01 (1.32e-02) (+)	2.66e-01 (1.35e-02) (+)	2.49e-01 (1.13e-02) (+)	4.49e-01 (1.64e-02) (+)	2.39e-01 (1.07e-02) (+)
40×25×20×10-7	**6.84e-01 (1.88e-02)**	5.28e-01 (1.43e-02) (+)	4.17e-01 (1.58e-02) (+)	4.28e-01 (2.42e-02) (+)	3.75e-01 (1.63e-02) (+)	2.24e-01 (4.38e-03) (+)	4.69e-01 (6.17e-03) (+)	3.96e-01 (1.02e-02) (+)	2.14e-01 (9.27e-03) (+)
40×25×20×10-8	**7.35e-01 (2.85e-02)**	6.95e-01 (2.11e-02) (+)	4.60e-01 (1.25e-02) (+)	3.58e-01 (7.15e-03) (+)	5.10e-01 (2.44e-02) (+)	1.21e-01 (4.59e-03) (+)	1.35e-01 (3.89e-03) (+)	3.43e-01 (1.77e-02) (+)	1.94e-01 (1.12e-02) (+)
40×25×20×10-9	**7.28e-01 (4.75e-02)**	6.44e-01 (3.40e-02) (+)	4.86e-01 (1.33e-02) (+)	3.61e-01 (6.71e-03) (+)	5.08e-01 (2.36e-02) (+)	1.13e-01 (3.28e-03) (+)	1.32e-01 (4.14e-03) (+)	3.57e-01 (1.72e-02) (+)	2.18e-01 (6.22e-03) (+)
40×25×20×10-10	**7.18e-01 (3.66e-02)**	6.31e-01 (2.88e-02) (+)	4.27e-01 (1.09e-02) (+)	3.58e-01 (1.18e-02) (+)	5.67e-01 (1.40e-02) (+)	1.21e-01 (5.73e-03) (+)	1.30e-01 (7.53e-03) (+)	3.15e-01 (1.49e-02) (+)	2.06e-01 (9.54e-03) (+)
40×25×20×10-11	**7.94e-01 (2.65e-02)**	6.17e-01 (4.12e-02) (+)	4.19e-01 (1.45e-02) (+)	3.58e-01 (1.41e-02) (+)	5.46e-01 (1.05e-02) (+)	1.10e-01 (6.20e-03) (+)	1.26e-01 (3.44e-03) (+)	3.05e-01 (1.76e-02) (+)	2.10e-01 (1.22e-02) (+)
40×25×20×10-12	**7.83e-01 (3.13e-02)**	6.25e-01 (3.16e-02) (+)	4.32e-01 (2.80e-02) (+)	3.63e-01 (1.09e-02) (+)	5.53e-01 (2.61e-02) (+)	1.10e-01 (3.80e-03) (+)	1.29e-01 (8.27e-03) (+)	2.93e-01 (1.29e-02) (+)	2.27e-01 (9.28e-03) (+)
40×25×20×10-13	**8.21e-01 (3.12e-02)**	6.60e-01 (1.58e-02) (+)	4.48e-01 (2.11e-02) (+)	3.73e-01 (9.72e-03) (+)	5.14e-01 (2.36e-02) (+)	1.19e-01 (6.36e-03) (+)	1.39e-01 (8.30e-03) (+)	3.53e-01 (1.24e-02) (+)	2.21e-01 (9.53e-03) (+)
40×25×20×10-14	**7.13e-01 (3.27e-02)**	6.84e-01 (1.34e-02) (+)	4.86e-01 (1.10e-02) (+)	3.58e-01 (1.33e-02) (+)	5.32e-01 (1.55e-02) (+)	1.00e-01 (6.01e-03) (+)	1.36e-01 (8.70e-03) (+)	3.05e-01 (1.98e-02) (+)	1.94e-01 (1.26e-02) (+)
40×25×20×10-15	**8.57e-01 (3.78e-02)**	6.02e-01 (2.13e-02) (+)	4.54e-01 (2.35e-02) (+)	5.43e-01 (1.64e-02) (+)	4.76e-01 (2.62e-02) (+)	2.55e-01 (1.33e-02) (+)	2.79e-01 (9.48e-03) (+)	4.33e-01 (1.80e-02) (+)	1.95e-01 (1.28e-02) (+)
40×25×20×10-16	**6.93e-01 (2.27e-02)**	6.60e-01 (2.16e-02) (+)	4.70e-01 (2.14e-02) (+)	5.14e-01 (2.86e-02) (+)	4.23e-01 (1.66e-02) (+)	2.48e-01 (1.16e-02) (+)	2.78e-01 (6.10e-03) (+)	4.04e-01 (1.38e-02) (+)	2.42e-01 (9.37e-03) (+)
40×25×20×10-17	**6.69e-01 (1.68e-02)**	6.51e-01 (1.41e-02) (+)	4.74e-01 (1.54e-02) (+)	4.06e-01 (1.77e-02) (+)	4.16e-01 (1.86e-02) (+)	2.71e-01 (1.02e-02) (+)	**7.40e-01 (7.66e-03) (-)**	3.94e-01 (1.94e-02) (+)	2.39e-01 (1.08e-02) (+)
40×25×20×10-18	**7.65e-01 (4.16e-02)**	6.61e-01 (1.82e-02) (+)	4.40e-01 (1.76e-02) (+)	4.75e-01 (3.05e-02) (+)	4.64e-01 (1.69e-02) (+)	2.84e-01 (1.49e-02) (+)	2.72e-01 (8.37e-03) (+)	4.48e-01 (1.37e-02) (+)	2.21e-01 (5.28e-03) (+)
40×25×20×10-19	**7.49e-01 (4.17e-02)**	6.42e-01 (1.75e-02) (+)	4.81e-01 (1.27e-02) (+)	5.24e-01 (2.58e-02) (+)	4.56e-01 (2.66e-02) (+)	2.73e-01 (1.45e-02) (+)	2.83e-01 (1.49e-02) (+)	4.02e-01 (1.95e-02) (+)	2.60e-01 (5.96e-03) (+)
40×25×20×10-20	**8.33e-01 (4.32e-02)**	6.56e-01 (3.15e-02) (+)	4.91e-01 (1.82e-02) (+)	3.10e-01 (1.43e-02) (+)	4.90e-01 (1.59e-02) (+)	1.12e-01 (5.44e-03) (+)	1.31e-01 (5.54e-03) (+)	3.05e-01 (1.63e-02) (+)	1.82e-01 (1.13e-02) (+)
+/ ≈ / −		80/0/0	80/0/0	77/2/1	80/0/0	80/0/0	78/1/1	80/0/0	79/0/1

Table 9

STATISTICAL RESULTS OF HGMAPPO VERSUS OTHER COMPARISON ALGORITHMS ON GD METRIC.

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
10×5×4×2-1	**0.00e+00 (0.00e+00)**	3.12e-01 (1.03e-02) (+)	3.86e-01 (2.06e-02) (+)	3.18e-01 (9.66e-03) (+)	1.80e-01 (4.54e-03) (+)	4.57e-01 (1.63e-02) (+)	5.99e-01 (1.42e-02) (+)	2.24e-01 (7.22e-03) (+)	5.38e-01 (2.02e-02) (+)
10×5×4×2-2	**0.00e+00 (0.00e+00)**	2.64e-01 (8.18e-03) (+)	3.32e-01 (1.78e-02) (+)	3.45e-01 (2.19e-02) (+)	2.93e-01 (1.32e-02) (+)	4.96e-01 (1.29e-02) (+)	5.61e-01 (1.93e-02) (+)	1.78e-01 (4.47e-03) (+)	4.69e-01 (1.17e-02) (+)
10×5×4×2-3	**0.00e+00 (0.00e+00)**	2.39e-01 (7.83e-03) (+)	3.18e-01 (7.39e-03) (+)	2.68e-01 (1.03e-02) (+)	2.49e-01 (8.85e-03) (+)	4.87e-01 (9.79e-03) (+)	6.29e-01 (2.40e-02) (+)	1.16e-01 (4.05e-03) (+)	5.40e-01 (2.74e-02) (+)
10×5×4×2-4	**0.00e+00 (0.00e+00)**	3.57e-01 (1.10e-02) (+)	1.75e-01 (9.48e-03) (+)	1.75e-01 (7.19e-03) (+)	1.90e-01 (7.66e-03) (+)	3.60e-01 (1.69e-02) (+)	4.81e-01 (2.55e-02) (+)	2.24e-01 (1.16e-02) (+)	5.50e-01 (1.38e-02) (+)
10×5×4×2-5	**0.00e+00 (0.00e+00)**	3.91e-01 (1.71e-02) (+)	2.33e-01 (5.40e-03) (+)	3.87e-01 (2.41e-02) (+)	2.65e-01 (1.27e-02) (+)	5.43e-01 (2.34e-02) (+)	6.29e-01 (2.40e-02) (+)	1.71e-01 (7.10e-03) (+)	6.58e-01 (2.38e-02) (+)
10×5×4×2-6	**0.00e+00 (0.00e+00)**	2.98e-01 (7.02e-03) (+)	4.20e-01 (9.97e-03) (+)	2.87e-01 (1.03e-02) (+)	2.30e-01 (4.52e-03) (+)	4.26e-01 (1.18e-02) (+)	5.77e-01 (2.76e-02) (+)	2.01e-01 (6.40e-03) (+)	5.36e-01 (2.08e-02) (+)
10×5×4×2-7	**0.00e+00 (0.00e+00)**	3.29e-01 (6.34e-03) (+)	3.73e-01 (2.18e-02) (+)	3.09e-01 (1.56e-02) (+)	2.32e-01 (1.04e-02) (+)	4.82e-01 (2.28e-02) (+)	5.77e-01 (2.40e-02) (+)	1.80e-01 (9.35e-03) (+)	5.45e-01 (9.88e-03) (+)
10×5×4×2-8	**0.00e+00 (0.00e+00)**	3.27e-01 (1.57e-02) (+)	3.62e-01 (1.94e-02) (+)	3.09e-01 (1.20e-02) (+)	2.37e-01 (1.23e-02) (+)	4.81e-01 (2.41e-02) (+)	5.73e-01 (1.24e-02) (+)	1.86e-01 (5.82e-03) (+)	5.12e-01 (2.04e-02) (+)
10×5×4×2-9	**0.00e+00 (0.00e+00)**	2.99e-01 (9.79e-03) (+)	1.54e-01 (6.07e-03) (+)	1.46e-01 (6.80e-03) (+)	1.66e-01 (3.47e-03) (+)	2.83e-01 (1.70e-02) (+)	3.73e-01 (1.91e-02) (+)	1.91e-01 (6.07e-03) (+)	4.39e-01 (1.17e-02) (+)
10×5×4×2-10	**0.00e+00 (0.00e+00)**	2.55e-01 (1.35e-02) (+)	1.64e-01 (5.37e-03) (+)	1.48e-01 (8.40e-03) (+)	1.75e-01 (9.10e-03) (+)	3.09e-01 (7.56e-03) (+)	3.36e-01 (6.92e-03) (+)	1.86e-01 (5.56e-03) (+)	3.86e-01 (2.12e-02) (+)
10×5×4×2-11	**1.21e-02 (3.33e-04)**	2.58e-01 (1.02e-02) (+)	1.45e-01 (9.30e-03) (+)	1.60e-01 (4.98e-03) (+)	1.58e-01 (6.82e-03) (+)	3.06e-02 (2.58e-04) (+)	3.59e-01 (1.40e-02) (+)	1.81e-01 (9.90e-03) (+)	4.25e-01 (1.82e-02) (+)
10×5×4×2-12	**5.79e-03 (2.24e-04)**	2.96e-01 (1.33e-02) (+)	1.54e-01 (3.50e-03) (+)	**5.47e-03 (5.66e-03) (≈)**	1.63e-01 (5.61e-03) (+)	3.21e-01 (1.55e-02) (+)	3.18e-01 (1.60e-02) (+)	1.82e-01 (7.93e-03) (+)	4.25e-01 (2.54e-02) (+)
10×5×4×2-13	**0.00e+00 (0.00e+00)**	2.61e-01 (6.58e-03) (+)	1.61e-01 (3.06e-03) (+)	1.32e-01 (8.40e-03) (+)	1.48e-01 (3.00e-03) (+)	2.50e-01 (8.86e-03) (+)	3.08e-01 (1.50e-02) (+)	1.74e-01 (9.54e-03) (+)	4.33e-01 (2.51e-02) (+)
10×5×4×2-14	**1.23e-02 (5.71e-04)**	2.37e-01 (1.27e-02) (+)	2.34e-02 (7.78e-03) (+)	1.44e-01 (7.17e-03) (+)	1.53e-01 (5.72e-03) (+)	2.19e-02 (1.27e-02) (+)	3.49e-01 (2.04e-02) (+)	1.89e-01 (8.60e-03) (+)	4.12e-01 (1.77e-02) (+)
10×5×4×2-15	**0.00e+00 (0.00e+00)**	3.68e-01 (1.93e-02) (+)	2.58e-01 (1.27e-02) (+)	4.34e-01 (2.10e-02) (+)	2.75e-01 (1.57e-02) (+)	5.30e-01 (1.35e-02) (+)	5.52e-01 (1.69e-02) (+)	1.65e-01 (5.57e-03) (+)	6.62e-01 (3.45e-02) (+)
10×5×4×2-16	**0.00e+00 (0.00e+00)**	3.38e-01 (9.69e-03) (+)	2.38e-01 (9.00e-03) (+)	3.85e-01 (1.55e-02) (+)	2.45e-01 (6.29e-03) (+)	5.02e-01 (1.76e-02) (+)	5.79e-01 (1.40e-02) (+)	2.00e-01 (4.58e-03) (+)	6.00e-01 (1.44e-02) (+)
10×5×4×2-17	**0.00e+00 (0.00e+00)**	3.42e-01 (1.16e-02) (+)	2.28e-01 (8.86e-03) (+)	3.99e-01 (1.50e-02) (+)	2.52e-01 (1.18e-02) (+)	4.93e-01 (2.78e-02) (+)	5.45e-01 (3.46e-02) (+)	1.91e-01 (1.08e-02) (+)	5.97e-01 (1.42e-02) (+)
10×5×4×2-18	**0.00e+00 (0.00e+00)**	3.12e-01 (1.57e-02) (+)	2.33e-01 (7.53e-03) (+)	3.33e-01 (1.27e-02) (+)	2.43e-01 (8.47e-03) (+)	4.75e-01 (2.40e-02) (+)	5.76e-01 (1.36e-02) (+)	2.14e-01 (4.90e-03) (+)	6.33e-01 (3.10e-02) (+)
10×5×4×2-19	2.52e-02 (6.60e-04)	3.07e-01 (9.06e-03) (+)	**2.17e-02 (7.89e-03) (-)**	3.65e-01 (2.26e-02) (+)	1.60e-01 (7.19e-03) (+)	4.88e-01 (2.41e-02) (+)	4.83e-01 (2.26e-02) (+)	1.79e-01 (6.07e-03) (+)	5.70e-01 (2.12e-02) (+)
10×5×4×2-20	**2.61e-02 (9.05e-04)**	2.42e-01 (7.18e-03) (+)	3.54e-02 (1.31e-02) (+)	2.96e-01 (1.14e-02) (+)	3.44e-02 (6.31e-03) (+)	4.33e-01 (1.59e-02) (+)	4.40e-01 (2.12e-02) (+)	1.73e-01 (5.21e-03) (+)	4.54e-01 (1.85e-02) (+)
20×15×10×5-1	**0.00e+00 (0.00e+00)**	2.57e-01 (5.78e-03) (+)	3.34e-01 (8.60e-03) (+)	3.11e-01 (8.73e-03) (+)	2.76e-01 (1.23e-02) (+)	4.62e-01 (9.63e-03) (+)	5.00e-01 (1.26e-02) (+)	1.44e-01 (5.10e-03) (+)	6.85e-01 (2.58e-02) (+)
20×15×10×5-2	**0.00e+00 (0.00e+00)**	3.10e-01 (1.74e-02) (+)	2.75e-01 (1.46e-02) (+)	3.14e-01 (1.72e-02) (+)	3.64e-01 (1.22e-02) (+)	4.89e-01 (1.29e-02) (+)	4.97e-01 (2.32e-02) (+)	1.62e-01 (3.89e-03) (+)	5.47e-01 (3.27e-02) (+)
20×15×10×5-3	**0.00e+00 (0.00e+00)**	3.21e-01 (1.75e-02) (+)	3.55e-01 (1.56e-02) (+)	4.57e-01 (2.05e-02) (+)	4.22e-01 (2.09e-02) (+)	5.89e-01 (1.72e-02) (+)	5.37e-01 (1.80e-02) (+)	1.37e-01 (5.75e-03) (+)	6.28e-01 (3.16e-02) (+)
20×15×10×5-4	**0.00e+00 (0.00e+00)**	3.37e-01 (1.13e-02) (+)	5.54e-01 (1.42e-02) (+)	2.98e-01 (1.49e-02) (+)	3.87e-01 (1.65e-02) (+)	5.32e-01 (1.46e-02) (+)	7.01e-01 (1.82e-02) (+)	2.05e-01 (9.92e-03) (+)	8.43e-01 (4.85e-02) (+)
20×15×10×5-5	**0.00e+00 (0.00e+00)**	2.95e-01 (1.62e-02) (+)	3.00e-01 (1.21e-02) (+)	3.52e-01 (1.04e-02) (+)	3.86e-01 (1.90e-02) (+)	6.00e-01 (2.11e-02) (+)	4.69e-01 (1.34e-02) (+)	1.85e-01 (4.95e-03) (+)	6.09e-01 (2.71e-02) (+)
20×15×10×5-6	**0.00e+00 (0.00e+00)**	1.30e-01 (5.65e-03) (+)	3.07e-01 (1.48e-02) (+)	2.78e-01 (1.25e-02) (+)	2.76e-01 (1.45e-02) (+)	2.84e-01 (1.11e-02) (+)	3.68e-01 (1.32e-02) (+)	1.12e-01 (6.26e-03) (+)	3.21e-01 (1.17e-02) (+)
20×15×10×5-7	**0.00e+00 (0.00e+00)**	2.75e-01 (1.02e-02) (+)	4.76e-01 (1.35e-02) (+)	5.06e-01 (2.48e-02) (+)	2.86e-01 (1.33e-02) (+)	6.76e-01 (2.08e-02) (+)	6.62e-01 (2.03e-02) (+)	3.14e-01 (1.48e-02) (+)	4.80e-01 (1.14e-02) (+)
20×15×10×5-8	**0.00e+00 (0.00e+00)**	3.33e-01 (1.51e-02) (+)	4.61e-01 (1.31e-02) (+)	2.97e-01 (1.20e-02) (+)	3.14e-01 (1.42e-02) (+)	5.07e-01 (2.38e-02) (+)	6.11e-01 (1.74e-02) (+)	1.67e-01 (4.27e-03) (+)	7.25e-01 (2.03e-02) (+)
20×15×10×5-9	**0.00e+00 (0.00e+00)**	2.94e-01 (1.43e-02) (+)	5.06e-01 (1.17e-02) (+)	2.90e-01 (8.37e-03) (+)	3.17e-01 (1.17e-02) (+)	5.20e-01 (2.66e-02) (+)	5.85e-01 (3.09e-02) (+)	1.72e-01 (7.37e-03) (+)	7.04e-01 (4.49e-02) (+)
20×15×10×5-10	2.58e-03 (9.51e-05)	3.31e-01 (1.40e-02) (+)	3.72e-03 (1.70e-04) (+)	**1.47e-03 (5.87e-03) (-)**	2.34e-02 (9.02e-03) (+)	4.82e-01 (2.21e-02) (+)	5.17e-01 (2.69e-02) (+)	1.35e-01 (4.70e-03) (+)	6.81e-01 (2.49e-02) (+)
20×15×10×5-11	**0.00e+00 (0.00e+00)**	3.03e-01 (1.59e-02) (+)	4.63e-01 (1.22e-02) (+)	2.92e-01 (8.46e-03) (+)	2.98e-01 (1.76e-02) (+)	5.25e-01 (2.97e-02) (+)	5.99e-01 (2.01e-02) (+)	1.74e-01 (1.06e-02) (+)	7.61e-01 (3.16e-02) (+)
20×15×10×5-12	**0.00e+00 (0.00e+00)**	2.26e-01 (1.30e-02) (+)	3.19e-01 (1.91e-02) (+)	1.88e-01 (4.68e-03) (+)	1.91e-01 (7.29e-03) (+)	3.45e-01 (1.70e-02) (+)	4.70e-01 (1.53e-02) (+)	8.97e-02 (2.29e-03) (+)	5.83e-01 (1.75e-02) (+)
20×15×10×5-13	**0.00e+00 (0.00e+00)**	2.66e-01 (1.21e-02) (+)	3.88e-01 (2.03e-02) (+)	1.90e-01 (7.70e-03) (+)	2.01e-01 (5.05e-03) (+)	3.47e-01 (8.07e-03) (+)	4.61e-01 (1.43e-02) (+)	1.12e-01 (3.22e-03) (+)	5.20e-01 (1.43e-02) (+)
20×15×10×5-14	**0.00e+00 (0.00e+00)**	2.30e-01 (9.75e-03) (+)	3.66e-01 (1.12e-02) (+)	1.70e-01 (6.36e-03) (+)	1.76e-01 (9.42e-03) (+)	3.60e-01 (1.10e-02) (+)	4.06e-01 (1.27e-02) (+)	1.05e-01 (3.58e-03) (+)	5.72e-01 (1.99e-02) (+)
20×15×10×5-15	**0.00e+00 (0.00e+00)**	2.93e-01 (5.87e-03) (+)	3.24e-01 (1.40e-02) (+)	2.02e-01 (5.69e-03) (+)	2.40e-01 (1.24e-02) (+)	3.53e-01 (1.76e-02) (+)	4.68e-01 (1.08e-02) (+)	1.69e-01 (4.46e-03) (+)	5.04e-01 (2.54e-02) (+)
20×15×10×5-16	**0.00e+00 (0.00e+00)**	3.41e-01 (1.79e-02) (+)	2.83e-01 (9.23e-03) (+)	3.22e-01 (1.64e-02) (+)	3.24e-01 (1.21e-02) (+)	5.03e-01 (1.59e-02) (+)	4.65e-01 (1.99e-02) (+)	1.70e-01 (5.97e-03) (+)	5.90e-01 (2.92e-02) (+)

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
20×15×10×5-17	**0.00e+00 (0.00e+00)**	3.11e-01 (8.48e-03) (+)	3.00e-01 (1.13e-02) (+)	3.00e-01 (7.22e-03) (+)	3.55e-01 (7.43e-03) (+)	5.05e-01 (2.30e-02) (+)	4.60e-01 (2.35e-02) (+)	1.77e-01 (8.98e-03) (+)	5.55e-01 (1.96e-02) (+)
20×15×10×5-18	**0.00e+00 (0.00e+00)**	3.02e-01 (7.29e-03) (+)	2.78e-01 (6.58e-03) (+)	3.18e-01 (1.62e-02) (+)	3.03e-01 (2.01e-02) (+)	5.21e-01 (1.10e-02) (+)	4.89e-01 (1.54e-02) (+)	1.45e-01 (4.95e-03) (+)	5.47e-01 (1.27e-02) (+)
20×15×10×5-19	**0.00e+00 (0.00e+00)**	3.32e-01 (9.64e-03) (+)	3.10e-01 (1.55e-02) (+)	3.39e-01 (1.01e-02) (+)	3.60e-01 (1.18e-02) (+)	5.00e-01 (2.87e-02) (+)	5.16e-01 (1.75e-02) (+)	1.81e-01 (5.65e-03) (+)	5.91e-01 (2.14e-02) (+)
20×15×10×5-20	**0.00e+00 (0.00e+00)**	2.91e-01 (1.63e-02) (+)	2.77e-01 (8.01e-03) (+)	3.40e-01 (7.13e-03) (+)	2.98e-01 (1.88e-02) (+)	5.23e-01 (1.77e-02) (+)	4.82e-01 (1.92e-02) (+)	1.82e-01 (9.94e-03) (+)	5.22e-01 (1.73e-02) (+)
30×20×15×7-1	**0.00e+00 (0.00e+00)**	3.87e-01 (2.19e-02) (+)	2.48e-01 (5.88e-03) (+)	4.00e-01 (1.86e-02) (+)	3.15e-01 (1.05e-02) (+)	6.55e-01 (1.58e-02) (+)	6.81e-01 (2.35e-02) (+)	4.57e-01 (2.52e-02) (+)	7.22e-01 (1.79e-02) (+)
30×20×15×7-2	**0.00e+00 (0.00e+00)**	1.13e-01 (3.57e-03) (+)	4.12e-01 (2.23e-02) (+)	2.30e-01 (6.34e-03) (+)	2.40e-01 (8.04e-03) (+)	4.87e-01 (1.28e-02) (+)	4.56e-01 (1.15e-02) (+)	3.91e-01 (1.86e-02) (+)	5.40e-01 (1.60e-02) (+)
30×20×15×7-3	**0.00e+00 (0.00e+00)**	1.23e-01 (3.02e-03) (+)	3.23e-01 (1.79e-02) (+)	2.40e-01 (9.08e-03) (+)	1.83e-01 (7.99e-03) (+)	5.45e-01 (2.74e-02) (+)	4.45e-01 (1.74e-02) (+)	5.29e-01 (2.74e-02) (+)	5.60e-01 (1.35e-02) (+)
30×20×15×7-4	8.99e-03 (2.56e-04)	1.73e-01 (5.08e-03) (+)	2.92e-01 (8.87e-03) (+)	**4.39e-03 (1.15e-02) (-)**	4.42e-01 (1.23e-02) (+)	5.47e-02 (1.53e-02) (+)	5.36e-01 (3.01e-02) (+)	3.82e-01 (1.51e-02) (+)	**5.47e-03 (1.96e-02) (-)**
30×20×15×7-5	**0.00e+00 (0.00e+00)**	1.36e-01 (8.68e-03) (+)	2.61e-01 (1.59e-02) (+)	2.31e-01 (1.10e-02) (+)	2.66e-01 (1.39e-02) (+)	5.06e-01 (2.05e-02) (+)	3.43e-01 (1.82e-02) (+)	2.89e-01 (1.39e-02) (+)	4.06e-01 (1.05e-02) (+)
30×20×15×7-6	**0.00e+00 (0.00e+00)**	1.41e-01 (6.94e-03) (+)	3.43e-01 (1.13e-02) (+)	2.72e-01 (6.29e-03) (+)	3.08e-01 (1.59e-02) (+)	6.45e-01 (1.83e-02) (+)	4.05e-01 (1.16e-02) (+)	3.38e-01 (1.53e-02) (+)	4.64e-01 (2.54e-02) (+)
30×20×15×7-7	**0.00e+00 (0.00e+00)**	1.49e-01 (4.57e-03) (+)	2.55e-01 (6.83e-03) (+)	2.21e-01 (8.33e-03) (+)	2.64e-01 (1.00e-02) (+)	4.41e-01 (1.55e-02) (+)	3.52e-01 (7.32e-03) (+)	2.96e-01 (8.50e-03) (+)	3.91e-01 (1.75e-02) (+)
30×20×15×7-8	**0.00e+00 (0.00e+00)**	1.35e-01 (7.02e-03) (+)	4.16e-01 (2.25e-02) (+)	2.44e-01 (1.17e-02) (+)	2.27e-01 (9.65e-03) (+)	5.11e-01 (3.22e-02) (+)	5.16e-01 (2.21e-02) (+)	4.13e-01 (1.07e-02) (+)	5.48e-01 (2.47e-02) (+)
30×20×15×7-9	**0.00e+00 (0.00e+00)**	1.11e-01 (3.69e-03) (+)	3.89e-01 (8.12e-03) (+)	2.40e-01 (6.88e-03) (+)	2.21e-01 (1.10e-02) (+)	5.42e-01 (2.88e-02) (+)	4.36e-01 (1.99e-02) (+)	4.28e-01 (2.34e-02) (+)	4.63e-01 (1.34e-02) (+)
30×20×15×7-10	**0.00e+00 (0.00e+00)**	1.20e-01 (2.77e-03) (+)	3.52e-01 (1.18e-02) (+)	1.98e-01 (1.13e-02) (+)	1.82e-01 (7.54e-03) (+)	4.87e-01 (2.57e-02) (+)	4.43e-01 (1.69e-02) (+)	3.81e-01 (1.11e-02) (+)	4.92e-01 (1.32e-02) (+)
30×20×15×7-11	**0.00e+00 (0.00e+00)**	1.53e-01 (7.02e-03) (+)	3.55e-01 (1.41e-02) (+)	2.21e-01 (9.50e-03) (+)	2.28e-01 (1.00e-02) (+)	5.70e-01 (2.51e-02) (+)	4.87e-01 (2.48e-02) (+)	4.41e-01 (2.21e-02) (+)	5.14e-01 (2.39e-02) (+)
30×20×15×7-12	**0.00e+00 (0.00e+00)**	1.52e-01 (7.73e-03) (+)	3.37e-01 (1.04e-02) (+)	2.01e-01 (8.47e-03) (+)	2.28e-01 (7.41e-03) (+)	5.47e-01 (2.54e-02) (+)	4.81e-01 (1.87e-02) (+)	3.58e-01 (1.87e-02) (+)	4.63e-01 (1.48e-02) (+)
30×20×15×7-13	**0.00e+00 (0.00e+00)**	1.77e-01 (5.13e-03) (+)	3.51e-01 (1.34e-02) (+)	2.32e-01 (6.47e-03) (+)	2.13e-01 (1.15e-02) (+)	5.49e-01 (2.33e-02) (+)	4.51e-01 (1.29e-02) (+)	3.98e-01 (1.54e-02) (+)	4.62e-01 (1.34e-02) (+)
30×20×15×7-14	**0.00e+00 (0.00e+00)**	1.42e-01 (7.65e-03) (+)	4.26e-01 (1.53e-02) (+)	2.50e-01 (9.07e-03) (+)	2.29e-01 (8.66e-03) (+)	5.24e-01 (1.49e-02) (+)	5.15e-01 (2.20e-02) (+)	4.09e-01 (1.66e-02) (+)	4.51e-01 (1.68e-02) (+)
30×20×15×7-15	**0.00e+00 (0.00e+00)**	1.67e-01 (9.18e-03) (+)	3.78e-01 (2.01e-02) (+)	2.78e-01 (9.42e-03) (+)	2.27e-01 (7.50e-03) (+)	5.27e-01 (2.56e-02) (+)	5.14e-01 (1.43e-02) (+)	4.41e-01 (1.35e-02) (+)	5.03e-01 (1.08e-02) (+)
30×20×15×7-16	**0.00e+00 (0.00e+00)**	1.44e-01 (3.95e-03) (+)	2.27e-01 (1.04e-02) (+)	2.37e-01 (9.54e-03) (+)	2.73e-01 (1.09e-02) (+)	4.71e-01 (2.13e-02) (+)	3.35e-01 (1.28e-02) (+)	3.15e-01 (1.39e-02) (+)	3.84e-01 (2.17e-02) (+)
30×20×15×7-17	**0.00e+00 (0.00e+00)**	1.36e-01 (8.19e-03) (+)	2.31e-01 (1.21e-02) (+)	2.47e-01 (1.29e-02) (+)	2.86e-01 (8.76e-03) (+)	4.12e-01 (1.22e-02) (+)	3.80e-01 (1.65e-02) (+)	3.26e-01 (1.33e-02) (+)	4.28e-01 (2.16e-02) (+)
30×20×15×7-18	**0.00e+00 (0.00e+00)**	1.25e-01 (6.54e-03) (+)	2.37e-01 (8.52e-03) (+)	2.11e-01 (4.31e-03) (+)	2.57e-01 (1.24e-02) (+)	4.06e-01 (2.19e-02) (+)	3.19e-01 (1.13e-02) (+)	2.86e-01 (8.55e-03) (+)	3.89e-01 (1.99e-02) (+)
30×20×15×7-19	**0.00e+00 (0.00e+00)**	1.39e-01 (4.59e-03) (+)	2.28e-01 (1.36e-02) (+)	1.95e-01 (9.83e-03) (+)	2.80e-01 (5.70e-03) (+)	4.20e-01 (1.67e-02) (+)	3.27e-01 (1.33e-02) (+)	2.70e-01 (9.24e-03) (+)	3.53e-01 (9.27e-03) (+)
30×20×15×7-20	**0.00e+00 (0.00e+00)**	1.35e-01 (6.85e-03) (+)	2.30e-01 (9.00e-03) (+)	2.03e-01 (9.01e-03) (+)	2.32e-01 (1.37e-02) (+)	4.48e-01 (2.03e-02) (+)	3.44e-01 (1.03e-02) (+)	2.75e-01 (5.81e-03) (+)	3.92e-01 (1.82e-02) (+)
40×25×20×10-1	**2.86e-02 (1.09e-03)**	8.36e-02 (2.90e-03) (+)	2.74e-01 (1.25e-02) (+)	3.51e-01 (1.42e-02) (+)	1.77e-01 (9.23e-03) (+)	6.55e-01 (3.03e-02) (+)	6.34e-01 (3.26e-02) (+)	3.94e-02 (3.48e-04) (+)	5.02e-01 (1.83e-02) (+)
40×25×20×10-2	**2.40e-02 (9.61e-04)**	5.14e-02 (3.10e-03) (+)	3.55e-02 (8.34e-03) (+)	3.31e-01 (1.75e-02) (+)	1.60e-01 (4.23e-03) (+)	6.21e-01 (1.66e-02) (+)	5.67e-01 (3.25e-02) (+)	3.16e-01 (1.82e-02) (+)	4.70e-01 (1.44e-02) (+)
40×25×20×10-3	2.71e-02 (1.59e-03)	1.07e-01 (6.58e-03) (+)	2.71e-01 (1.29e-02) (+)	3.51e-01 (1.19e-02) (+)	2.20e-01 (7.26e-03) (+)	5.58e-01 (1.27e-02) (+)	5.87e-01 (1.43e-02) (+)	**3.67e-03 (9.63e-03) (-)**	4.50e-01 (2.68e-02) (+)
40×25×20×10-4	**2.69e-02 (9.90e-04)**	8.68e-02 (4.07e-03) (+)	2.66e-01 (1.76e-02) (+)	3.86e-01 (2.07e-02) (+)	1.67e-01 (8.83e-03) (+)	4.98e-01 (2.69e-02) (+)	5.42e-01 (1.50e-02) (+)	3.40e-01 (1.65e-02) (+)	4.76e-01 (2.50e-02) (+)
40×25×20×10-5	**0.00e+00 (0.00e+00)**	1.93e-01 (1.27e-02) (+)	2.83e-01 (1.20e-02) (+)	2.39e-01 (8.21e-03) (+)	2.84e-01 (1.59e-02) (+)	4.77e-01 (1.41e-02) (+)	4.45e-01 (2.93e-02) (+)	3.36e-01 (1.78e-02) (+)	4.63e-01 (1.67e-02) (+)
40×25×20×10-6	**0.00e+00 (0.00e+00)**	1.33e-01 (6.21e-03) (+)	2.19e-01 (1.05e-02) (+)	2.43e-01 (7.14e-03) (+)	2.59e-01 (1.45e-02) (+)	3.70e-01 (2.09e-02) (+)	3.91e-01 (1.18e-02) (+)	2.27e-01 (7.66e-03) (+)	3.78e-01 (9.72e-03) (+)
40×25×20×10-7	**0.00e+00 (0.00e+00)**	1.76e-01 (4.85e-03) (+)	2.39e-01 (1.42e-02) (+)	2.37e-01 (9.70e-03) (+)	2.60e-01 (9.22e-03) (+)	3.91e-01 (2.05e-02) (+)	4.49e-01 (2.46e-02) (+)	2.37e-01 (9.37e-03) (+)	4.43e-01 (1.15e-02) (+)
40×25×20×10-8	**3.24e-02 (1.23e-03)**	9.26e-02 (4.07e-03) (+)	3.74e-02 (6.21e-03) (+)	3.48e-01 (2.10e-02) (+)	2.01e-01 (8.98e-03) (+)	5.79e-01 (2.20e-02) (+)	5.95e-01 (1.87e-02) (+)	3.93e-01 (1.34e-02) (+)	5.06e-01 (1.06e-02) (+)
40×25×20×10-9	**0.00e+00 (0.00e+00)**	9.54e-02 (4.08e-03) (+)	2.38e-01 (6.67e-03) (+)	2.78e-01 (1.31e-02) (+)	1.93e-01 (6.28e-03) (+)	6.15e-01 (2.09e-02) (+)	5.96e-01 (2.69e-02) (+)	3.28e-01 (1.51e-02) (+)	4.19e-01 (1.61e-02) (+)
40×25×20×10-10	**2.84e-02 (8.83e-04)**	1.03e-01 (2.86e-03) (+)	2.72e-01 (1.25e-02) (+)	3.30e-01 (7.36e-03) (+)	1.72e-01 (7.38e-03) (+)	5.54e-01 (2.36e-02) (+)	6.09e-01 (2.84e-02) (+)	3.42e-01 (1.02e-02) (+)	4.50e-01 (2.01e-02) (+)
40×25×20×10-11	**2.53e-02 (5.32e-04)**	9.88e-02 (5.12e-03) (+)	2.68e-01 (1.15e-02) (+)	**2.28e-02 (1.99e-02) (≈)**	1.72e-01 (7.34e-03) (+)	6.10e-01 (1.95e-02) (+)	6.21e-01 (1.50e-02) (+)	3.27e-01 (1.40e-02) (+)	4.50e-01 (1.38e-02) (+)
40×25×20×10-12	**2.82e-02 (9.46e-04)**	9.08e-02 (4.43e-03) (+)	2.54e-01 (4.73e-03) (+)	3.33e-01 (1.07e-02) (+)	1.72e-01 (1.10e-02) (+)	6.20e-01 (2.37e-02) (+)	5.99e-01 (2.99e-02) (+)	3.69e-01 (1.34e-02) (+)	4.81e-01 (1.96e-02) (+)
40×25×20×10-13	**3.99e-02 (2.46e-03)**	9.19e-02 (2.94e-03) (+)	2.73e-01 (1.56e-02) (+)	3.15e-01 (1.32e-02) (+)	2.11e-01 (1.22e-02) (+)	5.64e-01 (3.30e-02) (+)	6.01e-01 (1.61e-02) (+)	3.63e-01 (1.42e-02) (+)	4.32e-01 (1.57e-02) (+)
40×25×20×10-14	**1.74e-02 (6.47e-04)**	7.92e-02 (3.08e-03) (+)	2.14e-01 (4.94e-03) (+)	3.28e-01 (1.19e-02) (+)	1.52e-01 (4.71e-03) (+)	6.12e-01 (1.76e-02) (+)	5.77e-01 (2.72e-02) (+)	3.16e-01 (1.03e-02) (+)	4.58e-01 (2.33e-02) (+)
40×25×20×10-15	**0.00e+00 (0.00e+00)**	2.19e-01 (1.41e-02) (+)	3.09e-01 (8.34e-03) (+)	2.67e-01 (1.56e-02) (+)	3.02e-01 (1.16e-02) (+)	4.74e-01 (2.32e-02) (+)	4.67e-01 (1.16e-02) (+)	2.91e-01 (9.28e-03) (+)	5.20e-01 (2.46e-02) (+)
40×25×20×10-16	**0.00e+00 (0.00e+00)**	1.34e-01 (2.79e-03) (+)	2.24e-01 (7.77e-03) (+)	2.22e-01 (8.25e-03) (+)	2.63e-01 (7.61e-03) (+)	4.16e-01 (1.15e-02) (+)	3.98e-01 (2.50e-02) (+)	2.52e-01 (1.25e-02) (+)	3.90e-01 (8.49e-03) (+)
40×25×20×10-17	**0.00e+00 (0.00e+00)**	1.22e-01 (6.24e-03) (+)	2.07e-01 (1.09e-02) (+)	2.22e-01 (1.11e-02) (+)	2.50e-01 (1.18e-02) (+)	3.71e-01 (2.27e-02) (+)	4.28e-01 (2.26e-02) (+)	2.43e-01 (1.22e-02) (+)	3.66e-01 (8.75e-03) (+)
40×25×20×10-18	**1.50e-02 (3.95e-04)**	1.38e-01 (7.12e-03) (+)	2.28e-01 (9.59e-03) (+)	2.33e-01 (1.14e-02) (+)	2.24e-01 (9.15e-03) (+)	3.49e-01 (1.88e-02) (+)	3.82e-01 (1.87e-02) (+)	2.20e-01 (8.24e-03) (+)	4.00e-01 (1.58e-02) (+)
40×25×20×10-19	**2.19e-02 (7.98e-04)**	1.17e-01 (2.85e-03) (+)	2.07e-01 (1.12e-02) (+)	1.74e-01 (3.93e-03) (+)	2.23e-01 (8.08e-03) (+)	3.76e-01 (1.93e-02) (+)	3.85e-01 (2.58e-02) (+)	2.45e-01 (1.16e-02) (+)	3.68e-01 (1.76e-02) (+)
40×25×20×10-20	**0.00e+00 (0.00e+00)**	8.80e-02 (1.76e-03) (+)	2.61e-01 (5.96e-03) (+)	3.74e-01 (1.91e-02) (+)	1.77e-01 (8.47e-03) (+)	5.79e-01 (2.30e-02) (+)	6.09e-01 (1.88e-02) (+)	4.00e-01 (1.09e-02) (+)	4.91e-01 (2.16e-02) (+)
+/ ≈ / −		80/0/0	79/0/1	76/2/2	80/0/0	80/0/0	80/0/0	79/0/1	79/0/1

Table 10

STATISTICAL RESULTS OF HGMAPPO VERSUS OTHER COMPARISON ALGORITHMS ON IGD METRIC.

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
10×5×4×2-1	**0.00e+00 (0.00e+00)**	3.65e-01 (1.83e-02) (+)	3.60e-01 (1.31e-02) (+)	3.46e-01 (9.85e-03) (+)	2.21e-01 (1.32e-02) (+)	4.92e-01 (2.43e-02) (+)	6.21e-01 (3.21e-02) (+)	2.25e-01 (1.15e-02) (+)	5.64e-01 (2.29e-02) (+)
10×5×4×2-2	**0.00e+00 (0.00e+00)**	2.76e-01 (1.32e-02) (+)	3.35e-01 (1.46e-02) (+)	4.07e-01 (1.78e-02) (+)	3.07e-01 (1.57e-02) (+)	5.67e-01 (3.44e-02) (+)	5.76e-01 (2.55e-02) (+)	1.92e-01 (7.92e-03) (+)	4.30e-01 (2.14e-02) (+)
10×5×4×2-3	**0.00e+00 (0.00e+00)**	2.42e-01 (1.04e-02) (+)	3.13e-01 (9.20e-03) (+)	2.55e-01 (5.22e-03) (+)	2.12e-01 (1.10e-02) (+)	4.56e-01 (2.52e-02) (+)	6.18e-01 (1.86e-02) (+)	1.22e-01 (6.83e-03) (+)	5.65e-01 (2.80e-02) (+)
10×5×4×2-4	**2.75e-02 (1.65e-03)**	3.76e-01 (1.25e-02) (+)	1.72e-01 (4.52e-03) (+)	1.90e-01 (3.83e-03) (+)	3.08e-02 (7.79e-03) (+)	4.07e-01 (1.26e-02) (+)	5.06e-01 (1.16e-02) (+)	2.28e-01 (1.14e-02) (+)	6.03e-01 (3.11e-02) (+)
10×5×4×2-5	**0.00e+00 (0.00e+00)**	3.35e-01 (7.43e-03) (+)	2.20e-01 (5.93e-03) (+)	3.89e-01 (9.99e-03) (+)	2.77e-01 (8.75e-03) (+)	5.39e-01 (2.16e-02) (+)	6.35e-01 (1.78e-02) (+)	1.67e-01 (7.85e-03) (+)	6.59e-01 (1.74e-02) (+)
10×5×4×2-6	**5.95e-02 (1.36e-03)**	3.43e-01 (1.31e-02) (+)	3.98e-01 (1.29e-02) (+)	3.05e-01 (9.25e-03) (+)	2.58e-01 (6.34e-03) (+)	4.88e-01 (1.19e-02) (+)	6.21e-01 (3.70e-02) (+)	2.20e-01 (4.88e-03) (+)	6.13e-01 (2.61e-02) (+)
10×5×4×2-7	**5.50e-02 (1.67e-03)**	3.85e-01 (7.54e-03) (+)	3.59e-01 (1.97e-02) (+)	3.32e-01 (1.16e-02) (+)	2.57e-01 (1.40e-02) (+)	5.41e-01 (3.13e-02) (+)	6.12e-01 (1.42e-02) (+)	2.07e-01 (1.06e-02) (+)	6.09e-01 (2.38e-02) (+)
10×5×4×2-8	**8.04e-02 (2.54e-03)**	3.78e-01 (1.17e-02) (+)	3.55e-01 (1.19e-02) (+)	3.37e-01 (1.14e-02) (+)	2.38e-01 (9.66e-03) (+)	5.41e-01 (1.00e-02) (+)	6.08e-01 (1.40e-02) (+)	2.08e-01 (8.63e-03) (+)	6.10e-01 (2.19e-02) (+)
10×5×4×2-9	**9.81e-02 (4.07e-03)**	2.91e-01 (6.18e-03) (+)	2.03e-01 (5.55e-03) (+)	1.58e-01 (1.01e-02) (+)	1.88e-01 (9.97e-03) (+)	2.98e-01 (7.34e-03) (+)	4.04e-01 (1.86e-02) (+)	1.97e-01 (6.54e-03) (+)	4.53e-01 (1.84e-02) (+)
10×5×4×2-10	**5.90e-02 (1.86e-03)**	2.49e-01 (1.36e-02) (+)	1.77e-01 (4.09e-03) (+)	1.56e-01 (4.00e-03) (+)	1.68e-01 (4.60e-03) (+)	3.21e-01 (1.13e-02) (+)	3.88e-01 (1.52e-02) (+)	1.96e-01 (1.00e-02) (+)	4.41e-01 (1.98e-02) (+)
10×5×4×2-11	**6.24e-02 (3.11e-03)**	2.66e-01 (1.42e-02) (+)	1.73e-01 (4.82e-03) (+)	1.78e-01 (4.88e-03) (+)	1.60e-01 (4.63e-03) (+)	3.04e-01 (1.25e-02) (+)	3.82e-01 (1.81e-02) (+)	2.07e-01 (1.20e-02) (+)	4.45e-01 (1.52e-02) (+)
10×5×4×2-12	**5.44e-02 (2.86e-03)**	2.75e-01 (1.24e-02) (+)	1.85e-01 (1.03e-02) (+)	**5.35e-02 (6.14e-03) (≈)**	1.58e-01 (4.77e-03) (+)	3.29e-01 (1.56e-02) (+)	3.32e-01 (1.71e-02) (+)	2.09e-01 (6.84e-03) (+)	4.37e-01 (1.00e-02) (+)
10×5×4×2-13	**5.55e-02 (2.01e-03)**	2.62e-01 (8.95e-03) (+)	1.74e-01 (5.01e-03) (+)	1.67e-01 (8.45e-03) (+)	1.48e-01 (7.20e-03) (+)	2.74e-01 (1.44e-02) (+)	3.28e-01 (1.39e-02) (+)	1.90e-01 (9.85e-03) (+)	4.11e-01 (9.43e-03) (+)
10×5×4×2-14	**5.57e-02 (2.14e-03)**	2.37e-01 (7.05e-03) (+)	1.82e-01 (3.70e-03) (+)	1.81e-01 (1.01e-02) (+)	1.43e-01 (7.42e-03) (+)	3.10e-01 (1.46e-02) (+)	3.73e-01 (1.37e-02) (+)	2.21e-01 (5.81e-03) (+)	4.29e-01 (2.02e-02) (+)
10×5×4×2-15	**0.00e+00 (0.00e+00)**	3.28e-01 (8.36e-03) (+)	2.36e-01 (1.19e-02) (+)	4.08e-01 (2.03e-02) (+)	2.74e-01 (6.76e-03) (+)	5.37e-01 (3.33e-02) (+)	5.60e-01 (2.68e-02) (+)	1.54e-01 (8.66e-03) (+)	6.59e-01 (3.37e-02) (+)
10×5×4×2-16	**0.00e+00 (0.00e+00)**	3.25e-01 (1.85e-02) (+)	2.00e-01 (9.48e-03) (+)	3.90e-01 (1.52e-02) (+)	2.33e-01 (9.29e-03) (+)	5.20e-01 (2.67e-02) (+)	6.06e-01 (1.73e-02) (+)	1.94e-01 (4.63e-03) (+)	6.05e-01 (2.60e-02) (+)

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
10×5×4×2-17	**0.00e+00 (0.00e+00)**	2.90e-01 (7.65e-03) (+)	1.78e-01 (4.09e-03) (+)	3.62e-01 (1.57e-02) (+)	1.97e-01 (1.01e-02) (+)	5.02e-01 (1.89e-02) (+)	5.45e-01 (2.33e-02) (+)	1.89e-01 (6.68e-03) (+)	6.16e-01 (3.29e-02) (+)
10×5×4×2-18	**0.00e+00 (0.00e+00)**	2.70e-01 (1.55e-02) (+)	1.85e-01 (1.09e-02) (+)	3.56e-01 (1.11e-02) (+)	2.18e-01 (1.24e-02) (+)	4.90e-01 (1.01e-02) (+)	5.84e-01 (3.23e-02) (+)	2.02e-01 (7.20e-03) (+)	6.46e-01 (1.65e-02) (+)
10×5×4×2-19	**2.07e-02 (5.74e-04)**	**2.42e-02 (1.31e-02) (≈)**	1.85e-01 (5.21e-03) (+)	3.28e-01 (1.17e-02) (+)	1.74e-01 (3.64e-03) (+)	4.82e-01 (2.55e-02) (+)	5.08e-01 (1.13e-02) (+)	1.76e-01 (8.67e-03) (+)	5.93e-01 (2.99e-02) (+)
10×5×4×2-20	**2.09e-02 (7.57e-04)**	2.57e-01 (1.69e-02) (+)	2.12e-01 (9.69e-03) (+)	3.45e-01 (9.44e-03) (+)	3.81e-02 (9.01e-03) (+)	4.80e-01 (2.11e-02) (+)	4.73e-01 (1.42e-02) (+)	1.59e-01 (3.62e-03) (+)	4.82e-01 (1.00e-02) (+)
20×15×10×5-1	**0.00e+00 (0.00e+00)**	2.86e-01 (1.38e-02) (+)	3.36e-01 (1.22e-02) (+)	3.21e-01 (1.34e-02) (+)	2.85e-01 (9.41e-03) (+)	4.74e-01 (2.64e-02) (+)	5.01e-01 (1.55e-02) (+)	1.51e-01 (4.85e-03) (+)	7.28e-01 (3.88e-02) (+)
20×15×10×5-2	**0.00e+00 (0.00e+00)**	2.90e-01 (1.11e-02) (+)	2.59e-01 (1.54e-02) (+)	2.97e-01 (1.12e-02) (+)	3.18e-01 (1.29e-02) (+)	4.84e-01 (1.24e-02) (+)	5.06e-01 (1.92e-02) (+)	2.24e-01 (5.22e-03) (+)	5.72e-01 (2.09e-02) (+)
20×15×10×5-3	**4.55e-03 (1.21e-04)**	3.36e-01 (7.97e-03) (+)	3.82e-01 (9.91e-03) (+)	7.46e-03 (1.18e-02) (+)	4.51e-01 (1.76e-02) (+)	5.88e-01 (3.07e-02) (+)	5.91e-01 (2.91e-02) (+)	1.21e-01 (7.66e-03) (+)	6.57e-01 (2.67e-02) (+)
20×15×10×5-4	**0.00e+00 (0.00e+00)**	3.29e-01 (8.49e-03) (+)	5.29e-01 (1.73e-02) (+)	2.92e-01 (8.43e-03) (+)	3.10e-01 (1.61e-02) (+)	4.67e-01 (2.92e-02) (+)	6.68e-01 (4.02e-02) (+)	1.64e-01 (3.79e-03) (+)	8.08e-01 (3.22e-02) (+)
20×15×10×5-5	**0.00e+00 (0.00e+00)**	2.63e-01 (1.32e-02) (+)	2.79e-01 (1.06e-02) (+)	3.33e-01 (1.43e-02) (+)	3.68e-01 (9.95e-03) (+)	5.96e-01 (2.96e-02) (+)	4.84e-01 (2.80e-02) (+)	1.88e-01 (4.82e-03) (+)	5.44e-01 (2.08e-02) (+)
20×15×10×5-6	**1.58e-02 (7.68e-04)**	1.40e-01 (9.13e-03) (+)	3.27e-01 (9.71e-03) (+)	2.95e-01 (1.19e-02) (+)	2.96e-01 (1.43e-02) (+)	2.90e-01 (1.23e-02) (+)	4.13e-01 (1.04e-02) (+)	9.72e-02 (1.75e-03) (+)	3.37e-01 (2.00e-02) (+)
20×15×10×5-7	**0.00e+00 (0.00e+00)**	2.52e-01 (9.64e-03) (+)	5.22e-01 (1.57e-02) (+)	4.62e-01 (1.70e-02) (+)	2.94e-01 (1.55e-02) (+)	6.90e-01 (2.53e-02) (+)	7.74e-01 (2.56e-02) (+)	2.84e-01 (7.30e-03) (+)	5.71e-01 (1.63e-02) (+)
20×15×10×5-8	**2.52e-02 (1.17e-03)**	3.25e-01 (1.33e-02) (+)	4.49e-01 (1.58e-02) (+)	2.70e-01 (5.62e-03) (+)	3.28e-02 (1.13e-02) (+)	4.55e-01 (2.13e-02) (+)	5.85e-01 (2.40e-02) (+)	1.42e-01 (8.07e-03) (+)	7.22e-01 (2.71e-02) (+)
20×15×10×5-9	**1.14e-02 (3.07e-04)**	2.78e-01 (1.46e-02) (+)	5.02e-01 (3.05e-02) (+)	2.56e-01 (7.61e-03) (+)	2.91e-01 (9.24e-03) (+)	4.78e-01 (1.59e-02) (+)	5.88e-01 (2.00e-02) (+)	1.48e-01 (8.04e-03) (+)	6.96e-01 (4.12e-02) (+)
20×15×10×5-10	6.37e-02 (1.44e-03)	3.20e-01 (1.13e-02) (+)	4.30e-01 (1.65e-02) (+)	**2.34e-02 (1.21e-02) (-)**	2.48e-01 (5.96e-03) (+)	5.23e-01 (2.97e-02) (+)	5.50e-01 (3.26e-02) (+)	1.28e-01 (3.20e-03) (+)	7.12e-01 (3.13e-02) (+)
20×15×10×5-11	**2.50e-02 (7.18e-04)**	3.13e-01 (1.04e-02) (+)	4.88e-01 (2.82e-02) (+)	2.65e-01 (1.40e-02) (+)	2.39e-01 (6.34e-03) (+)	4.65e-01 (2.71e-02) (+)	5.59e-01 (2.65e-02) (+)	1.40e-01 (4.31e-03) (+)	7.71e-01 (3.25e-02) (+)
20×15×10×5-12	**2.82e-02 (8.28e-04)**	2.57e-01 (1.29e-02) (+)	3.64e-01 (1.35e-02) (+)	2.17e-01 (1.01e-02) (+)	1.95e-01 (4.86e-03) (+)	3.82e-01 (2.37e-02) (+)	5.12e-01 (1.39e-02) (+)	9.27e-02 (5.39e-03) (+)	6.48e-01 (1.88e-02) (+)
20×15×10×5-13	**8.12e-02 (4.32e-03)**	2.35e-01 (1.05e-02) (+)	3.92e-01 (9.75e-03) (+)	1.88e-01 (4.60e-03) (+)	2.19e-01 (5.56e-03) (+)	3.88e-01 (2.50e-02) (+)	5.12e-01 (1.86e-02) (+)	1.06e-01 (2.89e-03) (+)	5.75e-01 (2.91e-02) (+)
20×15×10×5-14	**6.47e-02 (3.05e-03)**	2.10e-01 (1.10e-02) (+)	4.03e-01 (1.61e-02) (+)	1.92e-01 (6.91e-03) (+)	1.90e-01 (4.80e-03) (+)	4.20e-01 (1.66e-02) (+)	4.51e-01 (1.05e-02) (+)	9.29e-02 (2.68e-03) (+)	6.25e-01 (2.35e-02) (+)
20×15×10×5-15	**7.58e-03 (2.47e-04)**	2.13e-01 (6.04e-03) (+)	3.26e-01 (1.61e-02) (+)	1.96e-01 (3.94e-03) (+)	2.09e-01 (1.04e-02) (+)	3.63e-01 (1.08e-02) (+)	4.47e-01 (1.95e-02) (+)	1.09e-01 (2.77e-03) (+)	5.31e-01 (1.31e-02) (+)
20×15×10×5-16	**0.00e+00 (0.00e+00)**	2.95e-01 (9.62e-03) (+)	2.68e-01 (9.54e-03) (+)	2.71e-01 (1.18e-02) (+)	3.05e-01 (1.47e-02) (+)	5.01e-01 (1.23e-02) (+)	4.73e-01 (2.52e-02) (+)	2.10e-01 (1.10e-02) (+)	5.41e-01 (2.02e-02) (+)
20×15×10×5-17	**0.00e+00 (0.00e+00)**	2.97e-01 (8.13e-03) (+)	3.08e-01 (7.47e-03) (+)	3.15e-01 (6.33e-03) (+)	3.26e-01 (1.18e-02) (+)	5.22e-01 (2.96e-02) (+)	4.39e-01 (1.52e-02) (+)	2.17e-01 (1.09e-02) (+)	5.17e-01 (1.94e-02) (+)
20×15×10×5-18	**0.00e+00 (0.00e+00)**	2.96e-01 (1.64e-02) (+)	2.75e-01 (1.19e-02) (+)	3.01e-01 (1.55e-02) (+)	2.98e-01 (1.69e-02) (+)	5.05e-01 (2.35e-02) (+)	4.96e-01 (1.92e-02) (+)	1.88e-01 (1.06e-02) (+)	4.89e-01 (2.72e-02) (+)
20×15×10×5-19	**0.00e+00 (0.00e+00)**	3.41e-01 (1.71e-02) (+)	3.16e-01 (1.79e-02) (+)	2.82e-01 (6.23e-03) (+)	3.17e-01 (1.20e-02) (+)	4.55e-01 (2.90e-02) (+)	4.92e-01 (2.71e-02) (+)	1.83e-01 (1.04e-02) (+)	5.56e-01 (3.22e-02) (+)
20×15×10×5-20	**0.00e+00 (0.00e+00)**	2.93e-01 (1.13e-02) (+)	2.82e-01 (9.63e-03) (+)	2.98e-01 (1.02e-02) (+)	2.90e-01 (1.58e-02) (+)	4.90e-01 (1.37e-02) (+)	4.82e-01 (1.15e-02) (+)	2.14e-01 (9.20e-03) (+)	5.02e-01 (1.54e-02) (+)
30×20×15×7-1	**0.00e+00 (0.00e+00)**	4.11e-01 (1.64e-02) (+)	2.74e-01 (1.29e-02) (+)	3.61e-01 (1.64e-02) (+)	3.58e-01 (1.99e-02) (+)	7.43e-01 (1.54e-02) (+)	7.67e-01 (4.06e-02) (+)	4.60e-01 (1.70e-02) (+)	7.41e-01 (2.80e-02) (+)
30×20×15×7-2	**1.23e-02 (5.02e-04)**	4.16e-02 (5.20e-03) (+)	4.53e-01 (1.96e-02) (+)	3.19e-02 (1.20e-02) (+)	2.37e-01 (1.41e-02) (+)	5.29e-01 (2.78e-02) (+)	4.77e-01 (1.57e-02) (+)	4.28e-01 (1.08e-02) (+)	4.77e-01 (9.78e-03) (+)
30×20×15×7-3	**5.17e-03 (2.24e-04)**	1.06e-01 (3.95e-03) (+)	3.43e-01 (1.59e-02) (+)	2.59e-01 (6.64e-03) (+)	2.32e-01 (5.59e-03) (+)	5.46e-01 (2.40e-02) (+)	4.81e-01 (2.79e-02) (+)	5.39e-01 (9.94e-03) (+)	5.76e-01 (2.20e-02) (+)
30×20×15×7-4	**8.96e-03 (2.29e-04)**	1.28e-01 (7.48e-03) (+)	3.07e-01 (7.45e-03) (+)	**8.79e-03 (1.22e-02) (≈)**	3.65e-01 (2.20e-02) (+)	4.72e-01 (1.09e-02) (+)	5.32e-01 (2.56e-02) (+)	3.33e-01 (9.36e-03) (+)	4.61e-01 (1.78e-02) (+)
30×20×15×7-5	**0.00e+00 (0.00e+00)**	1.33e-01 (8.08e-03) (+)	2.97e-01 (8.20e-03) (+)	2.84e-01 (1.60e-02) (+)	2.34e-01 (5.86e-03) (+)	5.62e-01 (1.51e-02) (+)	3.72e-01 (1.48e-02) (+)	2.85e-01 (1.05e-02) (+)	4.38e-01 (1.40e-02) (+)
30×20×15×7-6	4.53e-02 (1.77e-03)	1.30e-01 (6.52e-03) (+)	3.56e-01 (1.15e-02) (+)	2.60e-01 (9.99e-03) (+)	5.24e-02 (6.36e-03) (+)	6.82e-01 (2.80e-02) (+)	**3.17e-02 (1.17e-02) (-)**	3.10e-01 (1.30e-02) (+)	4.37e-01 (1.46e-02) (+)
30×20×15×7-7	**0.00e+00 (0.00e+00)**	1.55e-01 (3.09e-03) (+)	3.08e-01 (8.15e-03) (+)	2.57e-01 (6.73e-03) (+)	2.83e-01 (1.34e-02) (+)	4.55e-01 (2.51e-02) (+)	3.95e-01 (1.75e-02) (+)	3.25e-01 (1.45e-02) (+)	3.95e-01 (1.97e-02) (+)
30×20×15×7-8	**0.00e+00 (0.00e+00)**	1.41e-01 (3.09e-03) (+)	4.72e-01 (2.28e-02) (+)	2.50e-01 (1.00e-02) (+)	2.27e-01 (5.03e-03) (+)	5.46e-01 (3.19e-02) (+)	5.14e-01 (1.94e-02) (+)	4.40e-01 (1.54e-02) (+)	5.30e-01 (1.55e-02) (+)
30×20×15×7-9	**4.39e-02 (2.28e-03)**	1.20e-01 (5.48e-03) (+)	4.31e-01 (1.16e-02) (+)	2.63e-01 (1.04e-02) (+)	2.24e-01 (1.09e-02) (+)	6.09e-01 (3.19e-02) (+)	4.34e-01 (1.45e-02) (+)	4.56e-01 (1.76e-02) (+)	4.66e-01 (1.66e-02) (+)
30×20×15×7-10	**7.25e-02 (1.46e-03)**	1.24e-01 (5.22e-03) (+)	4.04e-01 (2.28e-02) (+)	2.32e-01 (1.03e-02) (+)	2.07e-01 (6.39e-03) (+)	5.63e-01 (2.13e-02) (+)	4.43e-01 (2.46e-02) (+)	4.51e-01 (2.37e-02) (+)	5.12e-01 (1.18e-02) (+)
30×20×15×7-11	**0.00e+00 (0.00e+00)**	1.46e-01 (7.47e-03) (+)	4.00e-01 (9.87e-03) (+)	2.42e-01 (9.21e-03) (+)	2.38e-01 (7.09e-03) (+)	6.08e-01 (2.08e-02) (+)	5.21e-01 (1.86e-02) (+)	4.66e-01 (1.04e-02) (+)	5.40e-01 (1.35e-02) (+)
30×20×15×7-12	**0.00e+00 (0.00e+00)**	1.45e-01 (7.20e-03) (+)	4.25e-01 (1.70e-02) (+)	2.35e-01 (1.01e-02) (+)	2.17e-01 (1.27e-02) (+)	6.01e-01 (2.61e-02) (+)	4.97e-01 (1.52e-02) (+)	4.08e-01 (1.39e-02) (+)	4.92e-01 (2.84e-02) (+)
30×20×15×7-13	**8.35e-02 (2.22e-03)**	**8.39e-02 (3.32e-03) (≈)**	4.04e-01 (1.26e-02) (+)	2.30e-01 (9.35e-03) (+)	2.23e-01 (1.08e-02) (+)	5.86e-01 (1.69e-02) (+)	5.30e-01 (1.62e-02) (+)	4.02e-01 (1.98e-02) (+)	5.01e-01 (1.27e-02) (+)
30×20×15×7-14	**6.38e-02 (3.68e-03)**	1.66e-01 (9.08e-03) (+)	4.32e-01 (2.39e-02) (+)	2.83e-01 (9.69e-03) (+)	2.24e-01 (1.23e-02) (+)	6.08e-01 (1.47e-02) (+)	5.48e-01 (2.53e-02) (+)	4.29e-01 (1.54e-02) (+)	5.15e-01 (1.93e-02) (+)
30×20×15×7-15	**5.79e-02 (1.95e-03)**	1.67e-01 (5.31e-03) (+)	3.83e-01 (1.41e-02) (+)	2.71e-01 (1.32e-02) (+)	2.27e-01 (1.30e-02) (+)	5.73e-01 (2.56e-02) (+)	5.38e-01 (2.23e-02) (+)	4.49e-01 (2.03e-02) (+)	5.77e-01 (2.40e-02) (+)
30×20×15×7-16	**0.00e+00 (0.00e+00)**	1.53e-01 (5.76e-03) (+)	2.88e-01 (1.51e-02) (+)	2.82e-01 (6.77e-03) (+)	2.80e-01 (7.68e-03) (+)	5.01e-01 (2.36e-02) (+)	3.53e-01 (1.62e-02) (+)	3.39e-01 (1.45e-02) (+)	3.92e-01 (2.28e-02) (+)
30×20×15×7-17	**0.00e+00 (0.00e+00)**	1.65e-01 (8.80e-03) (+)	2.84e-01 (7.36e-03) (+)	2.66e-01 (1.24e-02) (+)	2.48e-01 (5.65e-03) (+)	4.86e-01 (2.01e-02) (+)	3.84e-01 (1.19e-02) (+)	3.43e-01 (1.47e-02) (+)	4.28e-01 (1.16e-02) (+)
30×20×15×7-18	**8.06e-03 (2.57e-04)**	1.44e-01 (4.29e-03) (+)	2.87e-01 (6.78e-03) (+)	2.33e-01 (8.51e-03) (+)	2.62e-01 (1.51e-02) (+)	4.19e-01 (1.91e-02) (+)	3.26e-01 (1.21e-02) (+)	3.04e-01 (6.19e-03) (+)	4.11e-01 (2.45e-02) (+)
30×20×15×7-19	**0.00e+00 (0.00e+00)**	1.71e-01 (6.41e-03) (+)	2.84e-01 (7.45e-03) (+)	2.29e-01 (1.13e-02) (+)	2.65e-01 (1.37e-02) (+)	4.37e-01 (9.73e-03) (+)	3.36e-01 (1.57e-02) (+)	2.84e-01 (1.26e-02) (+)	3.75e-01 (1.10e-02) (+)
30×20×15×7-20	**0.00e+00 (0.00e+00)**	1.46e-01 (6.93e-03) (+)	2.81e-01 (1.22e-02) (+)	2.49e-01 (5.62e-03) (+)	2.47e-01 (1.45e-02) (+)	4.64e-01 (1.24e-02) (+)	3.68e-01 (1.07e-02) (+)	2.88e-01 (1.74e-02) (+)	4.15e-01 (1.87e-02) (+)
40×25×20×10-1	4.56e-02 (1.32e-03)	1.03e-01 (3.85e-03) (+)	2.92e-01 (8.68e-03) (+)	3.20e-01 (1.79e-02) (+)	1.63e-01 (6.31e-03) (+)	6.86e-01 (2.06e-02) (+)	**3.26e-02 (2.02e-02) (-)**	4.39e-01 (1.44e-02) (+)	5.56e-01 (2.29e-02) (+)
40×25×20×10-2	**3.17e-02 (1.41e-03)**	1.08e-01 (3.74e-03) (+)	2.56e-01 (9.49e-03) (+)	3.22e-01 (6.11e-03) (+)	1.73e-01 (8.78e-03) (+)	6.45e-01 (1.94e-02) (+)	5.66e-01 (3.43e-02) (+)	3.61e-01 (1.03e-02) (+)	5.27e-01 (1.19e-02) (+)
40×25×20×10-3	3.45e-02 (1.43e-03)	1.14e-01 (2.38e-03) (+)	2.81e-01 (1.56e-02) (+)	3.13e-01 (9.22e-03) (+)	1.94e-01 (6.51e-03) (+)	5.54e-01 (2.39e-02) (+)	5.97e-01 (2.50e-02) (+)	**2.53e-02 (2.23e-02) (-)**	4.57e-01 (1.07e-02) (+)
40×25×20×10-4	**4.43e-02 (2.07e-03)**	1.04e-01 (3.15e-03) (+)	2.56e-01 (8.99e-03) (+)	3.82e-01 (2.06e-02) (+)	1.64e-01 (4.14e-03) (+)	4.95e-01 (1.08e-02) (+)	5.38e-01 (1.17e-02) (+)	3.21e-01 (7.78e-03) (+)	4.84e-01 (1.84e-02) (+)
40×25×20×10-5	**7.07e-02 (3.36e-03)**	2.00e-01 (9.68e-03) (+)	3.22e-01 (6.85e-03) (+)	2.53e-01 (1.56e-02) (+)	2.92e-01 (1.22e-02) (+)	5.04e-01 (1.95e-02) (+)	4.35e-01 (1.97e-02) (+)	3.55e-01 (6.86e-03) (+)	5.70e-01 (1.83e-02) (+)
40×25×20×10-6	**6.55e-02 (2.94e-03)**	1.31e-01 (4.21e-03) (+)	2.12e-01 (1.19e-02) (+)	2.23e-01 (7.46e-03) (+)	2.60e-01 (7.36e-03) (+)	3.91e-01 (2.06e-02) (+)	3.83e-01 (8.37e-03) (+)	2.62e-01 (1.31e-02) (+)	4.28e-01 (2.52e-02) (+)
40×25×20×10-7	**1.63e-01 (8.07e-03)**	4.69e-01 (4.29e-03) (+)	2.36e-01 (6.55e-03) (+)	2.42e-01 (1.17e-02) (+)	2.59e-01 (9.70e-03) (+)	3.97e-01 (1.60e-02) (+)	4.40e-01 (1.20e-02) (+)	2.60e-01 (1.33e-02) (+)	4.26e-01 (1.04e-02) (+)
40×25×20×10-8	**4.57e-02 (1.13e-03)**	1.10e-01 (4.44e-03) (+)	2.87e-01 (6.16e-03) (+)	3.30e-01 (1.24e-02) (+)	1.99e-01 (1.19e-02) (+)	6.14e-01 (3.20e-02) (+)	5.74e-01 (2.74e-02) (+)	4.14e-01 (1.15e-02) (+)	5.37e-01 (2.25e-02) (+)
40×25×20×10-9	**4.46e-02 (8.77e-04)**	1.71e-01 (8.37e-03) (+)	2.42e-01 (1.15e-02) (+)	2.97e-01 (1.21e-02) (+)	1.76e-01 (6.73e-03) (+)	6.35e-01 (1.67e-02) (+)	6.12e-02 (1.46e-02) (+)	3.32e-01 (1.98e-02) (+)	4.54e-01 (2.20e-02) (+)
40×25×20×10-10	**4.42e-02 (1.12e-03)**	1.19e-01 (4.02e-03) (+)	2.78e-01 (5.94e-03) (+)	3.51e-01 (1.64e-02) (+)	1.62e-01 (1.03e-02) (+)	5.78e-01 (2.43e-02) (+)	5.56e-01 (1.71e-02) (+)	3.88e-01 (1.31e-02) (+)	4.89e-01 (2.91e-02) (+)
40×25×20×10-11	**4.20e-02 (1.34e-03)**	9.96e-02 (5.28e-03) (+)	2.90e-01 (8.99e-03) (+)	**3.57e-02 (1.62e-02) (≈)**	1.72e-01 (8.19e-03) (+)	6.42e-01 (3.29e-02) (+)	6.09e-01 (2.05e-02) (+)	3.53e-01 (1.28e-02) (+)	4.70e-01 (2.13e-02) (+)
40×25×20×10-12	**4.45e-02 (1.37e-03)**	9.75e-02 (2.71e-03) (+)	2.82e-01 (9.76e-03) (+)	3.14e-01 (1.02e-02) (+)	1.73e-01 (7.72e-03) (+)	6.37e-01 (1.33e-02) (+)	6.03e-01 (1.38e-02) (+)	3.97e-01 (2.14e-02) (+)	5.08e-01 (2.85e-02) (+)
40×25×20×10-13	**5.04e-02 (1.47e-03)**	9.22e-02 (2.53e-03) (+)	2.47e-01 (1.09e-02) (+)	2.97e-01 (1.67e-02) (+)	1.78e-01 (7.44e-03) (+)	6.57e-01 (2.44e-02) (+)	5.96e-01 (3.16e-02) (+)	3.91e-01 (9.36e-03) (+)	4.75e-01 (1.63e-02) (+)
40×25×20×10-14	**4.70e-02 (1.10e-03)**	7.63e-02 (3.07e-03) (+)	2.25e-01 (9.52e-03) (+)	3.17e-01 (1.33e-02) (+)	1.59e-01 (5.01e-03) (+)	6.48e-01 (3.07e-02) (+)	5.74e-01 (2.68e-02) (+)	3.33e-01 (1.19e-02) (+)	4.86e-01 (1.62e-02) (+)
40×25×20×10-15	**1.07e-01 (2.88e-03)**	2.14e-01 (7.29e-03) (+)	3.12e-01 (7.41e-03) (+)	2.69e-01 (1.39e-02) (+)	3.17e-01 (7.02e-03) (+)	5.01e-01 (1.37e-02) (+)	4.82e-01 (2.10e-02) (+)	3.27e-01 (2.09e-02) (+)	5.58e-01 (2.32e-02) (+)
40×25×20×10-16	**7.20e-02 (1.97e-03)**	1.18e-01 (7.48e-03) (+)	2.22e-01 (1.07e-02) (+)	2.11e-01 (4.66e-03) (+)	2.46e-01 (1.02e-02) (+)	4.29e-01 (2.25e-02) (+)	3.87e-01 (1.87e-02) (+)	2.86e-01 (1.50e-02) (+)	4.33e-01 (1.05e-02) (+)
40×25×20×10-17	**7.36e-02 (2.37e-03)**	1.20e-01 (4.60e-03) (+)	2.05e-01 (5.43e-03) (+)	2.06e-01 (1.18e-02) (+)	2.43e-01 (1.18e-02) (+)	3.80e-01 (1.75e-02) (+)	4.28e-01 (1.72e-02) (+)	2.76e-01 (1.00e-02) (+)	4.21e-01 (2.51e-02) (+)
40×25×20×10-18	**7.34e-02 (3.78e-03)**	1.41e-01 (6.45e-03) (+)	2.22e-01 (1.29e-02) (+)	2.26e-01 (1.13e-02) (+)	2.11e-01 (9.00e-03) (+)	3.60e-01 (1.25e-02) (+)	3.80e-01 (1.84e-02) (+)	2.54e-01 (1.45e-02) (+)	4.57e-01 (1.91e-02) (+)
40×25×20×10-19	**6.22e-02 (3.40e-03)**	1.33e-01 (5.70e-03) (+)	2.08e-01 (6.77e-03) (+)	2.26e-01 (7.38e-03) (+)	2.15e-01 (6.61e-03) (+)	3.90e-01 (1.84e-02) (+)	3.64e-01 (2.23e-02) (+)	2.91e-01 (1.31e-02) (+)	4.15e-01 (1.26e-02) (+)
40×25×20×10-20	**4.61e-02 (1.95e-03)**	1.17e-01 (2.25e-03) (+)	2.92e-01 (1.72e-02) (+)	3.46e-01 (1.40e-02) (+)	1.90e-01 (1.01e-02) (+)	6.04e-01 (2.18e-02) (+)	6.32e-01 (2.89e-02) (+)	4.31e-01 (1.36e-02) (+)	5.55e-01 (2.75e-02) (+)
+/ ≈ / –		78/2/0	80/0/0	76/3/1	80/0/0	80/0/0	78/0/2	79/0/1	80/0/0

PART 5. STATISTICAL RESULTS OF HGMAPPO VERSUS OTHER COMPARISON ALGORITHMS

As shown in Table 11, the results of the Wilcoxon signed-rank test further verify the superiority of HGMAPPO over the competing baseline algorithms, where $p < 0.05$ indicates that the difference is statistically significant. Here, R^+ denotes the sum of positive ranks, corresponding to the instances in which HGMAPPO outperforms the compared algorithm, whereas R^- denotes the sum of negative ranks, reflecting the instances in which the baseline algorithm performs better than HGMAPPO. Across all three performance indicators (HV, GD, and IGD), the consistently higher R^+ values and lower R^- values demonstrate the overall advantage of HGMAPPO. Moreover, the standardized test statistic Z further corroborates this conclusion; under the adopted ranking convention, a larger positive Z value provides stronger evidence of the superior performance of HGMAPPO.

Table 11
RESULTS OF WILCOXON TEST FOR HGMAPPO WITH ALL COMPARED ALGORITHMS ON HV, GD, AND IGD METRICS AT α= 0.05.

Algorithms	HV				GD				IGD			
	R^+	R^-	Z	*p*-value	R^+	R^-	Z	*p*-value	R^+	R^-	Z	*p*-value
DQN	3240	0	7.769996	7.848834e-15	3240	0	7.770019	7.847450e-15	3240	0	7.770008	7.848142e-15
TD3	3240	0	7.769985	7.849527e-15	3238	2	7.760471	8.461470e-15	3240	0	7.770019	7.847450e-15
SAC	3237	3	7.755608	8.792154e-15	3230	10	7.722133	1.143986e-14	3227	13	7.707645	1.281611e-14
DDPG	3240	0	7.770075	7.843990e-15	3239	1	7.765267	8.147333e-15	3240	0	7.769985	7.849527e-15
AMOSA	3240	0	7.770097	7.842606e-15	3240	0	7.770041	7.846066e-15	3240	0	7.770030	7.846758e-15
MOACO	3238	2	7.760426	8.464448e-15	3240	0	7.770052	7.845374e-15	3237	3	7.755596	8.792926e-15
MOPSO	3240	0	7.770008	7.848142e-15	3238	2	7.760460	8.462214e-15	3239	1	7.765211	8.150923e-15
NSGA-II	3239	1	7.765267	8.147333e-15	3239	1	7.765267	8.147333e-15	3240	0	7.770019	7.847450e-15

PART 6. SCALABILITY EVALUATION UNDER INCREASING OPERATION-GRAPH DEPTH

To evaluate scalability with respect to operation-graph depth, four depth groups are considered under the fixed $40\times25\times20\times10$ configuration: D0 with $n_s \in [4,7]$, D1 with $n_s \in [8,12]$, D2 with $n_s \in [13,18]$, and D3 with $n_s \in [19,25]$. All other instance parameters remain unchanged. The experiments use the final revised framework and the same evaluation protocol as the main comparison experiments. Each algorithm is independently executed 20 times on every evaluated instance, and the results are reported as the mean with the standard deviation in parentheses.

The same normalized reference-point rule as that used in Section V-B is adopted for HV, while the HV value is recalculated for each depth instance. For GD and IGD, the reference Pareto front of each instance is constructed by aggregating the nondominated solutions obtained by all compared algorithms. The resulting performance distributions are presented in Fig. 1. The four rows correspond to D0–D3, while the three columns report HV, GD, and IGD, respectively.

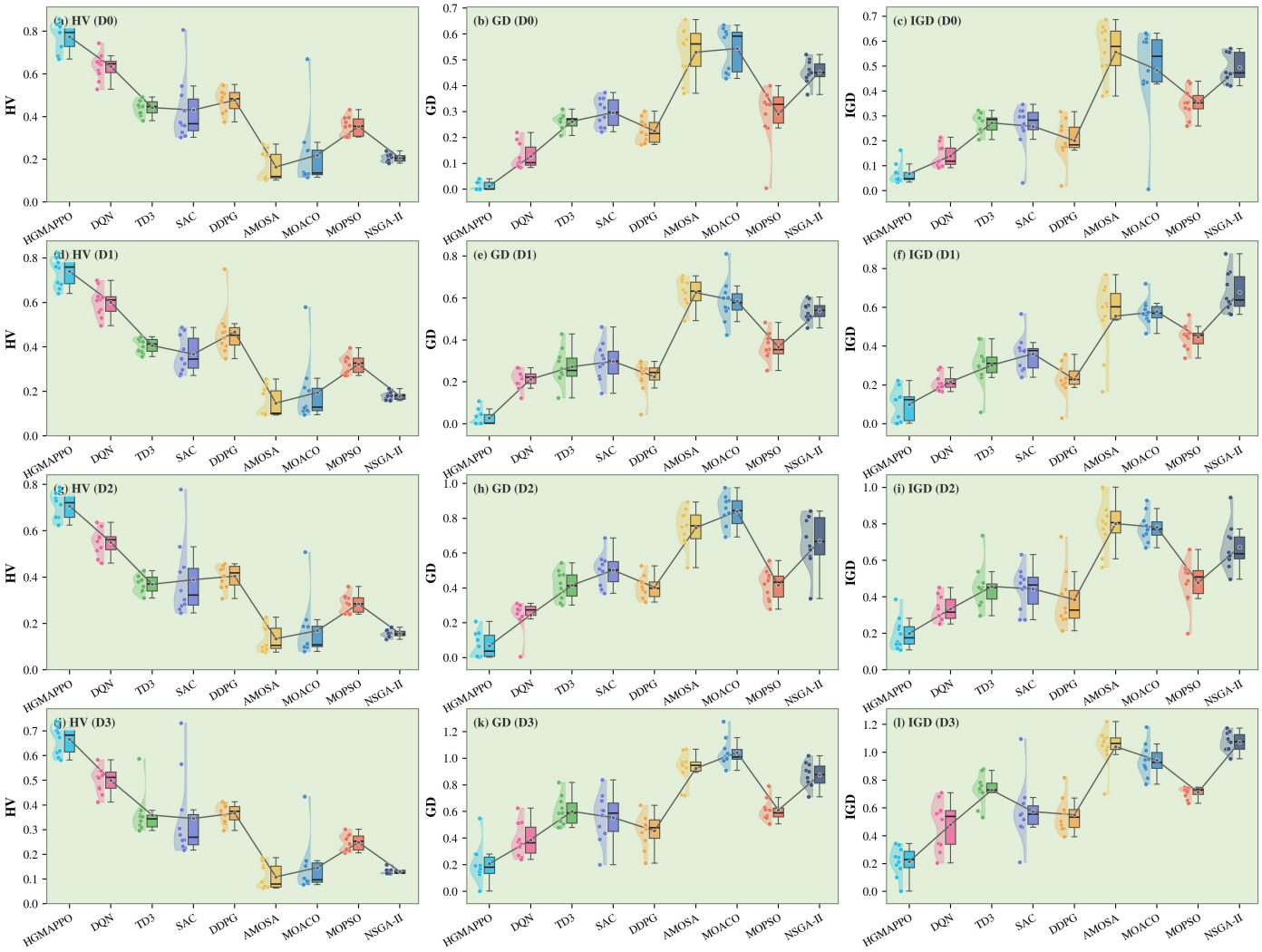


Fig. 1. Performance distributions under increasing operation-graph depth: (a)-(c) HV, GD, and IGD for D0; (d)-(f) HV, GD, and IGD for D1; (g)-(i) HV, GD, and IGD for D2; and (j)-(l) HV, GD, and IGD for D3.

As shown in Fig. 1, HGMAPPO consistently exhibits the highest HV distributions and the lowest GD and IGD distributions across D0–D3. Although the scheduling difficulty increases with operation depth, its performance remains clearly superior to that of the compared algorithms. Even under D3, HGMAPPO retains the highest median HV and the lowest median GD and IGD. Its relatively concentrated distributions also indicate stable performance across different instances. These results demonstrate that HGMAPPO can preserve convergence quality and solution diversity when handling deeper operation graphs.

The complete instance-level statistical results are reported in Tables 12-14.

Table 12

STATISTICAL RESULTS OF HGMAPPO VERSUS OTHER COMPARISON ALGORITHMS ON HV METRIC UNDER DIFFERENT OPERATION-GRAPH DEPTHS.

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
D0-I361	**8.25e-01 (2.30e-02)**	7.44e-01 (3.91e-02) (+)	4.46e-01 (2.57e-02) (+)	3.26e-01 (1.00e-02) (+)	5.51e-01 (2.62e-02) (+)	1.02e-01 (6.50e-03) (+)	1.21e-01 (7.02e-03) (+)	3.04e-01 (7.01e-03) (+)	1.89e-01 (1.06e-02) (+)
D0-I374	**7.31e-01 (1.67e-02)**	6.85e-01 (2.46e-02) (+)	4.05e-01 (2.35e-02) (+)	3.03e-01 (1.78e-02) (+)	4.69e-01 (1.61e-02) (+)	1.18e-01 (2.51e-03) (+)	1.15e-01 (4.68e-03) (+)	3.15e-01 (1.60e-02) (+)	2.00e-01 (1.01e-02) (+)
D0-I387	**7.96e-01 (3.71e-02)**	5.63e-01 (1.23e-02) (+)	3.81e-01 (1.77e-02) (+)	5.00e-01 (1.34e-02) (+)	4.27e-01 (1.02e-02) (+)	2.18e-01 (1.10e-02) (+)	2.43e-01 (7.00e-03) (+)	3.74e-01 (9.66e-03) (+)	1.88e-01 (9.71e-03) (+)
D0-I401	**6.84e-01 (1.88e-02)**	5.28e-01 (1.43e-02) (+)	4.17e-01 (1.58e-02) (+)	4.28e-01 (2.42e-02) (+)	3.75e-01 (1.63e-02) (+)	2.24e-01 (4.38e-03) (+)	**6.69e-01 (6.17e-03) (≈)**	3.96e-01 (1.02e-02) (+)	2.14e-01 (9.27e-03) (+)
D0-I414	**7.28e-01 (4.75e-02)**	6.44e-01 (3.40e-02) (+)	4.86e-01 (1.33e-02) (+)	3.61e-01 (6.71e-03) (+)	5.08e-01 (2.36e-02) (+)	1.13e-01 (3.28e-03) (+)	1.32e-01 (4.14e-03) (+)	3.57e-01 (1.72e-02) (+)	2.18e-01 (6.22e-03) (+)
D0-I427	**7.94e-01 (2.65e-02)**	6.17e-01 (4.12e-02) (+)	4.19e-01 (1.45e-02) (+)	3.58e-01 (1.41e-02) (+)	5.46e-01 (1.05e-02) (+)	1.10e-01 (6.20e-03) (+)	1.26e-01 (3.44e-03) (+)	3.05e-01 (1.76e-02) (+)	2.10e-01 (1.22e-02) (+)
D0-I440	**8.21e-01 (3.12e-02)**	6.60e-01 (1.58e-02) (+)	4.48e-01 (2.11e-02) (+)	3.73e-01 (9.72e-03) (+)	5.14e-01 (2.36e-02) (+)	1.19e-01 (6.36e-03) (+)	1.39e-01 (8.30e-03) (+)	3.53e-01 (1.24e-02) (+)	2.21e-01 (9.53e-03) (+)
D0-I454	**8.57e-01 (3.78e-02)**	6.02e-01 (2.13e-02) (+)	4.54e-01 (2.35e-02) (+)	5.43e-01 (1.64e-02) (+)	4.76e-01 (2.62e-02) (+)	2.55e-01 (1.33e-02) (+)	2.79e-01 (9.48e-03) (+)	4.33e-01 (1.80e-02) (+)	1.95e-01 (1.28e-02) (+)
D0-I467	6.69e-01 (1.68e-02)	6.51e-01 (1.41e-02) (+)	4.74e-01 (1.54e-02) (+)	**8.06e-01 (1.77e-02) (-)**	4.16e-01 (1.86e-02) (+)	2.71e-01 (1.02e-02) (+)	2.40e-01 (7.66e-03) (+)	3.94e-01 (1.94e-02) (+)	2.39e-01 (1.08e-02) (+)
D0-I480	**8.33e-01 (4.32e-02)**	6.56e-01 (3.15e-02) (+)	4.91e-01 (1.82e-02) (+)	3.10e-01 (1.43e-02) (+)	4.90e-01 (1.59e-02) (+)	1.12e-01 (5.44e-03) (+)	1.31e-01 (5.54e-03) (+)	3.05e-01 (1.63e-02) (+)	1.82e-01 (1.13e-02) (+)
D1-DI481	**7.79e-01 (2.52e-02)**	6.87e-01 (4.33e-02) (+)	4.02e-01 (2.72e-02) (+)	2.98e-01 (1.11e-02) (+)	5.05e-01 (3.27e-02) (+)	9.40e-02 (8.00e-03) (+)	9.50e-02 (8.44e-03) (+)	2.71e-01 (8.71e-03) (+)	1.63e-01 (1.21e-02) (+)
D1-DI494	6.82e-01 (2.04e-02)	**7.99e-01 (2.80e-02) (-)**	3.69e-01 (2.54e-02) (+)	2.72e-01 (2.20e-02) (+)	4.17e-01 (1.90e-02) (+)	1.01e-01 (8.00e-03) (+)	1.11e-01 (8.00e-03) (+)	2.91e-01 (1.95e-02) (+)	1.69e-01 (1.08e-02) (+)
D1-DI507	**7.56e-01 (3.97e-02)**	5.31e-01 (1.30e-02) (+)	3.56e-01 (1.91e-02) (+)	4.55e-01 (1.63e-02) (+)	4.05e-01 (1.08e-02) (+)	2.07e-01 (1.12e-02) (+)	2.04e-01 (8.00e-03) (+)	3.48e-01 (1.03e-02) (+)	1.58e-01 (1.22e-02) (+)
D1-DI521	**6.62e-01 (2.17e-02)**	4.96e-01 (1.60e-02) (+)	3.79e-01 (1.90e-02) (+)	3.90e-01 (2.45e-02) (+)	3.47e-01 (1.65e-02) (+)	1.90e-01 (8.00e-03) (+)	5.79e-01 (8.00e-03) (+)	3.50e-01 (1.10e-02) (+)	1.83e-01 (1.04e-02) (+)
D1-DI534	**6.90e-01 (4.83e-02)**	6.13e-01 (3.58e-02) (+)	4.42e-01 (1.63e-02) (+)	3.44e-01 (8.00e-03) (+)	4.68e-01 (2.45e-02) (+)	1.03e-01 (8.00e-03) (+)	1.15e-01 (8.00e-03) (+)	3.19e-01 (1.88e-02) (+)	1.88e-01 (8.00e-03) (+)
D1-DI547	**7.61e-01 (3.12e-02)**	5.68e-01 (4.25e-02) (+)	3.83e-01 (1.54e-02) (+)	3.25e-01 (1.74e-02) (+)	**7.50e-01 (1.30e-02) (≈)**	9.37e-02 (8.00e-03) (+)	1.12e-01 (8.00e-03) (+)	2.83e-01 (1.90e-02) (+)	1.81e-01 (1.47e-02) (+)
D1-DI560	**8.03e-01 (3.71e-02)**	6.27e-01 (1.98e-02) (+)	4.23e-01 (2.46e-02) (+)	3.47e-01 (1.09e-02) (+)	4.92e-01 (2.55e-02) (+)	1.01e-01 (8.00e-03) (+)	1.32e-01 (8.44e-03) (+)	3.35e-01 (1.50e-02) (+)	1.96e-01 (1.12e-02) (+)
D1-DI574	**8.22e-01 (4.12e-02)**	5.57e-01 (2.50e-02) (+)	4.26e-01 (2.69e-02) (+)	4.87e-01 (2.01e-02) (+)	4.41e-01 (2.81e-02) (+)	2.32e-01 (1.46e-02) (+)	2.59e-01 (1.06e-02) (+)	3.96e-01 (2.14e-02) (+)	1.80e-01 (1.48e-02) (+)
D1-DI587	**6.40e-01 (1.78e-02)**	6.09e-01 (1.52e-02) (+)	4.35e-01 (1.79e-02) (+)	4.76e-01 (2.07e-02) (+)	3.83e-01 (1.95e-02) (+)	2.56e-01 (1.16e-02) (+)	2.17e-01 (8.65e-03) (+)	3.54e-01 (2.16e-02) (+)	2.13e-01 (1.19e-02) (+)
D1-DI600	**8.02e-01 (5.26e-02)**	6.23e-01 (3.84e-02) (+)	4.46e-01 (2.06e-02) (+)	2.79e-01 (1.55e-02) (+)	4.65e-01 (1.67e-02) (+)	9.67e-02 (8.00e-03) (+)	1.26e-01 (8.00e-03) (+)	2.72e-01 (1.98e-02) (+)	1.61e-01 (1.21e-02) (+)
D2-DI601	**7.41e-01 (3.15e-02)**	6.36e-01 (4.61e-02) (+)	3.60e-01 (3.11e-02) (+)	2.77e-01 (1.33e-02) (+)	4.55e-01 (3.40e-02) (+)	7.58e-02 (8.00e-03) (+)	1.02e-01 (9.56e-03) (+)	2.48e-01 (9.88e-03) (+)	1.32e-01 (1.36e-02) (+)
D2-DI614	**6.58e-01 (2.13e-02)**	5.68e-01 (2.92e-02) (+)	3.36e-01 (2.76e-02) (+)	2.45e-01 (2.35e-02) (+)	3.83e-01 (2.05e-02) (+)	1.02e-01 (8.00e-03) (+)	7.93e-02 (8.00e-03) (+)	2.42e-01 (2.04e-02) (+)	1.45e-01 (1.33e-02) (+)
D2-DI627	**7.30e-01 (5.22e-02)**	4.76e-01 (1.59e-02) (+)	3.10e-01 (2.53e-02) (+)	4.22e-01 (1.56e-02) (+)	3.56e-01 (1.21e-02) (+)	1.81e-01 (1.57e-02) (+)	1.86e-01 (9.57e-03) (+)	3.09e-01 (1.12e-02) (+)	1.52e-01 (1.24e-02) (+)
D2-DI641	**6.26e-01 (2.26e-02)**	4.60e-01 (1.69e-02) (+)	3.39e-01 (1.92e-02) (+)	3.42e-01 (3.16e-02) (+)	3.07e-01 (2.07e-02) (+)	1.76e-01 (8.00e-03) (+)	5.08e-01 (8.00e-03) (+)	3.11e-01 (1.23e-02) (+)	1.58e-01 (1.18e-02) (+)
D2-DI654	**6.59e-01 (5.84e-02)**	5.54e-01 (4.07e-02) (+)	4.04e-01 (1.90e-02) (+)	3.03e-01 (9.42e-03) (+)	4.37e-01 (3.09e-02) (+)	9.05e-02 (8.00e-03) (+)	1.02e-01 (8.00e-03) (+)	2.88e-01 (2.42e-02) (+)	1.69e-01 (8.15e-03) (+)
D2-DI667	**7.13e-01 (3.56e-02)**	5.15e-01 (5.74e-02) (+)	3.42e-01 (1.90e-02) (+)	2.84e-01 (1.88e-02) (+)	4.57e-01 (1.38e-02) (+)	9.67e-02 (8.00e-03) (+)	9.65e-02 (8.00e-03) (+)	2.51e-01 (2.20e-02) (+)	1.51e-01 (1.70e-02) (+)
D2-DI680	**7.62e-01 (3.96e-02)**	5.75e-01 (2.22e-02) (+)	3.80e-01 (2.46e-02) (+)	**7.79e-01 (1.34e-02) (≈)**	4.50e-01 (3.04e-02) (+)	1.08e-01 (8.00e-03) (+)	9.90e-02 (9.67e-03) (+)	2.82e-01 (1.60e-02) (+)	1.71e-01 (1.29e-02) (+)
D2-DI694	**7.88e-01 (4.56e-02)**	5.28e-01 (3.00e-02) (+)	3.88e-01 (2.87e-02) (+)	4.43e-01 (2.25e-02) (+)	4.05e-01 (3.69e-02) (+)	2.04e-01 (1.76e-02) (+)	2.16e-01 (1.22e-02) (+)	3.59e-01 (2.35e-02) (+)	1.57e-01 (1.68e-02) (+)
D2-DI707	**6.24e-01 (2.27e-02)**	**6.20e-01 (1.81e-02) (≈)**	4.04e-01 (1.86e-02) (+)	5.31e-01 (2.53e-02) (+)	3.60e-01 (2.59e-02) (+)	2.27e-01 (1.29e-02) (+)	1.88e-01 (1.05e-02) (+)	3.16e-01 (2.36e-02) (+)	1.84e-01 (1.50e-02) (+)
D2-DI720	**7.72e-01 (5.24e-02)**	5.71e-01 (4.21e-02) (+)	4.28e-01 (2.15e-02) (+)	2.62e-01 (1.78e-02) (+)	4.32e-01 (2.00e-02) (+)	8.08e-02 (8.00e-03) (+)	1.13e-01 (8.00e-03) (+)	2.40e-01 (2.01e-02) (+)	1.43e-01 (1.55e-02) (+)
D3-DI721	**7.01e-01 (3.78e-02)**	5.83e-01 (5.22e-02) (+)	3.18e-01 (3.45e-02) (+)	2.25e-01 (1.52e-02) (+)	4.13e-01 (3.81e-02) (+)	6.47e-02 (9.29e-03) (+)	8.31e-02 (1.15e-02) (+)	2.16e-01 (1.10e-02) (+)	1.22e-01 (1.50e-02) (+)
D3-DI734	**6.13e-01 (2.32e-02)**	5.39e-01 (3.60e-02) (+)	2.97e-01 (3.35e-02) (+)	2.17e-01 (2.63e-02) (+)	3.59e-01 (2.51e-02) (+)	6.89e-02 (8.00e-03) (+)	7.76e-02 (8.00e-03) (+)	2.06e-01 (2.54e-02) (+)	1.36e-01 (1.56e-02) (+)
D3-DI747	**6.92e-01 (5.44e-02)**	4.41e-01 (1.74e-02) (+)	2.97e-01 (2.45e-02) (+)	3.80e-01 (2.01e-02) (+)	3.26e-01 (1.65e-02) (+)	1.54e-01 (1.61e-02) (+)	1.70e-01 (9.68e-03) (+)	2.70e-01 (1.60e-02) (+)	1.22e-01 (1.47e-02) (+)
D3-DI761	5.82e-01 (2.65e-02)	4.12e-01 (2.01e-02) (+)	**5.87e-01 (2.48e-02) (≈)**	3.05e-01 (3.80e-02) (+)	2.97e-01 (2.32e-02) (+)	1.46e-01 (8.00e-03) (+)	4.34e-01 (9.02e-03) (+)	2.76e-01 (1.67e-02) (+)	1.25e-01 (1.50e-02) (+)
D3-DI774	**6.20e-01 (6.50e-02)**	5.14e-01 (4.92e-02) (+)	3.60e-01 (2.03e-02) (+)	2.59e-01 (1.09e-02) (+)	3.90e-01 (3.20e-02) (+)	6.73e-02 (8.00e-03) (+)	9.15e-02 (8.00e-03) (+)	2.43e-01 (2.49e-02) (+)	1.35e-01 (9.44e-03) (+)
D3-DI787	**6.74e-01 (3.76e-02)**	4.72e-01 (6.50e-02) (+)	3.09e-01 (1.98e-02) (+)	2.57e-01 (2.12e-02) (+)	4.06e-01 (1.59e-02) (+)	6.37e-02 (8.53e-03) (+)	8.46e-02 (8.00e-03) (+)	2.15e-01 (2.51e-02) (+)	1.22e-01 (1.66e-02) (+)
D3-DI800	**7.18e-01 (4.57e-02)**	5.38e-01 (2.31e-02) (+)	3.54e-01 (3.32e-02) (+)	2.79e-01 (1.38e-02) (+)	3.94e-01 (3.48e-02) (+)	8.80e-02 (9.57e-03) (+)	1.03e-01 (1.27e-02) (+)	2.62e-01 (1.94e-02) (+)	1.36e-01 (1.53e-02) (+)
D3-DI814	**7.37e-01 (5.20e-02)**	4.67e-01 (3.27e-02) (+)	3.35e-01 (3.54e-02) (+)	**7.32e-01 (2.42e-02) (≈)**	3.72e-01 (3.65e-02) (+)	1.75e-01 (1.88e-02) (+)	1.74e-01 (1.42e-02) (+)	3.02e-01 (2.84e-02) (+)	1.23e-01 (1.84e-02) (+)
D3-DI827	**5.92e-01 (2.32e-02)**	5.10e-01 (2.14e-02) (+)	3.53e-01 (2.45e-02) (+)	5.66e-01 (2.76e-02) (+)	3.33e-01 (3.07e-02) (+)	1.87e-01 (1.53e-02) (+)	1.53e-01 (1.05e-02) (+)	2.80e-01 (2.89e-02) (+)	1.58e-01 (1.52e-02) (+)
D3-DI840	**7.39e-01 (6.50e-02)**	5.20e-01 (4.39e-02) (+)	3.78e-01 (2.64e-02) (+)	2.33e-01 (2.23e-02) (+)	3.77e-01 (2.30e-02) (+)	7.19e-02 (8.91e-03) (+)	9.20e-02 (8.00e-03) (+)	2.20e-01 (2.55e-02) (+)	1.23e-01 (1.57e-02) (+)
+ / ≈ / −		38/1/1	39/1/0	37/2/1	39/1/0	40/0/0	39/1/0	40/0/0	40/0/0

Table 13

STATISTICAL RESULTS OF HGMAPPO VERSUS OTHER COMPARISON ALGORITHMS ON GD METRIC UNDER DIFFERENT OPERATION-GRAPH DEPTHS.

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
D0-I361	2.86e-02 (1.09e-03)	8.36e-02 (2.90e-03) (+)	2.74e-01 (1.25e-02) (+)	3.51e-01 (1.42e-02) (+)	1.77e-01 (9.23e-03) (+)	6.55e-01 (3.03e-02) (+)	6.34e-01 (3.26e-02) (+)	**3.94e-03 (3.48e-04) (-)**	5.02e-01 (1.83e-02) (+)
D0-I374	**2.71e-02 (1.59e-03)**	1.07e-01 (6.58e-03) (+)	2.71e-01 (1.29e-02) (+)	3.51e-01 (1.19e-02) (+)	2.20e-01 (7.26e-03) (+)	5.58e-01 (1.27e-02) (+)	5.87e-01 (1.43e-02) (+)	3.67e-01 (9.63e-03) (+)	4.50e-01 (2.68e-02) (+)
D0-I387	**0.00e+00 (0.00e+00)**	1.93e-01 (1.27e-02) (+)	2.83e-01 (1.20e-02) (+)	2.39e-01 (8.21e-03) (+)	2.84e-01 (1.59e-02) (+)	4.77e-01 (1.41e-02) (+)	4.45e-01 (2.93e-02) (+)	3.36e-01 (1.78e-02) (+)	4.63e-01 (1.67e-02) (+)
D0-I401	**0.00e+00 (0.00e+00)**	1.76e-01 (4.85e-03) (+)	2.39e-01 (1.42e-02) (+)	2.37e-01 (9.70e-03) (+)	2.60e-01 (9.22e-03) (+)	3.91e-01 (2.05e-02) (+)	4.49e-01 (2.46e-02) (+)	2.37e-01 (9.37e-03) (+)	4.43e-01 (1.15e-02) (+)
D0-I414	**0.00e+00 (0.00e+00)**	9.54e-02 (4.08e-03) (+)	2.38e-01 (6.67e-03) (+)	2.78e-01 (1.31e-02) (+)	1.93e-01 (6.28e-03) (+)	6.15e-01 (2.09e-02) (+)	5.96e-01 (2.69e-02) (+)	3.28e-01 (1.51e-02) (+)	4.19e-01 (1.61e-02) (+)
D0-I427	**2.53e-02 (5.32e-04)**	9.88e-02 (5.12e-03) (+)	2.68e-01 (1.15e-02) (+)	3.28e-01 (1.99e-02) (+)	1.72e-01 (7.34e-03) (+)	6.10e-01 (1.95e-02) (+)	6.21e-01 (1.50e-02) (+)	3.27e-01 (1.40e-02) (+)	4.50e-01 (1.38e-02) (+)
D0-I440	**3.99e-02 (2.46e-03)**	9.19e-02 (2.94e-03) (+)	2.73e-01 (1.56e-02) (+)	3.15e-01 (1.32e-02) (+)	2.11e-01 (1.22e-02) (+)	5.64e-01 (3.30e-02) (+)	6.01e-01 (1.61e-02) (+)	3.63e-01 (1.42e-02) (+)	4.32e-01 (1.57e-02) (+)
D0-I454	**0.00e+00 (0.00e+00)**	2.19e-01 (1.41e-02) (+)	3.09e-01 (8.34e-03) (+)	2.67e-01 (1.56e-02) (+)	3.02e-01 (1.16e-02) (+)	4.74e-01 (2.32e-02) (+)	4.67e-01 (1.16e-02) (+)	2.91e-01 (9.28e-03) (+)	5.20e-01 (2.46e-02) (+)
D0-I467	**0.00e+00 (0.00e+00)**	1.22e-01 (6.24e-03) (+)	2.07e-01 (1.09e-02) (+)	2.22e-01 (1.11e-02) (+)	2.50e-01 (1.18e-02) (+)	3.71e-01 (2.27e-02) (+)	4.28e-01 (2.26e-02) (+)	2.43e-01 (1.22e-02) (+)	3.66e-01 (8.75e-03) (+)
D0-I480	**0.00e+00 (0.00e+00)**	8.80e-02 (1.76e-03) (+)	2.61e-01 (5.96e-03) (+)	3.74e-01 (1.91e-02) (+)	1.77e-01 (8.47e-03) (+)	5.79e-01 (2.30e-02) (+)	6.09e-01 (1.88e-02) (+)	4.00e-01 (1.09e-02) (+)	4.91e-01 (2.16e-02) (+)
D1-DI481	**1.09e-01 (1.38e-02)**	1.69e-01 (3.61e-02) (+)	1.24e-01 (2.58e-02) (+)	1.45e-01 (1.69e-02) (+)	2.16e-01 (2.68e-02) (+)	5.85e-01 (8.73e-02) (+)	6.00e-01 (5.42e-02) (+)	2.54e-01 (4.41e-02) (+)	5.13e-01 (1.18e-01) (+)
D1-DI494	**3.00e-03 (3.84e-04)**	2.16e-01 (3.41e-02) (+)	3.18e-01 (2.22e-02) (+)	3.11e-01 (3.01e-02) (+)	2.50e-01 (2.52e-02) (+)	6.78e-01 (5.83e-02) (+)	5.51e-01 (6.85e-02) (+)	4.84e-01 (6.15e-02) (+)	5.27e-01 (6.06e-02) (+)
D1-DI507	**3.00e-03 (5.45e-04)**	2.65e-01 (2.74e-02) (+)	2.25e-01 (3.85e-02) (+)	2.27e-01 (1.92e-02) (+)	1.71e-01 (1.06e-02) (+)	5.77e-01 (4.09e-02) (+)	5.39e-01 (3.80e-02) (+)	4.07e-01 (2.53e-02) (+)	5.11e-01 (5.34e-02) (+)
D1-DI521	**7.07e-02 (9.63e-03)**	2.68e-01 (3.64e-02) (+)	2.36e-01 (3.62e-02) (+)	3.27e-01 (5.79e-02) (+)	2.88e-01 (3.83e-02) (+)	6.72e-01 (1.18e-01) (+)	6.02e-01 (7.90e-02) (+)	3.88e-01 (6.19e-02) (+)	6.04e-01 (7.37e-02) (+)
D1-DI534	**4.54e-02 (5.68e-03)**	1.23e-01 (1.25e-02) (+)	2.45e-01 (3.36e-02) (+)	3.84e-01 (2.95e-02) (+)	**4.47e-02 (2.03e-03) (≈)**	7.05e-01 (8.80e-02) (+)	6.27e-01 (8.62e-02) (+)	3.55e-01 (3.28e-02) (+)	5.65e-01 (9.32e-02) (+)
D1-DI547	**3.00e-03 (1.64e-04)**	2.29e-01 (3.71e-02) (+)	2.64e-01 (4.44e-02) (+)	2.73e-01 (2.63e-02) (+)	2.40e-01 (3.66e-02) (+)	5.94e-01 (1.26e-01) (+)	5.57e-01 (5.23e-02) (+)	3.38e-01 (3.23e-02) (+)	4.57e-01 (6.83e-02) (+)
D1-DI560	**3.00e-03 (2.59e-04)**	1.92e-01 (1.46e-02) (+)	4.28e-01 (4.24e-02) (+)	4.62e-01 (5.44e-02) (+)	2.06e-01 (3.54e-02) (+)	6.40e-01 (7.70e-02) (+)	4.87e-01 (4.02e-02) (+)	3.32e-01 (6.35e-02) (+)	4.97e-01 (7.46e-02) (+)
D1-DI574	**3.60e-02 (3.15e-03)**	2.35e-01 (3.67e-02) (+)	2.19e-01 (1.72e-02) (+)	2.14e-01 (3.71e-02) (+)	2.98e-01 (3.00e-02) (+)	4.92e-01 (7.26e-02) (+)	4.23e-01 (5.78e-02) (+)	4.31e-01 (6.28e-02) (+)	5.56e-01 (6.50e-02) (+)
D1-DI587	**3.00e-03 (4.17e-04)**	1.93e-01 (9.82e-03) (+)	2.99e-01 (3.75e-02) (+)	2.86e-01 (4.28e-02) (+)	2.68e-01 (4.48e-02) (+)	6.92e-01 (9.84e-02) (+)	6.57e-01 (1.40e-01) (+)	3.25e-01 (3.36e-02) (+)	5.60e-01 (9.75e-02) (+)
D1-DI600	**3.00e-03 (4.81e-04)**	2.38e-01 (5.84e-02) (+)	3.61e-01 (2.02e-02) (+)	3.53e-01 (4.76e-02) (+)	2.63e-01 (3.36e-02) (+)	6.26e-01 (7.69e-02) (+)	8.12e-01 (1.33e-01) (+)	3.53e-01 (5.12e-02) (+)	5.98e-01 (1.10e-01) (+)
D2-DI601	**3.00e-03 (3.37e-04)**	2.22e-01 (9.59e-03) (+)	5.44e-01 (6.46e-02) (+)	3.91e-01 (3.62e-02) (+)	4.29e-01 (4.41e-02) (+)	6.64e-01 (8.52e-02) (+)	8.16e-01 (8.23e-02) (+)	4.23e-01 (5.07e-02) (+)	8.42e-01 (1.39e-01) (+)
D2-DI614	**2.07e-01 (2.44e-02)**	3.00e-01 (4.72e-02) (+)	3.01e-01 (3.40e-02) (+)	3.69e-01 (4.99e-02) (+)	5.27e-01 (9.03e-02) (+)	6.73e-01 (4.72e-02) (+)	9.01e-01 (1.05e-01) (+)	5.58e-01 (6.15e-02) (+)	5.81e-01 (5.33e-02) (+)
D2-DI627	7.14e-03 (1.01e-03)	**5.57e-03 (4.00e-04) (-)**	3.81e-01 (4.16e-02) (+)	6.87e-01 (1.06e-01) (+)	3.90e-01 (2.31e-02) (+)	8.53e-01 (1.29e-01) (+)	8.29e-01 (9.73e-02) (+)	4.94e-01 (7.29e-02) (+)	3.39e-01 (4.49e-02) (+)
D2-DI641	**1.03e-01 (8.99e-03)**	2.77e-01 (2.20e-02) (+)	4.76e-01 (4.79e-02) (+)	5.56e-01 (6.11e-02) (+)	4.36e-01 (7.33e-02) (+)	8.18e-01 (6.55e-02) (+)	8.96e-01 (1.15e-01) (+)	4.57e-01 (4.97e-02) (+)	8.12e-01 (7.39e-02) (+)
D2-DI654	**1.36e-01 (1.04e-02)**	2.52e-01 (3.41e-02) (+)	3.44e-01 (3.51e-02) (+)	4.97e-01 (6.69e-02) (+)	4.71e-01 (5.40e-02) (+)	8.16e-01 (6.53e-02) (+)	9.76e-01 (1.13e-01) (+)	3.37e-01 (2.01e-02) (+)	8.10e-01 (4.46e-02) (+)
D2-DI667	**3.00e-03 (2.68e-04)**	2.38e-01 (4.14e-02) (+)	3.13e-01 (4.22e-02) (+)	5.36e-01 (1.04e-01) (+)	4.06e-01 (2.49e-02) (+)	7.61e-01 (9.77e-02) (+)	7.50e-01 (1.15e-01) (+)	2.78e-01 (4.66e-02) (+)	6.19e-01 (1.06e-01) (+)
D2-DI680	**7.55e-03 (5.87e-04)**	2.72e-01 (2.32e-02) (+)	3.98e-01 (4.49e-02) (+)	4.15e-01 (6.71e-02) (+)	3.37e-01 (4.88e-02) (+)	8.93e-01 (8.48e-02) (+)	9.23e-01 (2.19e-01) (+)	4.71e-01 (7.07e-02) (+)	6.93e-01 (1.08e-01) (+)
D2-DI694	**1.41e-01 (8.32e-03)**	3.11e-01 (3.49e-02) (+)	4.27e-01 (4.76e-02) (+)	5.09e-01 (5.61e-02) (+)	3.19e-01 (2.32e-02) (+)	5.16e-01 (3.37e-02) (+)	8.58e-01 (1.26e-01) (+)	4.42e-01 (3.68e-02) (+)	6.41e-01 (5.56e-02) (+)
D2-DI707	**6.43e-02 (1.04e-02)**	3.01e-01 (6.64e-02) (+)	4.73e-01 (8.38e-02) (+)	5.65e-01 (5.77e-02) (+)	3.42e-01 (3.71e-02) (+)	7.50e-01 (1.47e-01) (+)	6.93e-01 (6.65e-02) (+)	3.76e-01 (4.31e-02) (+)	5.71e-01 (4.16e-02) (+)
D2-DI720	**8.40e-03 (9.88e-04)**	2.78e-01 (3.33e-02) (+)	4.99e-01 (8.46e-02) (+)	4.95e-01 (1.07e-01) (+)	3.77e-01 (4.97e-02) (+)	7.09e-01 (4.57e-02) (+)	7.53e-01 (1.02e-01) (+)	3.24e-01 (5.79e-02) (+)	7.86e-01 (1.18e-01) (+)
D3-DI721	**1.27e-01 (1.74e-02)**	2.71e-01 (4.36e-02) (+)	6.75e-01 (1.06e-01) (+)	3.91e-01 (3.87e-02) (+)	4.67e-01 (6.28e-02) (+)	1.07e+00 (1.77e-01) (+)	1.07e+00 (1.16e-01) (+)	5.69e-01 (7.30e-02) (+)	1.02e+00 (1.27e-01) (+)
D3-DI734	**2.81e-01 (4.91e-02)**	**2.53e-01 (4.49e-02) (≈)**	8.18e-01 (1.02e-01) (+)	5.71e-01 (5.98e-02) (+)	4.90e-01 (7.99e-02) (+)	7.26e-01 (1.16e-01) (+)	1.15e+00 (1.44e-01) (+)	7.92e-01 (9.96e-02) (+)	8.09e-01 (1.46e-01) (+)
D3-DI747	**1.28e-01 (1.92e-02)**	3.73e-01 (3.48e-02) (+)	4.79e-01 (3.71e-02) (+)	4.88e-01 (3.55e-02) (+)	3.04e-01 (6.31e-02) (+)	9.70e-01 (8.13e-02) (+)	1.01e+00 (1.30e-01) (+)	5.06e-01 (2.77e-02) (+)	7.99e-01 (6.52e-02) (+)
D3-DI761	**1.50e-01 (1.53e-02)**	5.17e-01 (1.11e-01) (+)	5.27e-01 (7.74e-02) (+)	6.09e-01 (5.50e-02) (+)	6.46e-01 (5.57e-02) (+)	9.43e-01 (1.48e-01) (+)	1.00e+00 (1.63e-01) (+)	6.30e-01 (7.99e-02) (+)	9.68e-01 (1.21e-01) (+)
D3-DI774	**1.70e-01 (2.66e-02)**	3.58e-01 (5.19e-02) (+)	5.88e-01 (9.87e-02) (+)	7.18e-01 (5.34e-02) (+)	2.11e-01 (3.98e-02) (+)	9.49e-01 (1.42e-01) (+)	1.04e+00 (1.58e-01) (+)	7.06e-01 (1.09e-01) (+)	7.10e-01 (1.38e-01) (+)
D3-DI787	**2.78e-01 (3.26e-02)**	3.89e-01 (4.09e-02) (+)	4.80e-01 (7.01e-02) (+)	4.36e-01 (6.69e-02) (+)	5.02e-01 (8.45e-02) (+)	9.72e-01 (1.36e-01) (+)	9.91e-01 (1.61e-01) (+)	5.70e-01 (7.04e-02) (+)	8.54e-01 (1.66e-01) (+)
D3-DI800	**3.00e-03 (3.98e-04)**	2.40e-01 (4.14e-02) (+)	7.19e-01 (1.63e-01) (+)	6.81e-01 (3.93e-02) (+)	3.84e-01 (4.24e-02) (+)	8.93e-01 (1.81e-01) (+)	9.08e-01 (9.60e-02) (+)	5.57e-01 (6.87e-02) (+)	9.08e-01 (1.36e-01) (+)
D3-DI814	**1.94e-01 (1.67e-02)**	5.13e-01 (1.15e-01) (+)	5.07e-01 (3.64e-02) (+)	**2.01e-01 (2.55e-02) (≈)**	5.52e-01 (9.72e-02) (+)	7.19e-01 (7.98e-02) (+)	9.95e-01 (1.00e-01) (+)	6.10e-01 (5.30e-02) (+)	9.52e-01 (1.51e-01) (+)
D3-DI827	**5.49e-01 (9.08e-02)**	6.26e-01 (4.45e-02) (+)	6.26e-01 (6.39e-02) (+)	6.06e-01 (5.98e-02) (+)	**5.49e-01 (2.94e-02) (≈)**	9.30e-01 (7.02e-02) (+)	1.28e+00 (1.20e-01) (+)	**5.54e-01 (5.75e-02) (≈)**	8.22e-01 (1.38e-01) (+)
D3-DI840	**1.93e-01 (2.17e-02)**	3.34e-01 (2.85e-02) (+)	5.90e-01 (5.24e-02) (+)	8.36e-01 (7.10e-02) (+)	4.42e-01 (5.79e-02) (+)	1.06e+00 (1.65e-01) (+)	9.80e-01 (1.65e-01) (+)	6.14e-01 (5.61e-02) (+)	8.99e-01 (1.72e-01) (+)
+ / ≈ / −		38/1/1	40/0/0	39/1/0	38/2/0	40/0/0	40/0/0	38/1/1	40/0/0

Table 14

STATISTICAL RESULTS OF HGMAPPO VERSUS OTHER COMPARISON ALGORITHMS ON IGD METRIC UNDER DIFFERENT OPERATION-GRAPH DEPTHS.

Instances	HGMAPPO	DQN	TD3	SAC	DDPG	AMOSA	MOACO	MOPSO	NSGA-II
D0-I361	**4.56e-02 (1.32e-03)**	1.03e-01 (3.85e-03) (+)	2.92e-01 (8.68e-03) (+)	3.20e-01 (1.79e-02) (+)	1.63e-01 (6.31e-03) (+)	6.86e-01 (2.06e-02) (+)	6.26e-01 (2.02e-02) (+)	4.39e-01 (1.44e-02) (+)	5.56e-01 (2.29e-02) (+)
D0-I374	3.45e-02 (1.43e-03)	1.14e-01 (2.38e-03) (+)	2.81e-01 (1.56e-02) (+)	**3.13e-02 (9.22e-03) (≈)**	**1.94e-02 (6.51e-03) (-)**	5.54e-01 (2.39e-02) (+)	5.97e-01 (2.50e-02) (+)	3.53e-01 (2.23e-02) (+)	4.57e-01 (1.07e-02) (+)
D0-I387	**7.07e-02 (3.36e-03)**	2.00e-01 (9.68e-03) (+)	3.22e-01 (6.85e-03) (+)	2.53e-01 (1.56e-02) (+)	2.92e-01 (1.22e-02) (+)	5.04e-01 (1.95e-02) (+)	4.35e-01 (1.97e-02) (+)	3.55e-01 (6.86e-03) (+)	5.70e-01 (1.83e-02) (+)
D0-I401	**1.63e-01 (8.07e-03)**	**1.69e-01 (4.29e-03) (≈)**	2.36e-01 (6.55e-03) (+)	2.42e-01 (1.17e-02) (+)	2.59e-01 (9.70e-03) (+)	3.97e-01 (1.60e-02) (+)	4.40e-01 (1.20e-02) (+)	2.60e-01 (1.33e-02) (+)	4.26e-01 (1.04e-02) (+)
D0-I414	4.46e-02 (8.77e-04)	1.71e-01 (8.37e-03) (+)	2.42e-01 (1.15e-02) (+)	2.97e-01 (1.21e-02) (+)	1.76e-01 (6.73e-03) (+)	6.35e-01 (1.67e-02) (+)	**6.12e-03 (1.46e-02) (-)**	3.32e-01 (1.98e-02) (+)	4.54e-01 (2.20e-02) (+)
D0-I427	**4.20e-02 (1.34e-03)**	9.96e-02 (5.28e-03) (+)	2.90e-01 (8.99e-03) (+)	3.20e-01 (1.62e-02) (+)	1.72e-01 (8.19e-03) (+)	6.42e-01 (3.29e-02) (+)	6.09e-01 (2.05e-02) (+)	3.53e-01 (1.28e-02) (+)	4.70e-01 (2.13e-02) (+)
D0-I440	**5.04e-02 (1.47e-03)**	9.22e-02 (2.53e-03) (+)	2.47e-01 (1.09e-02) (+)	2.97e-01 (1.67e-02) (+)	1.78e-01 (7.44e-03) (+)	6.57e-01 (2.44e-02) (+)	5.96e-01 (3.16e-02) (+)	3.91e-01 (9.36e-03) (+)	4.75e-01 (1.63e-02) (+)
D0-I454	**1.07e-01 (2.88e-03)**	2.14e-01 (7.29e-03) (+)	3.12e-01 (7.41e-03) (+)	2.69e-01 (1.39e-02) (+)	3.17e-01 (7.02e-03) (+)	5.01e-01 (1.37e-02) (+)	4.82e-01 (2.10e-02) (+)	3.27e-01 (2.09e-02) (+)	5.58e-01 (2.32e-02) (+)
D0-I467	**7.36e-02 (2.37e-03)**	1.20e-01 (4.60e-03) (+)	2.05e-01 (5.43e-03) (+)	2.06e-01 (1.18e-02) (+)	2.43e-01 (1.18e-02) (+)	3.80e-01 (1.75e-02) (+)	4.28e-01 (1.72e-02) (+)	2.76e-01 (1.00e-02) (+)	4.21e-01 (2.51e-02) (+)
D0-I480	**4.61e-02 (1.95e-03)**	1.17e-01 (2.25e-03) (+)	2.92e-01 (1.72e-02) (+)	3.46e-01 (1.40e-02) (+)	1.90e-01 (1.01e-02) (+)	6.04e-01 (2.18e-02) (+)	6.32e-01 (2.89e-02) (+)	4.31e-01 (1.36e-02) (+)	5.55e-01 (2.75e-02) (+)
D1-DI481	**2.02e-01 (3.66e-02)**	**2.31e-01 (4.62e-02) (≈)**	2.38e-01 (1.85e-02) (+)	5.67e-01 (4.30e-02) (+)	**2.11e-01 (3.20e-02) (≈)**	6.88e-01 (1.11e-01) (+)	5.89e-01 (8.55e-02) (+)	3.85e-01 (7.26e-02) (+)	7.81e-01 (1.08e-01) (+)
D1-DI494	**3.00e-03 (3.66e-04)**	2.07e-01 (2.52e-02) (+)	2.90e-01 (2.20e-02) (+)	3.00e-01 (2.31e-02) (+)	2.23e-01 (6.01e-02) (+)	6.02e-01 (7.95e-02) (+)	5.60e-01 (6.08e-02) (+)	5.61e-01 (7.01e-02) (+)	7.18e-01 (8.64e-02) (+)
D1-DI507	**1.39e-01 (2.22e-02)**	1.87e-01 (2.55e-02) (+)	4.36e-01 (5.63e-02) (+)	2.75e-01 (7.78e-02) (+)	2.01e-01 (1.87e-02) (+)	1.66e-01 (1.25e-02) (+)	6.21e-01 (1.07e-01) (+)	4.02e-01 (2.50e-02) (+)	7.72e-01 (1.37e-01) (+)
D1-DI521	**1.25e-01 (1.06e-02)**	2.31e-01 (2.36e-02) (+)	2.54e-01 (4.30e-02) (+)	3.79e-01 (5.50e-02) (+)	3.57e-01 (7.17e-02) (+)	5.54e-01 (5.00e-02) (+)	5.41e-01 (1.02e-01) (+)	4.63e-01 (4.32e-02) (+)	6.18e-01 (5.98e-02) (+)
D1-DI534	**1.24e-01 (1.35e-02)**	2.79e-01 (5.36e-02) (+)	4.38e-01 (6.05e-02) (+)	2.83e-01 (2.61e-02) (+)	2.83e-01 (2.27e-02) (+)	7.68e-01 (1.04e-01) (+)	4.66e-01 (3.82e-02) (+)	5.02e-01 (5.63e-02) (+)	6.01e-01 (4.58e-02) (+)
D1-DI547	3.59e-02 (8.29e-03)	2.07e-01 (2.83e-02) (+)	5.92e-02 (4.71e-03) (+)	3.78e-01 (4.43e-02) (+)	**3.01e-02 (2.59e-03) (-)**	5.34e-01 (7.07e-02) (+)	6.09e-01 (9.21e-02) (+)	3.38e-01 (4.45e-02) (+)	6.45e-01 (7.97e-02) (+)
D1-DI560	**8.80e-03 (1.01e-03)**	1.66e-01 (3.21e-02) (+)	3.47e-01 (4.34e-02) (+)	2.40e-01 (4.49e-02) (+)	2.38e-01 (2.92e-02) (+)	6.20e-01 (3.19e-02) (+)	5.74e-01 (6.62e-02) (+)	4.67e-01 (6.72e-02) (+)	5.68e-01 (6.35e-02) (+)
D1-DI574	**2.23e-01 (2.18e-02)**	2.90e-01 (2.70e-02) (+)	3.33e-01 (2.54e-02) (+)	3.75e-01 (5.04e-02) (+)	3.26e-01 (7.03e-02) (+)	6.04e-01 (1.05e-01) (+)	7.22e-01 (1.24e-01) (+)	4.61e-01 (3.74e-02) (+)	8.77e-01 (9.12e-02) (+)
D1-DI587	**3.00e-03 (2.65e-04)**	1.92e-01 (1.57e-02) (+)	2.98e-01 (4.44e-02) (+)	3.91e-01 (2.83e-02) (+)	2.33e-01 (2.55e-02) (+)	3.02e-01 (4.91e-02) (+)	5.66e-01 (8.65e-02) (+)	4.41e-01 (4.44e-02) (+)	6.30e-01 (5.80e-02) (+)
D1-DI600	**1.37e-01 (1.27e-02)**	1.70e-01 (2.53e-02) (+)	3.22e-01 (2.12e-02) (+)	4.18e-01 (2.85e-02) (+)	1.86e-01 (4.23e-02) (+)	7.05e-01 (1.19e-01) (+)	5.28e-01 (8.19e-02) (+)	4.52e-01 (7.48e-02) (+)	5.64e-01 (4.15e-02) (+)
D2-DI601	**2.24e-01 (5.24e-02)**	2.58e-01 (1.50e-02) (+)	4.51e-01 (3.46e-02) (+)	4.43e-01 (8.66e-02) (+)	2.76e-01 (3.65e-02) (+)	5.62e-01 (4.68e-02) (+)	7.74e-01 (1.21e-01) (+)	4.72e-01 (4.50e-02) (+)	4.96e-01 (8.56e-02) (+)
D2-DI614	**1.10e-01 (1.83e-02)**	4.20e-01 (3.06e-02) (+)	4.77e-01 (3.07e-02) (+)	5.12e-01 (8.99e-02) (+)	2.79e-01 (4.78e-02) (+)	6.08e-01 (5.31e-02) (+)	9.28e-01 (1.67e-01) (+)	5.57e-01 (1.40e-01) (+)	6.16e-01 (6.09e-02) (+)
D2-DI627	**1.53e-01 (1.13e-02)**	2.77e-01 (1.73e-02) (+)	5.38e-01 (9.59e-02) (+)	6.31e-01 (5.40e-02) (+)	7.30e-01 (1.36e-01) (+)	8.44e-01 (7.24e-02) (+)	7.36e-01 (2.08e-01) (+)	6.59e-01 (6.23e-02) (+)	7.73e-01 (7.59e-02) (+)
D2-DI641	**1.40e-01 (2.94e-02)**	2.96e-01 (5.40e-02) (+)	3.70e-01 (6.78e-02) (+)	4.76e-01 (7.73e-02) (+)	5.38e-01 (5.41e-02) (+)	9.91e-01 (1.80e-01) (+)	6.98e-01 (1.02e-01) (+)	4.89e-01 (4.35e-02) (+)	6.05e-01 (9.45e-02) (+)
D2-DI654	**1.17e-01 (1.00e-02)**	3.98e-01 (7.83e-02) (+)	3.54e-01 (4.94e-02) (+)	2.74e-01 (2.31e-02) (+)	4.16e-01 (9.37e-02) (+)	1.00e+00 (1.34e-01) (+)	7.46e-01 (1.10e-01) (+)	5.47e-01 (4.23e-02) (+)	7.29e-01 (1.26e-01) (+)
D2-DI667	**2.82e-01 (2.84e-02)**	3.50e-01 (3.61e-02) (+)	4.53e-01 (6.92e-02) (+)	3.33e-01 (2.30e-02) (+)	3.15e-01 (1.65e-02) (+)	8.00e-01 (6.82e-02) (+)	7.67e-01 (1.24e-01) (+)	5.33e-01 (3.74e-02) (+)	9.44e-01 (9.49e-02) (+)
D2-DI680	**1.41e-01 (1.13e-02)**	2.52e-01 (5.23e-02) (+)	4.40e-01 (7.34e-02) (+)	5.38e-01 (4.69e-02) (+)	2.14e-01 (3.76e-02) (+)	7.72e-01 (6.81e-02) (+)	7.91e-01 (9.05e-02) (+)	3.91e-01 (6.16e-02) (+)	7.23e-01 (5.01e-02) (+)
D2-DI694	**3.86e-01 (4.75e-02)**	4.51e-01 (6.12e-02) (+)	4.51e-01 (7.08e-02) (+)	4.54e-01 (6.01e-02) (+)	4.40e-01 (4.46e-02) (+)	7.43e-01 (1.18e-01) (+)	8.84e-01 (1.62e-01) (+)	5.30e-01 (6.79e-02) (+)	5.66e-01 (6.24e-02) (+)
D2-DI707	**1.97e-01 (1.91e-02)**	3.34e-01 (4.27e-02) (+)	7.36e-01 (7.67e-02) (+)	2.75e-01 (3.29e-02) (+)	3.39e-01 (3.72e-02) (+)	8.14e-01 (5.51e-02) (+)	8.17e-01 (1.42e-01) (+)	**1.99e-01 (1.90e-02) (≈)**	6.26e-01 (9.82e-02) (+)
D2-DI720	**2.39e-01 (4.84e-02)**	2.97e-01 (3.84e-02) (+)	2.97e-01 (6.00e-02) (+)	4.97e-01 (5.68e-02) (+)	2.97e-01 (3.68e-02) (+)	8.77e-01 (1.76e-01) (+)	6.69e-01 (6.34e-02) (+)	3.99e-01 (7.87e-02) (+)	6.45e-01 (5.88e-02) (+)
D3-DI721	**3.01e-01 (5.79e-02)**	3.43e-01 (5.30e-02) (+)	5.33e-01 (5.95e-02) (+)	5.63e-01 (1.06e-01) (+)	6.72e-01 (1.57e-01) (+)	7.00e-01 (1.23e-01) (+)	8.80e-01 (1.15e-01) (+)	7.48e-01 (9.00e-02) (+)	1.08e+00 (1.90e-01) (+)
D3-DI734	**1.90e-01 (2.47e-02)**	6.83e-01 (1.23e-01) (+)	7.16e-01 (8.91e-02) (+)	5.59e-01 (1.24e-01) (+)	3.92e-01 (6.04e-02) (+)	1.11e+00 (2.11e-01) (+)	9.08e-01 (1.12e-01) (+)	7.35e-01 (1.30e-01) (+)	9.53e-01 (9.68e-02) (+)
D3-DI747	**2.44e-01 (4.33e-02)**	5.72e-01 (9.80e-02) (+)	7.83e-01 (9.09e-02) (+)	4.62e-01 (8.12e-02) (+)	4.52e-01 (6.76e-02) (+)	1.04e+00 (1.06e-01) (+)	9.49e-01 (1.73e-01) (+)	6.55e-01 (5.86e-02) (+)	1.10e+00 (1.24e-01) (+)
D3-DI761	**1.64e-01 (3.03e-02)**	5.59e-01 (3.57e-02) (+)	7.18e-01 (1.10e-01) (+)	4.69e-01 (6.89e-02) (+)	5.89e-01 (7.37e-02) (+)	1.10e+00 (7.22e-02) (+)	1.18e+00 (1.71e-01) (+)	7.29e-01 (8.22e-02) (+)	1.07e+00 (1.84e-01) (+)
D3-DI774	2.13e-01 (3.02e-02)	**2.04e-01 (2.37e-02) (≈)**	8.79e-01 (8.96e-02) (+)	6.32e-01 (5.96e-02) (+)	5.06e-01 (4.55e-02) (+)	1.01e+00 (1.89e-01) (+)	1.06e+00 (1.22e-01) (+)	7.43e-01 (1.30e-01) (+)	1.14e+00 (7.71e-02) (+)
D3-DI787	**3.35e-01 (3.91e-02)**	**3.37e-01 (8.98e-02) (≈)**	5.79e-01 (1.17e-01) (+)	5.18e-01 (9.12e-02) (+)	4.50e-01 (5.25e-02) (+)	1.12e+00 (1.36e-01) (+)	9.47e-01 (2.09e-01) (+)	7.40e-01 (1.19e-01) (+)	1.02e+00 (1.25e-01) (+)
D3-DI800	2.50e-01 (3.23e-02)	5.18e-01 (3.60e-02) (+)	7.59e-01 (1.36e-01) (+)	**2.10e-01 (1.63e-02) (-)**	5.61e-01 (4.26e-02) (+)	1.05e+00 (1.55e-01) (+)	7.71e-01 (1.02e-01) (+)	7.29e-01 (1.58e-01) (+)	1.17e+00 (7.82e-02) (+)
D3-DI814	**3.43e-01 (1.67e-02)**	7.09e-01 (1.08e-01) (+)	7.38e-01 (1.68e-01) (+)	1.10e+00 (1.65e-01) (+)	4.80e-01 (1.25e-01) (+)	9.84e-01 (5.48e-02) (+)	8.95e-01 (1.38e-01) (+)	7.17e-01 (9.82e-02) (+)	1.02e+00 (8.05e-02) (+)
D3-DI827	**1.03e-01 (1.27e-02)**	5.83e-01 (7.00e-02) (+)	8.69e-01 (1.71e-01) (+)	6.74e-01 (1.79e-01) (+)	5.86e-01 (7.18e-02) (+)	1.08e+00 (7.71e-02) (+)	8.14e-01 (1.15e-01) (+)	6.90e-01 (8.16e-02) (+)	1.15e+00 (1.46e-01) (+)
D3-DI840	**3.00e-03 (6.50e-04)**	2.81e-01 (6.20e-02) (+)	7.08e-01 (1.35e-01) (+)	5.50e-01 (1.29e-01) (+)	8.18e-01 (1.28e-01) (+)	1.22e+00 (1.89e-01) (+)	1.01e+00 (8.97e-02) (+)	6.33e-01 (1.25e-01) (+)	1.02e+00 (1.09e-01) (+)
+ / ≈ / −		36/4/0	40/0/0	38/1/1	37/1/2	40/0/0	39/0/1	39/1/0	40/0/0

PART 7. IMPLEMENTATION DETAILS OF COMPARISON ALGORITHMS

1) **NSGA-II chromosome representation:** For the NSGA-II baseline, a scheduling solution is represented by a three-component chromosome consisting of an operation-sequence component, a device-assignment component, and a server-assignment component. The operation-sequence and device-assignment components have the same length as the total number of manufacturing operations, whereas the server-assignment component has the same length as the number of computing tasks. In the operation-sequence component, the index of manufacturing task M_i appears as many times as the number of its operations. Its r-th occurrence is decoded as the r-th operation of M_i, thereby preserving the precedence relations among $O_{i,j}$ belonging to the same manufacturing task. The device-assignment component is indexed by the fixed operation identity (i, j), rather than by the current position of $O_{i,j}$ in the operation-sequence component. Each gene records an eligible device M_k for the corresponding operation, resulting in $X_{i,j,k} = 1$. The server-assignment component is indexed by computing task C_i, and each gene records an eligible edge server $\mathcal{E}_l$ or cloud server $\mathcal{C}_{l'}$, from which B_i is determined. Therefore, modifying the operation sequence does not change the associations represented by the other two components.

2) **Crossover operators:** A precedence operation crossover (POX) is applied to the operation-sequence component. The manufacturing-task set is randomly partitioned into two subsets. All occurrences associated with the first subset are inherited from one parent at their original positions, whereas the unfilled positions are filled with the occurrences associated with the second subset in the order in which they appear in the other parent. This procedure preserves the required number of occurrences of every manufacturing-task index and prevents operations from being omitted or duplicated. Together with occurrence-based decoding, it also maintains the internal precedence relations among the operations of each manufacturing task. Uniform crossover is independently applied to the device- and server-assignment components. For each fixed operation or computing-task index, the corresponding gene is inherited from either parent with equal probability. Because the parental genes at the same index represent eligible assignments for the same operation or computing task, the resulting assignment remains eligible.

3) **Mutation operators:** Each chromosome component is independently selected for mutation with a probability of 0.1. When mutation is triggered for the operation-sequence component, one manufacturing-task index is removed and reinserted at another position. Because operations are identified according to the occurrence order of their manufacturing-task indices, this insertion does not violate the internal precedence relations. When mutation is triggered for the device- or server-assignment component, one gene is randomly selected and reset to another candidate in its corresponding eligible set. Thus, only one insertion or one feasible random-reset operation is performed when a component is selected for mutation. The value 0.1 consequently denotes the component-wise mutation probability rather than the mutation probability of each gene.

4) **Feasibility handling and decoding:** The device- and server-assignment components are initialized and mutated only within their corresponding eligible candidate sets; therefore, no additional assignment-repair procedure is required. A device is eligible when it can process the corresponding operation $O_{i,j}$. A server is eligible when its total CPU and memory capacities can accommodate the corresponding computing task C_i. Instantaneous CPU and memory availability is checked by the decoder only when C_i becomes ready for execution. The decoder processes the operation sequence from left to right. Each operation is assigned its earliest feasible start time subject to the precedence, device-availability, and transportation constraints in Section III. The start time of each computing task is subsequently determined according to the manufacturing-completion, data-arrival, predecessor-completion, and instantaneous CPU and memory availability constraints. If the selected server has insufficient available resources, C_i waits until sufficient resources have been released.

5) **NSGA-II selection:** Binary tournament selection based on nondomination rank and crowding distance is used for parent selection. After generating the offspring, the parent and offspring populations are combined, and the next population is selected using the standard elitist NSGA-II environmental-selection procedure. The makespan and TEC defined in the manuscript are directly used as the two optimization objectives.

All learning-based baselines were trained for 1,200 episodes using a discount factor of 0.99, a batch size of 128, and a replay-buffer capacity of 10^5. The same feasible-action masks and training instances were used for all learning-based methods. All multi-objective metaheuristic baselines were terminated after 20,000 objective evaluations in each independent run.

Table 15
PARAMETER SETTINGS OF THE COMPARISON ALGORITHMS.

Algorithm	Parameter settings
DQN	Learning rate = 1×10^{-4}; ε decreases linearly from 1.0 to 0.05; target-network update interval = 200 gradient steps
DDPG	Actor learning rate = 1×10^{-4}; critic learning rate = 1×10^{-3}; soft-update coefficient = 0.005; Gaussian exploration-noise standard deviation decreases from 0.20 to 0.05
TD3	Actor/critic learning rate = 3×10^{-4}; soft-update coefficient = 0.005; policy-update delay = 2; target-policy noise = 0.20; noise clip = 0.50; exploration-noise standard deviation = 0.10
SAC	Actor/critic learning rate = 1×10^{-4}; soft-update coefficient = 0.005; entropy coefficient = automatically adjusted
AMOSA	Initial temperature = 1000; cooling rate = 0.995; archive size = 100
MOACO	Number of ants = 100; pheromone factor = 1; heuristic factor = 2; evaporation rate = 0.10; initial pheromone = 1; archive size = 100
MOPSO	Swarm size = 100; inertia weight decreases linearly from 0.9 to 0.4; $c_1=c_2=2.0$; archive size = 100
NSGA-II	Population size = 200; POX probability for the operation-sequence component = 0.9; uniform-crossover probability for the device- and server-assignment components = 0.9; component-wise mutation probability = 0.1; tournament size = 2